

# **Cellular Expert Desktop SAT for ArcGIS Pro User Guide**

Version 5.0



# Table of Contents

|       |                                              |    |
|-------|----------------------------------------------|----|
| 1.    | Software Purpose and Functionality           | 6  |
| 2.    | System requirements                          | 8  |
| 2.1   | Minimal Requirements for Hardware            | 8  |
| 2.2   | Minimal Requirements for Software            | 8  |
| 3.    | License types                                | 9  |
| 3.1   | Single-User Environment                      | 9  |
| 4.    | Getting Started                              | 10 |
| 4.1   | Installation                                 | 10 |
| 4.2   | Activation                                   | 10 |
| 4.3   | Tools                                        | 11 |
| 5.    | Geographic data                              | 13 |
| 5.1   | Geographic data requirements                 | 15 |
| 5.1.1 | Digital Terrain Model (DTM) Grid (Mandatory) | 16 |
| 5.1.2 | Clutter classes grid                         | 19 |
| 5.1.3 | Clutter heights                              | 25 |
| 5.1.4 | Buildings                                    | 29 |
| 6.    | Workspace                                    | 33 |
| 6.1   | Workspace Tool                               | 33 |
| 6.1.1 | Workspace Table                              | 33 |
| 6.1.2 | Create Workspace                             | 35 |
| 6.1.3 | Open Workspace                               | 40 |
| 6.1.4 | Remove Workspace                             | 40 |
| 6.1.5 | Workspace Upgrade                            | 40 |
| 6.1.6 | Workspace Properties                         | 42 |
| 6.2   | Docs Manager                                 | 48 |
| 6.3   | CE Express Connection                        | 53 |
| 6.3.1 | Properties                                   | 53 |
| 7.    | Data Management                              | 55 |
| 7.1   | Network Objects                              | 55 |
| 7.2   | Add Object                                   | 55 |

|        |                                                |     |
|--------|------------------------------------------------|-----|
| 7.2.1  | Add Satellite                                  | 56  |
| 7.2.2  | Add Ground Station                             | 56  |
| 7.2.3  | Feature Set Templates                          | 59  |
| 7.2.4  | Batch Creation                                 | 60  |
| 7.3    | Object Editor                                  | 61  |
| 7.3.1  | Move Objects                                   | 63  |
| 7.3.2  | Duplicate Objects                              | 67  |
| 7.4    | Calculate Cells Area                           | 70  |
| 7.5    | Clutter Classes                                | 72  |
| 7.6    | Antenna Viewer                                 | 76  |
| 7.6.1  | Preview Antenna Patterns                       | 76  |
| 7.7    | Prediction Model Manager                       | 77  |
| 7.7.1  | Models                                         | 78  |
| 7.8    | Template Manager                               | 103 |
| 7.8.1  | Edit Network Objects template                  | 104 |
| 7.8.2  | Manage Network Object Template                 | 105 |
| 7.8.3  | Modulations                                    | 105 |
| 7.8.4  | Modulation Editor                              | 107 |
| 7.9    | Import Objects                                 | 109 |
| 7.9.1  | Import Cells                                   | 110 |
| 7.9.2  | Import Sites, Radar, CPE, Sirens, or Repeaters | 113 |
| 7.10   | Import/Export Antenna Files                    | 114 |
| 7.10.1 | Import Antennas                                | 114 |
| 7.10.2 | Export Antennas                                | 114 |
| 7.10.3 | Create Antennas                                | 115 |
| 7.11   | Model Tuning                                   | 117 |
| 8.     | Profile                                        | 122 |
| 8.1    | Profile Tool                                   | 122 |
| 8.1.1  | Properties                                     | 122 |
| 8.1.2  | Draw Profile                                   | 127 |
| 8.1.3  | Tools                                          | 134 |
| 8.1.4  | Import                                         | 138 |
| 8.1.5  | Export (Profile Report)                        | 139 |
| 8.1.6  | Settings                                       | 141 |

|      |                          |     |
|------|--------------------------|-----|
| 9.   | Coverage Prediction      | 142 |
| 9.1  | CE Calculation Task List | 142 |
| 9.2  | Quick SAT Link Budget    | 143 |
| 9.3  | Sky Clearance            | 143 |
| 10.  | About                    | 148 |
| 10.1 | License information      | 148 |
| 10.2 | Help                     | 149 |
|      | 10.2.1 Documentation     | 149 |
|      | 10.2.2 What's New        | 149 |
|      | 10.2.3 Technical Support | 150 |
|      | 10.2.4 Error log         | 150 |
| 11.  | Technical Support        | 152 |



## About Cellular Expert

Cellular Expert UAB (CE) developed ultra-fast wave propagation, communication systems deployment planning, and radio/optical visibility calculation software for ESRI's ArcGIS mapping environment, which is widely used within Telecom, Defense, IoT, and other companies and organizations.

CE's communication network planning, network asset management, operational support software, and customer-tailored solutions enhance the intelligence and business efficiency of more than 170 communication network companies, regulators, and defense organizations in more than 50 countries.

Copyright © 2026 UAB CELLULAR EXPERT. All rights reserved. Cellular Expert and Cellular Expert logo are registered trademarks, @cellular-expert.com and www.cellular-expert.com are service marks of UAB CELLULAR EXPERT in Lithuania and some other countries.

## 1. Software Purpose and Functionality

Cellular Expert Desktop for ArcGIS Pro (CE Pro) is a professional radio coverage planning software tool designed on top of the Esri ArcGIS Pro environment. The CE Pro is a highly versatile and functional tool, covering the network design functionality needs of most professional radio communication planners and developers. It could support network planning and optimization for the entire range of wireless technologies in frequencies from 10 kHz to 350 GHz.

The key features and benefits of utilizing Cellular Expert software are as follows:

1. CE Pro is a purpose-built extension of ESRI software, a world-renowned standard in GIS environments. This means that along with wireless network planning and optimization, it gives users various value-added GIS/mapping functionalities to manage, visualize feature classes or rosters, create reports, and graphs, or automate workflows,
2. CE Pro is made of several processing engines and functional modules and therefore may be easily configured for various business functions: network planning, network engineering, technology information support to sales and marketing, network assets management, and network operational maintenance and supervision,
3. CE Pro is maintained and constantly developed by a dedicated agile team of software developers and radio planners, allowing attentive responsiveness to customer needs and developing custom-tailored solutions.

CE Pro performs mobile network coverage analysis based on:

1. Site and cell level details of radio equipment complement, specifications, cell and component carriers configuration,
2. Subscriber characteristics – geographic distribution density, generated traffic to estimate network/cell loading,
3. DTM, buildings, and clutter-based deterministic point-to-area radio wave propagation modeling.

CE Pro allows the user to input and use a broad variety of GIS and network data to support the simulations, as the overall quality of coverage calculations is dependent on the completeness and detail of technical network data and the resolution and quality of the GIS data. CE Pro can efficiently simulate wide area network coverage using the GIS data with a resolution of down to sub-meter.

Calculated results of coverage predictions could be presented as coverage raster maps.

CE Pro is capable of modeling various wireless technologies: cellular (2G/3G/4G/5G), PMR/PAMR (TETRA, APCO, others), FWA/BWA, IoT (LoRa/SigFox, others), as well as fixed microwave links. Therefore, it may be used as a radio planning tool in various industries: Mobile Operators, Integrated Telecom Companies, Wireless Internet Service providers, Regulatory authorities, Utilities, Broadband Infrastructure providers, Defense organizations, as well as any other users of radiocommunication systems.

In the following, we summarize the key functionalities of the tool.

### Data Management

The tool allows the importation, storage, and management of detailed technical data on network nodes, such as sites, cells, RF transmitters, and antennas.

### Signal Strength Prediction

The tool contains several in-built path loss prediction models that allow the user to easily start simulations based on the evaluation of the most essential pathloss contributing factors. The following two models constitute the starting set:

- Free space – typically used for modeling short-range mobile communications, fixed links, or other radiocommunications applications with prevalent Line-of-Sight conditions on propagation paths,

- UniMacro – a proprietary universal model for wide area mobile communication systems that flexibly accounts for a variety of propagation paths as determined for each specific reception point based on terrain and clutter data vs. configuration of the modeled system.

Further details on path loss models and their configuration options are provided in the relevant section of this manual.

### **Model Tuning**

The tool provides an automated method for fine-tuning propagation model parameters to fit the specific scenario and type of area of network deployment based on analyzing real field strength measurement results.

### **Radio Coverage Calculation for Different Mobile Technologies**

Radio coverage calculation is assisted in the tool by the possibility to use pre-loaded cell configuration templates that are tailored to typical technical parameters of base stations in different Radio Access Technologies. Accordingly, the model settings and outputs will be adjusted to suit the scenario pertinent to that technology, i.e.:

- 2G – radio coverage is calculated in dBm as receive power level of narrow-band (200 kHz) signal,
- 3G – radio coverage is calculated in dBm as the receive power level of a single broadband (3.85 MHz) carrier,
- 4G/5G – radio coverage is calculated in dBm as the equivalent RSRP of a single sub-carrier component in the complex OFDM broadband signal.
- Wi-Fi - wireless communication technology based on the IEEE 802.11 standards, used for setting up local area networks (WLANs) and providing internet access in various settings without requiring cable connections.

### **Profile Analysis**

The tool provides powerful GIS analytical features to analyze the terrain and clutter on the fixed link path, such as allowing to estimate of the Fresnel zone clearance condition, Power Budget, Path loss and Angles between Tx and Rx.

## 2. System requirements

This chapter will guide you through the minimal hardware and software requirements.

Note: requirements can vary significantly, depending on the acceptable calculation time and task complexity.

### 2.1 Minimal Requirements for Hardware

Processor (CPU):

- Minimum: 8 cores, hyperthreaded
- Recommended: 16 cores

(Optional) Requirements for GPU-accelerated calculations

- GPU – any NVIDIA GPU with CUDA capabilities (<https://developer.nvidia.com/cuda-gpus>)
- Driver version: 456.38 or later
- CUDA Toolkit 11.0 or later

Memory/RAM:

- Minimum: 16 GB
- Recommended: 32 GB

Storage:

- Minimum: 500 GB of free space
- Recommended: 2TB or more of free space on a solid-state drive (SSD)

### 2.2 Minimal Requirements for Software

Cellular Expert runs on Microsoft Windows 10 or higher operating system.

It requires:

- Microsoft .NET 10.0  
[Download .NET 10.0 Desktop Runtime \(v10.0.11\) - Windows x64 Installer](#)
- Microsoft Visual C++ Redistributable packages 2015-2022  
[https://aka.ms/vs/17/release/vc\\_redist.x64.exe](https://aka.ms/vs/17/release/vc_redist.x64.exe)
- .NET support for ArcGIS libraries
- ArcGIS Pro from version 3.7.x



### 3. License types

Only a Single Use Cellular Expert license is available. The license type is annual and dedicated to one workstation connected with ArcGIS Online, which is used for ArcGIS Pro.

#### 3.1 Single-User Environment

For the Single-User configuration of Cellular Expert, all information about radio network objects is stored in a personal geodatabase (GDB format) or locally on the disc (calculation results, raster data in GeoTIFF format, etc.).

Geographical data can be stored:

- Locally on a disc

An ArcGIS Pro license and an active ArcGIS Named User or ArcGIS Pro Standalone are required to operate in the Single-User environment.

## 4. Getting Started

Welcome to Cellular Expert.

This chapter will guide you through project creation and analysis. It includes the installation and preparation of the Cellular Expert extension, as well as using it for the first time and creating a project.

Note: the tools are not explained in this chapter. They are referenced in other chapters of this guide.

### 4.1 Installation

- If applicable, uninstall the previous version of Cellular Expert for ArcGIS Pro
- Make sure that *.NET 10.0* is installed
- Make sure that *ArcGIS Pro 3.7.x* is installed
- Run the *ceProSetup.msi* file to install the new version of Cellular Expert for ArcGIS Pro
- After successful installation, open Task Manager > Services, and run CE\_Pro\_Prediction\_Service

### 4.2 Activation

This step is required for new users. If the Cellular Expert for ArcGIS Pro license has already been activated, please skip this step.

You can activate the license in two ways:

First way:

1. Open ArcGIS Pro and select **Settings**
2. Navigate to **Licensing** → **External Extensions**
3. Find the **Cellular Expert** entry in the extensions table
4. Click the checkmark in the **Enabled** field. The **License Activation** dialog will appear
5. Copy the **User Key** and send it to [support@cellular-expert.com](mailto:support@cellular-expert.com). You will be provided with a **License Activation Key** which must be entered into the same dialogue
6. Press **Activate License** and the license will be activated, enabling the acquired versions of the add-on

Second way:

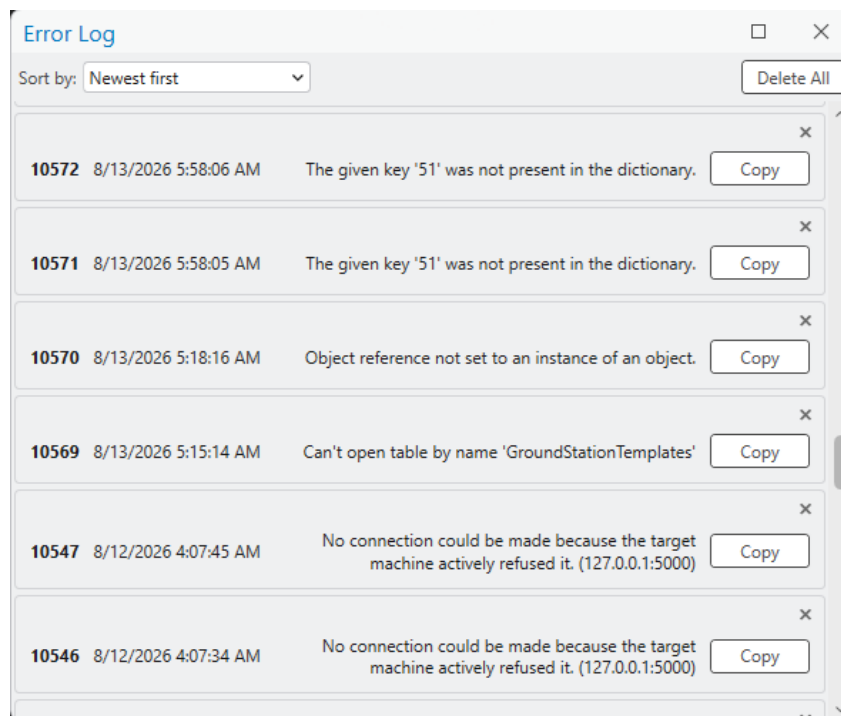
1. Create an empty ArcGIS Pro project
2. Navigate to **Insert** and select **New Map**. After the map insertion, the Cellular Expert tabs will be enabled
3. Navigate to any of the Cellular Expert tabs and select **License Information** in the **About** section.
4. Copy the **User Key** and send it to [support@cellular-expert.com](mailto:support@cellular-expert.com). You will be provided with a **License Activation Key** which must be entered into the same dialogue
5. Press **Activate License** and the license will be activated, enabling the acquired versions of the add-on

If you want to check the expiration date of the license you can:

1. Open ArcGIS Pro and start/open a project with an active map
2. Navigate to the **Cellular Expert** tab
3. Select **License Information** in the **About** section

If you encounter any problems or want additional details about the license, please contact Cellular Expert at [support@cellular-expert.com](mailto:support@cellular-expert.com). For more information, see the chapter **Click the**  **Error Log** button in the **Help** dropdown list to open the **Error Log** dialogue.

Here you will find information about errors that have occurred during certain processes of the *CE for ArcGIS Pro* extension's lifetime. These logs are crucial to improving the overall quality of the user experience. When an error happens and you decide to contact a Cellular Expert, you will be asked to send these logs so that the problems you encounter can be patched as soon as possible. The error logs can be sorted by either newest first (default option) or lowest first.



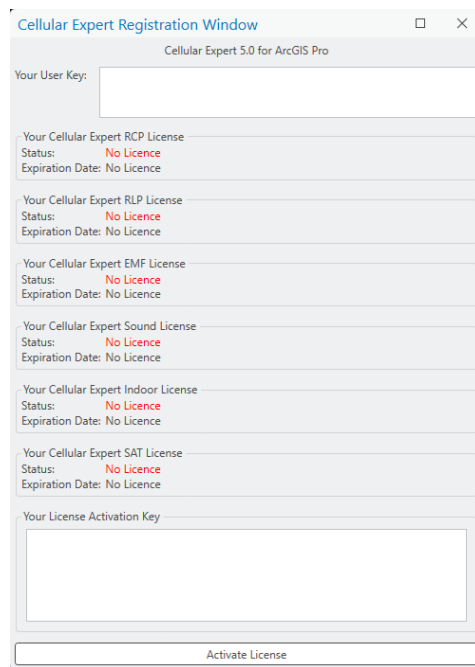
### Copy

Copy the necessary error information into your clipboard and send it to Cellular Expert. A notification appears on the screen indicating that the information was copied to the clipboard successfully.

### Delete All

If the error log ever gets too crowded, you can delete all errors from it.

## Technical Support.

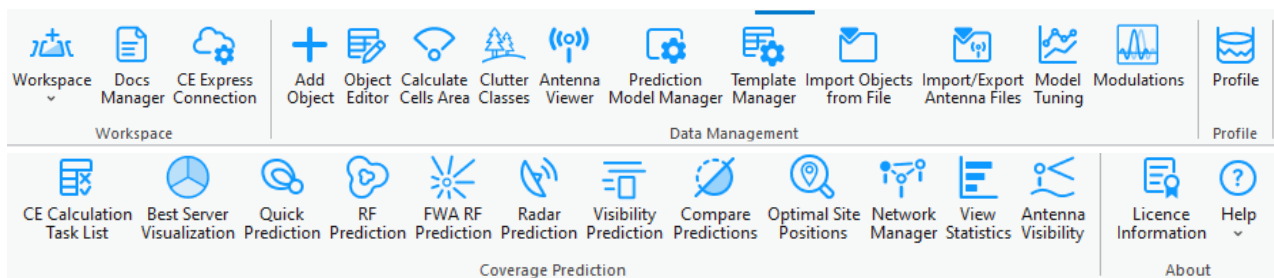


## 4.3 Tools

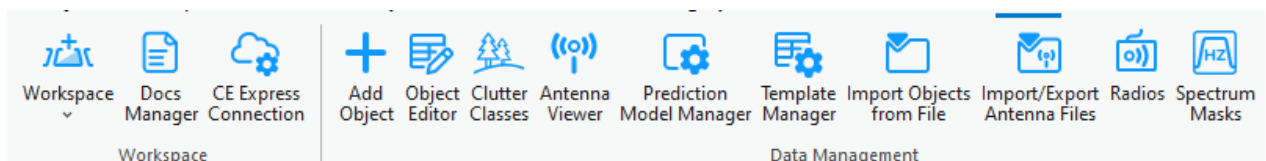
The Cellular Expert tools are in the Cellular Expert add-on. They appear automatically after installation of CE for ArcGIS Pro and will be found in the menu ribbon.

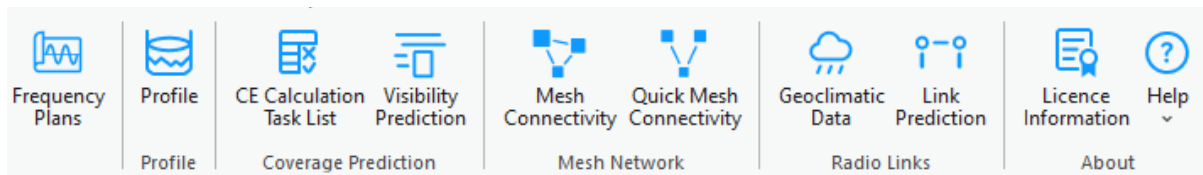
There are 6 types of licenses and therefore 6 different tabs with various tool configurations:

- RCP

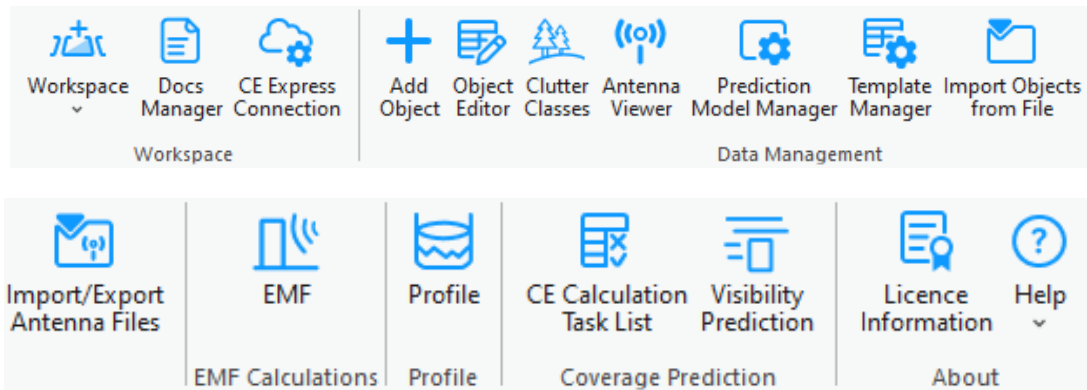


- RLP

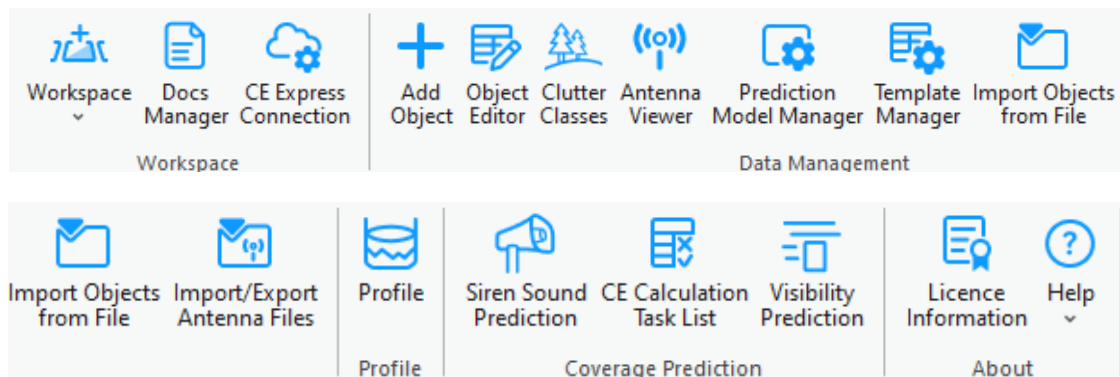




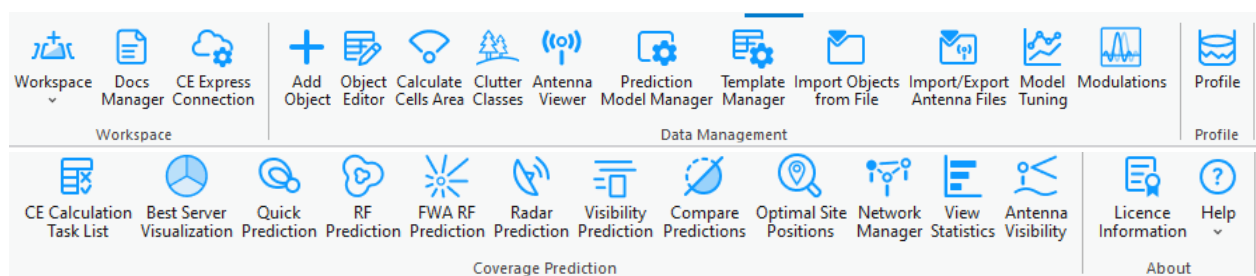
- EMF



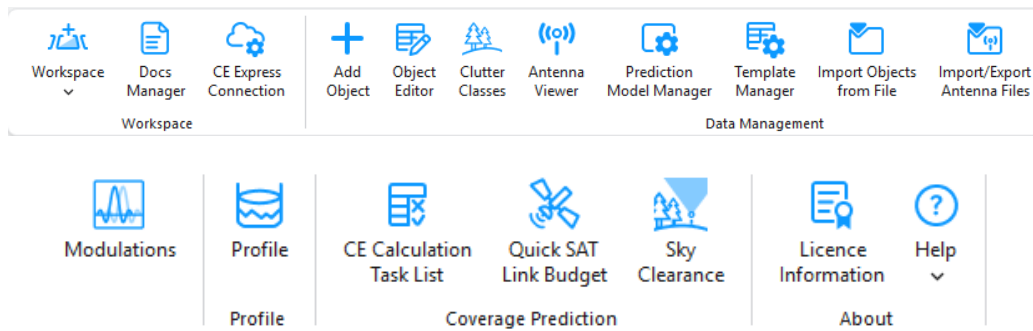
- Sound



- Indoor



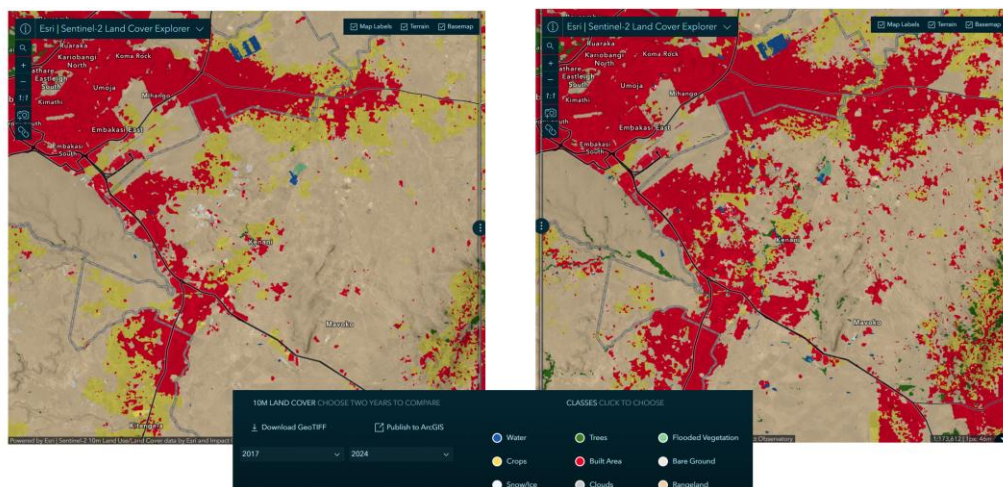
- SAT



This User Guide describes SAT tools.

## 5. Geographic data

CE Desktop is designed to work with any geospatial data available to the customer and fully exploit its precision for the most accurate coverage and QoS calculations. The platform supports multi-resolution input datasets — from freely available global sources such as Sentinel-2 10 m land cover and ASTER DEM, to premium high-resolution terrain and 3D building models when provided by the customer or government agencies.



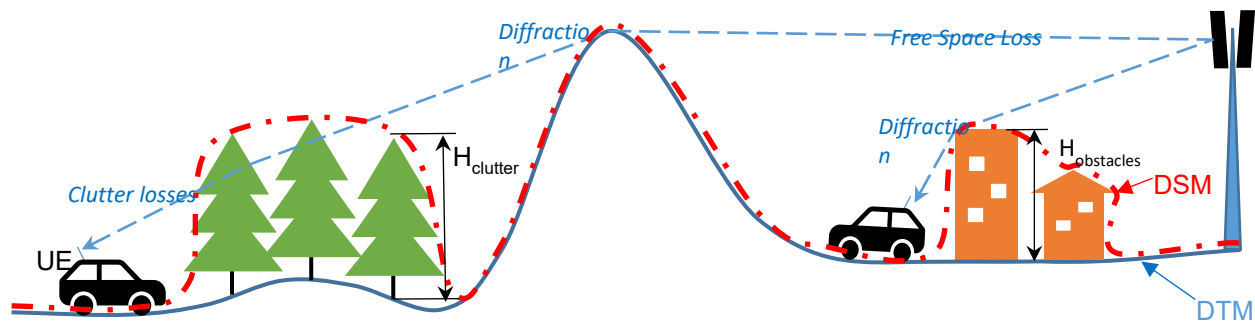
Source: <https://livingatlas.arcgis.com/landcoverexplorer/>

By leveraging whatever data is available locally, CE Express performs nationwide calculations at the maximum feasible resolution, accurately modeling signal propagation even in dense urban environments. Support for 3D multi-height calculations ensures that coverage predictions reflect street-level, indoor, and rooftop conditions, providing regulators with a realistic representation of service availability.

This flexibility ensures that NRAs can use their existing GIS assets, open datasets, or commercial data they already license, turning them into actionable broadband maps without additional data procurement requirements from the solution provider.

By using terrain elevation, obstacles, and clutter classification in every calculation, Cellular Expert accurately models:

- **Line-of-Sight and Non-Line-of-Sight Conditions** – Determining diffraction, reflection, and shadowing effects over hills, valleys, and urban obstacles.
- **Coverage Footprints** – Generating precise signal strength maps at national, regional, and local levels.
- **Capacity and Interference Analysis** – Modeling realistic signal overlaps and interference zones for multi-operator, multi-technology environments.



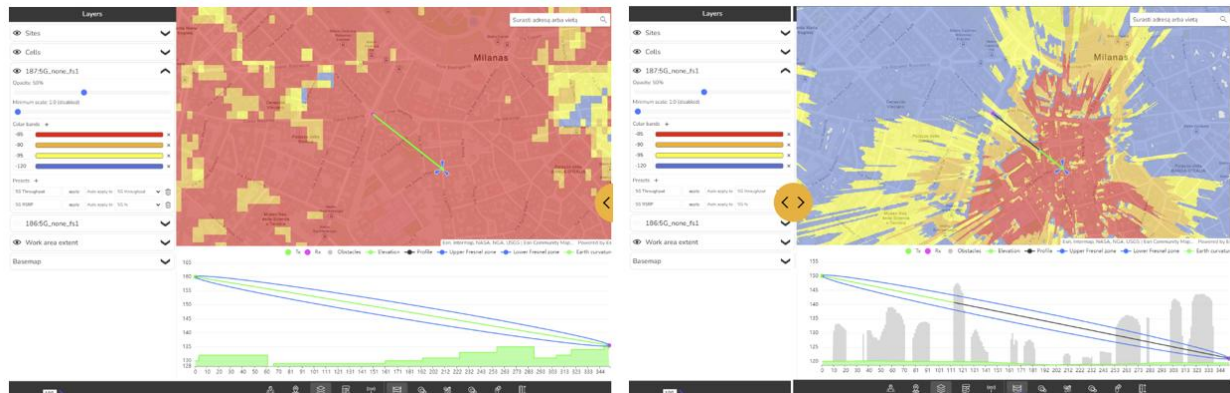
The CE tools make use of three distinct GIS data layers to obtain high precision modelling of radio wave propagation losses:

1. **Digital Terrain Model (DTM)**, also known as Digital Elevation Model (DEM), which describes Earth surface, i.e., path terrain profile in terms of ground elevation above uniform sea level.
2. **Obstacles layer**, delineating buildings and other such objects above Earth surface that may be considered to be principal impediments for radio wave propagation.
3. **Clutter layer**, delineating natural occurring or human cultivated ground cover that may be partially penetrable by radio waves, such as natural vegetation (e.g., forests, trees, bushes) or various crops, gardens, parks, etc.

The image above illustrates how Cellular Expert uses different resolutions of topographical data to significantly improve coverage prediction accuracy.

- Left image: Coverage calculated using 25 m resolution ASTER DEM data, showing a general view of signal distribution but with limited detail, especially in dense urban areas.
- Right image: Coverage calculated using 1 m resolution data, revealing a much more precise propagation pattern, including building-level shadowing and accurate street-by-street coverage.





More information: <https://blog.maxar.com/earth-intelligence/2022/benefits-of-using-maxars-precision3d-telco-suite-for-5g>

Cellular Expert can easily integrate and process 1 m or even sub-meter topographical data, providing highly detailed RF calculations. This level of precision is essential for:

- Modeling 2G/3G/4G/5G, small cells and mmWave networks.
- Identifying exact coverage gaps at the building and street level.
- Supporting regulatory-grade broadband mapping and planning.



By using high-resolution terrain and clutter data, Cellular Expert ensures that its calculations match real-world conditions as closely as possible — resulting in better network design decisions and more reliable broadband planning outcomes.

## 5.1 Geographic data requirements

The supported geographical data types:



Only GeoTIFF is supported. Topographical data must have specific names:

- The Digital terrain model must be named *elevation.tif*
- The land use (or clutter) grid must be named *clutterClasses.tif*
- The clutter heights (typically building, vegetation height) grid must be named *clutterHeight.tif*

#### **Mandatory geographical data:**

- Digital Terrain Model (DTM) grid

All geodata must be located in one catalog.



##### **5.1.1 Digital Terrain Model (DTM) Grid (Mandatory)**

The Digital Terrain Model (DTM), also known as Digital Elevation Model (DEM), represents the Earth's ground level above sea level. Each raster pixel has its height value.

A sample DTM raster is presented below. Each pixel represents 5 square meters with its height value. In reality, within a one-pixel area, the height is not the same everywhere. Thus, the pixel's height value is the height in its center or the maximum. The smaller the pixels, the more accurate is the grid - but also more data to calculate.



##### **5.1.1.1 Prepare DTM raster**

The Digital Terrain Model (DTM) has several requirements, which are listed below.

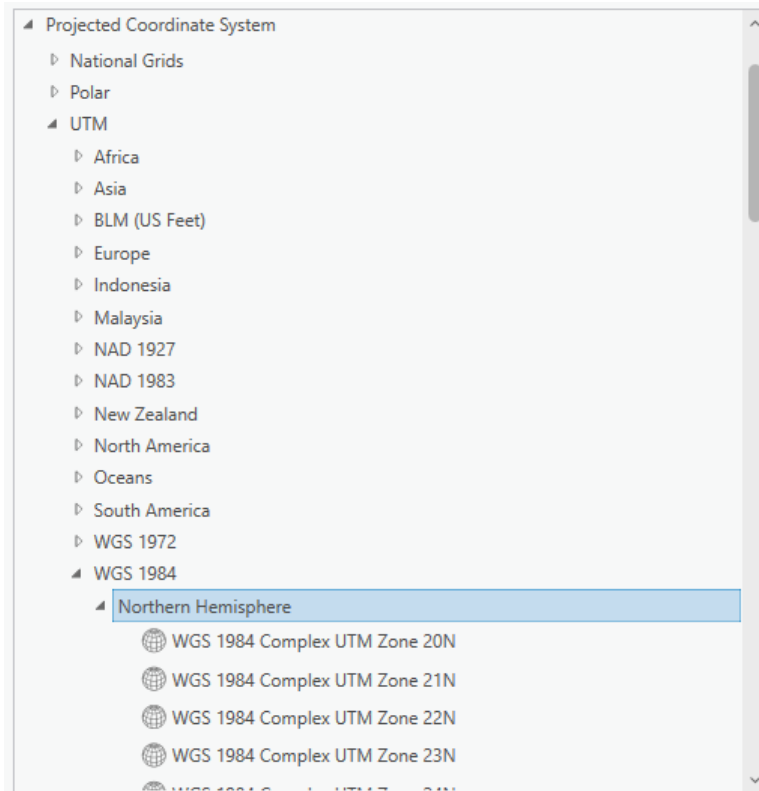
#### **Projection**

The raster must use a Projected Coordinate System. To check the coordinate system of your raster, use the Properties function in ArcGIS Pro. Add the raster to your project, right-click on it, and select Properties. Then, go to the Source tab > Spatial Reference and check the Coordinate System type parameter to confirm it is in a Projected Coordinate System.

|                             |                       |
|-----------------------------|-----------------------|
| Projected Coordinate System | LKS 1994 Lithuania TM |
| Projection                  | Transverse Mercator   |
| WKID                        | 3346                  |
| Previous WKID               | 2600                  |
| Authority                   | EPSG                  |
| Linear Unit                 | Meters (1.0)          |
| False Easting               | 500000.0              |
| False Northing              | 0.0                   |
| Central Meridian            | 24.0                  |
| Scale Factor                | 0.9998                |
| Latitude Of Origin          | 0.0                   |

If your raster is in a Geographic Coordinate System or needs a different projection, use the **Geoprocessing > Project Raster** tool to update it.

In the Output Coordinate System, specify a new coordinate system. It is recommended to use a UTM coordinate system under the WGS 1984 projection.



You can find the appropriate UTM zone for your area here:

<https://www.arcgis.com/apps/mapviewer/index.html?layers=b294795270aa4fb3bd25286bf09edc51>

### Correct No Data value and raster name

After setting the correct projection, assign the NoData attribute and specify the appropriate name for the DTM raster. To do this, use the **Copy Raster tool in Geoprocessing**.

The screenshot shows the 'Copy Raster' tool in the Geoprocessing pane. The 'Parameters' tab is selected. The 'Input Raster' field is empty. The 'Output Raster Dataset' field is empty. The 'Ignore Background Value' checkbox is unchecked. The 'NoData Value' field is empty. The 'Pixel Type' dropdown is set to '32 bit signed'. The 'Format' dropdown is set to 'TIFF format'. The 'Apply Transformation' checkbox is checked.

Configure the following settings:

- **Input Raster:** Select your newly projected DTM raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to elevation.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.

The screenshot shows the 'Copy Raster' tool with the following settings: 'Input Raster' is 'DTM raster', 'Output Raster Dataset' is 'elevation.tif', 'Ignore Background Value' is unchecked, 'NoData Value' is '-9999', 'Pixel Type' is '32 bit signed', 'Format' is 'TIFF format', and 'Apply Transformation' is checked.

### 5.1.2 Clutter classes grid

Land use or clutter refers to the classification of the earth's surface into categories such as urban, suburban, rural, forest, water, and open land, each of which affects radio propagation differently. Clutter data is crucial because it determines how signals are absorbed, reflected, or diffracted by the environment, directly

influencing coverage, interference, and quality of service. The naming and classification of land use types may vary. An example is the Sentinel-2 Land Cover dataset from the Living Atlas: [Living Atlas Sentinel-2 Land Cover](#)



This data is freely available worldwide and is detailed in Cellular Expert databases. Once the workspace is created, a default clutter class table is automatically applied for each land use class.

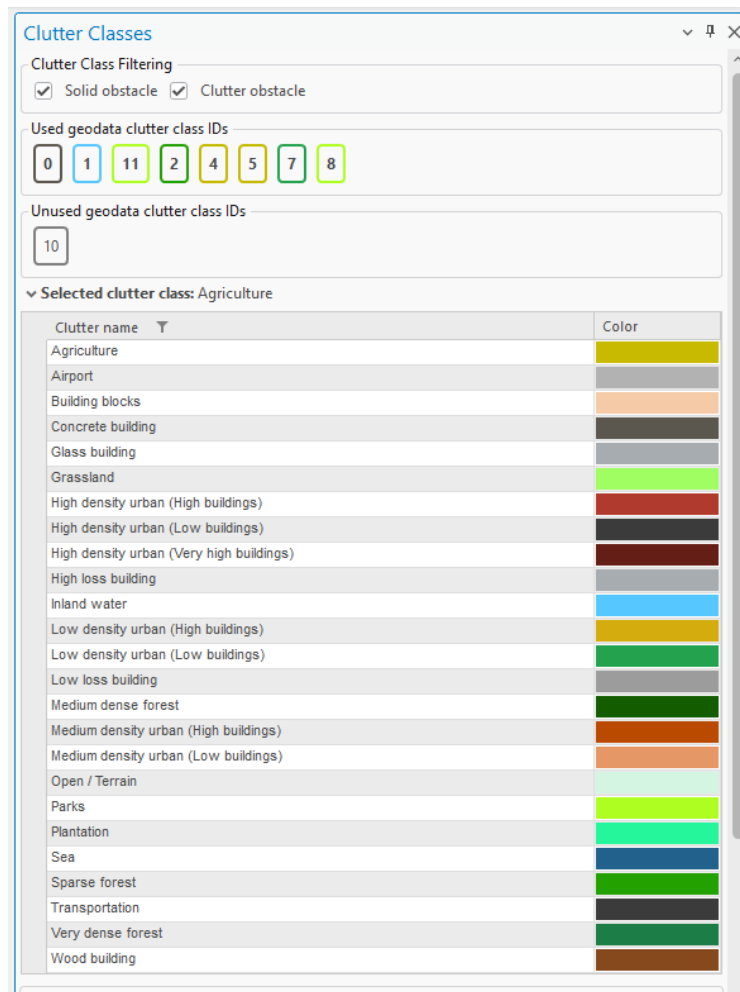


Table:

| clutter                                                                                                 |                        |        |                  |                      |                       |                      |            |         |        |                           |
|---------------------------------------------------------------------------------------------------------|------------------------|--------|------------------|----------------------|-----------------------|----------------------|------------|---------|--------|---------------------------|
| Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy Rows: Insert Rows |                        |        |                  |                      |                       |                      |            |         |        |                           |
| OBJECTID *                                                                                              | clutter_class_name     | height | nominal_distance | surface_refractivity | relative_permittivity | surface_conductivity | express_id | color   | ids    | clutter_class_alias       |
| 1                                                                                                       | concreteBuilding       | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #5B574E | 0      | Concrete building         |
| 2                                                                                                       | sparseForest           | 0      | 27               | 315                  | 13                    | 0.004                | 36         | #23A100 | 2      | Sparse forest             |
| 3                                                                                                       | mediumDenseForest      | 0      | 27               | 315                  | 13                    | 0.004                | 36         | #145C00 | <Null> | Medium dense forest       |
| 4                                                                                                       | lowDensityUrbanLow...  | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #24A24E | 7      | Low density urban (Lo...  |
| 5                                                                                                       | highDensityUrbanLow... | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #3B3B3B | <Null> | High density urban (Lo... |
| 6                                                                                                       | parks                  | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #AEFF21 | 8, 11  | Parks                     |
| 7                                                                                                       | agriculture            | 0      | 27               | 315                  | 20                    | 0.03                 | 36         | #C7BA00 | 4, 5   | Agriculture               |
| 8                                                                                                       | inlandWater            | 0      | 27               | 365                  | 80                    | 1                    | 36         | #57c7ff | 1      | Inland water              |
| 9                                                                                                       | transportation         | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #3B3B3B | <Null> | Transportation            |
| 10                                                                                                      | open                   | 0      | 1                | 315                  | 5                     | 0.001                | 0          | #d5f5e3 | <Null> | Open / Terrain            |
| 11                                                                                                      | grassland              | 0      | 27               | 315                  | 20                    | 0.03                 | 0          | #A0FE62 | <Null> | Grassland                 |
| 12                                                                                                      | veryDenseForest        | 0      | 15               | 315                  | 13                    | 0.004                | 0          | #1C7D46 | <Null> | Very dense forest         |
| 13                                                                                                      | lowDensityUrbanHinh... | 0      | 27               | 315                  | 5                     | 0.001                | 0          | #d4ac0d | <Null> | Low density urban (Hi...  |

These are standard clutter types in the default workspace database, which cannot be edited. You must map your clutter raster to these predefined clutter types. Standard mapping has already been configured for the Sentinel-2 Land Cover dataset from the Living Atlas: [Living Atlas Sentinel-2 Land Cover](#)

**Clutter Class Properties**

IDs in geodata raster:  
2

Height, m:  
0

Color:  
[Green]

Surface refractivity, N-Units:  
315

Relative permittivity:  
13

Surface conductivity, S/m:  
0.004

Apply OK Dismiss

If you have a different clutter class layer, it can be used for predictions by remapping it using the Clutter Classes tool and specifying the IDs in the geodata raster parameter. If multiple clutter classes correspond to a single default clutter type, separate the ID values with commas.

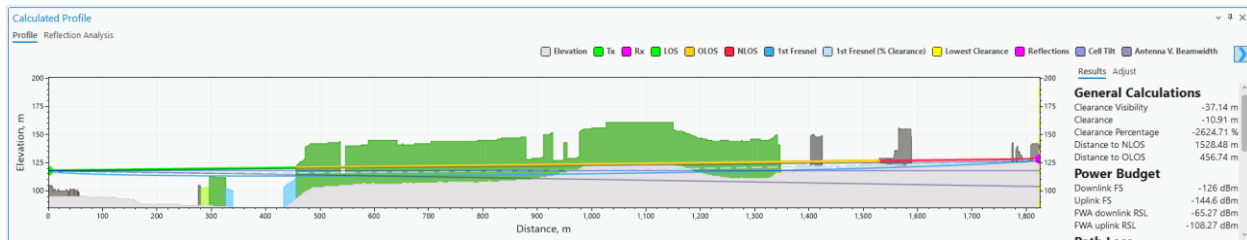
IDs in geodata raster:

8, 11

This mapping can also be adjusted in the Clutter table.

| OBJECTID | clutter_class_name     | height | nominal_distance | surface_refractivity | relative_permittivity | surface_conductivity | express_id | color   | ids    | clutter_class_alias      |
|----------|------------------------|--------|------------------|----------------------|-----------------------|----------------------|------------|---------|--------|--------------------------|
| 1        | concreteBuilding       | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #5B574E | 0      | Concrete building        |
| 2        | sparseForest           | 0      | 27               | 315                  | 13                    | 0.004                | 36         | #23A100 | 2      | Sparse forest            |
| 3        | mediumDenseForest      | 0      | 27               | 315                  | 13                    | 0.004                | 36         | #145C00 | <Null> | Medium dense forest      |
| 4        | lowDensityUrbanLow...  | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #24A24E | 7      | Low density urban (Lo... |
| 5        | highDensityUrbanLow... | 0      | 27               | 315                  | 5                     | 0.001                | 36         | #3B3B3B | <Null> | High density urban (L... |
| 6        | parks                  | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #AEFF21 | 8, 11  | Parks                    |
| 7        | agriculture            | 0      | 27               | 315                  | 20                    | 0.03                 | 36         | #C7BA00 | 4, 5   | Agriculture              |
| 8        | inlandWater            | 0      | 27               | 365                  | 80                    | 1                    | 36         | #57C7FF | 1      | Inland water             |
| 9        | transportation         | 0      | 27               | 315                  | 10                    | 0.002                | 36         | #3B3B3B | <Null> | Transportation           |
| 10       | open                   | 0      | 1                | 315                  | 5                     | 0.001                | 0          | #d5f5e3 | <Null> | Open / Terrain           |
| 11       | grassland              | 0      | 27               | 315                  | 20                    | 0.03                 | 0          | #A0FE62 | <Null> | Grassland                |
| 12       | veryDenseForest        | 0      | 15               | 315                  | 13                    | 0.004                | 0          | #1C7D46 | <Null> | Very dense forest        |
| 13       | lowDensityUrbanHigh... | 0      | 27               | 315                  | 5                     | 0.001                | 0          | #d4a0d4 | <Null> | Low density urban (Hi... |

Once clutter classes are successfully mapped, the prediction algorithms will recognize the clutter types, apply distinct symbols, and adjust path loss calculations accordingly, based on the parameters set in the prediction model.



### 5.1.2.1 Prepare Clutter Classes raster

The Clutter Raster has several requirements, which are the same as for DTM raster listed above.

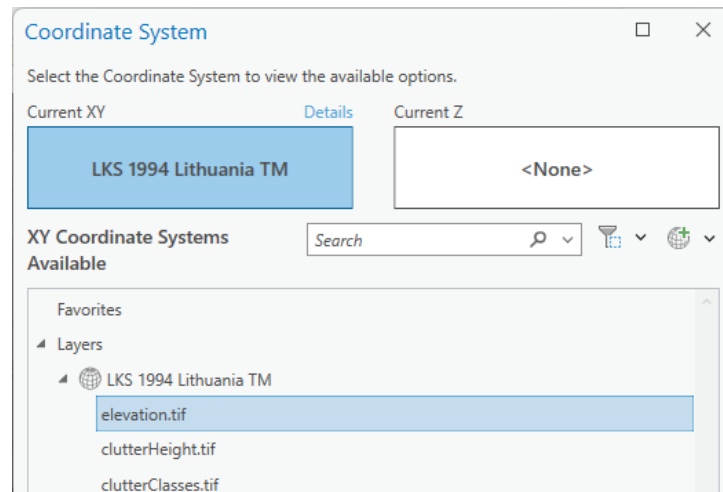
### Projection

It must have the same coordinate system as your elevation.tif raster. If your raster has different coordinate system, then use the **Geoprocessing tool** → **Project Raster** to fix it.

In the Output Coordinate System you would need to define the same coordinate system as your elevation.tif raster. Click on Select Coordinate System button.

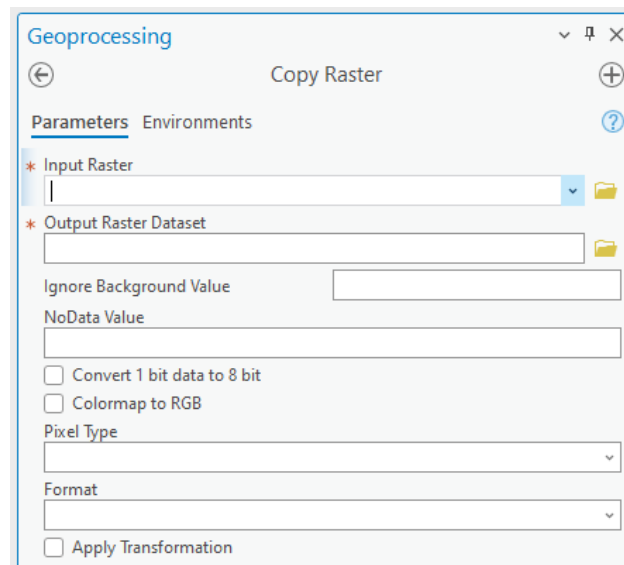
And choose the same coordinate system as your elevation.tif.





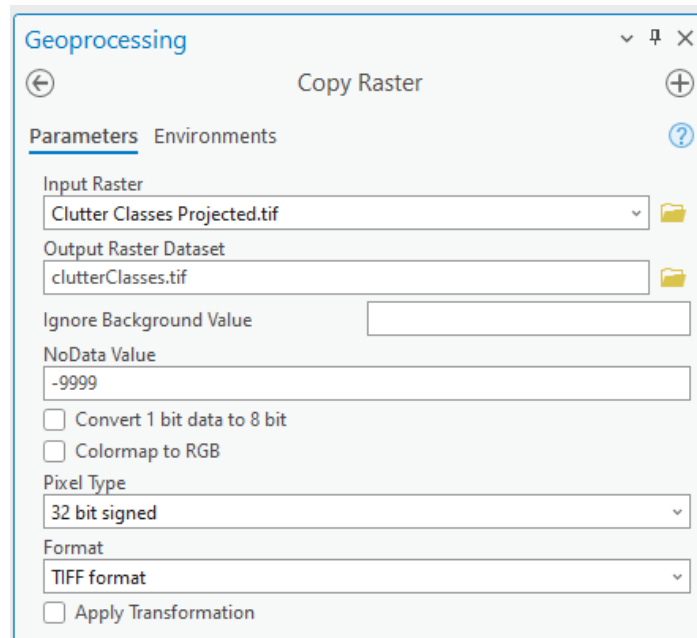
### Correct No Data value and raster name

After setting the correct projection, assign the NoData attribute and specify the appropriate name for the Clutter Class raster. To do this, use the **Copy Raster tool in Geoprocessing**.



Configure the following settings:

- **Input Raster:** Select your newly projected Clutter Class raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to clutterClasses.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.



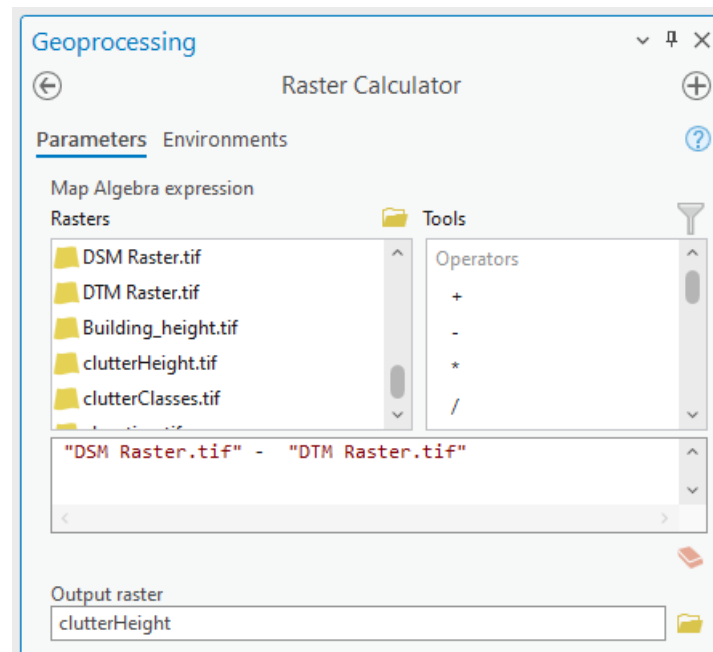
### 5.1.3 Clutter heights

Represents actual clutter heights, which override the default heights specified in the Clutter table. The clutter heights raster requires the accompanying clutterClasses.tif raster and cannot be used independently.



A clutter height raster can be derived from a Digital Surface Model (DSM) raster and a Digital Terrain Model (DTM) raster using the ArcGIS Raster Calculator tool. To access this tool, open Geoprocessing tools and navigate to Spatial Analyst > Map Algebra > Raster Calculator. Use the following formula:

$$\text{DSM} - \text{DTM}$$



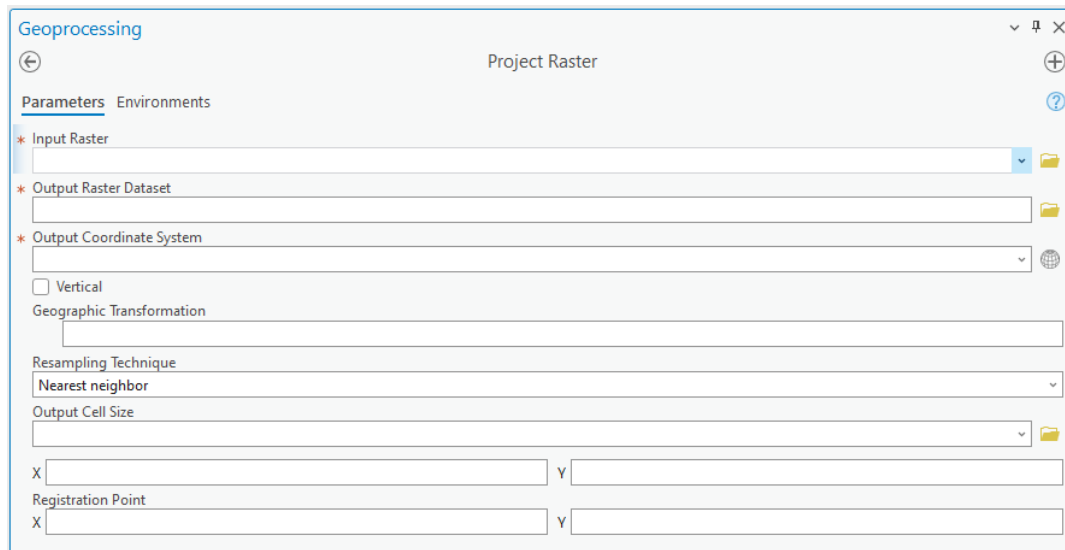
The calculation output will be the difference between the DSM and DTM grids, representing the clutter heights.

#### 5.1.3.1 Prepare Clutter Height raster

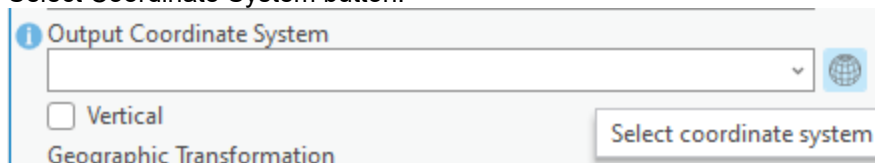
The Clutter Height has several requirements, which are the same as for DTM raster listed above.

#### Projection

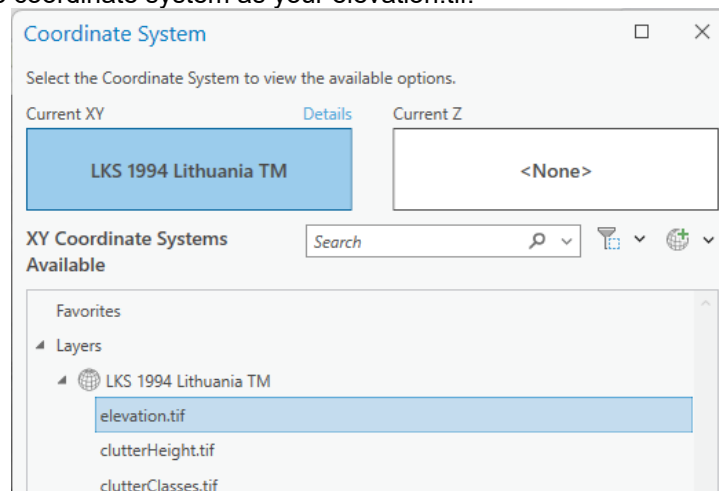
It must have the same coordinate system as your elevation.tif raster. If your raster has different coordinate system, then use the **Geoprocessing tool** → **Project Raster** to fix it.



In the Output Coordinate System you would need to define the same coordinate system as your elevation.tif raster. Click on Select Coordinate System button.

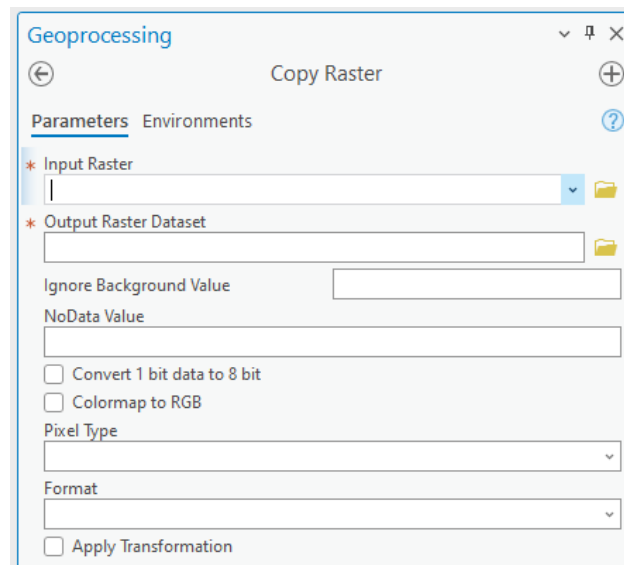


And choose the same coordinate system as your elevation.tif.



### Correct No Data value and raster name

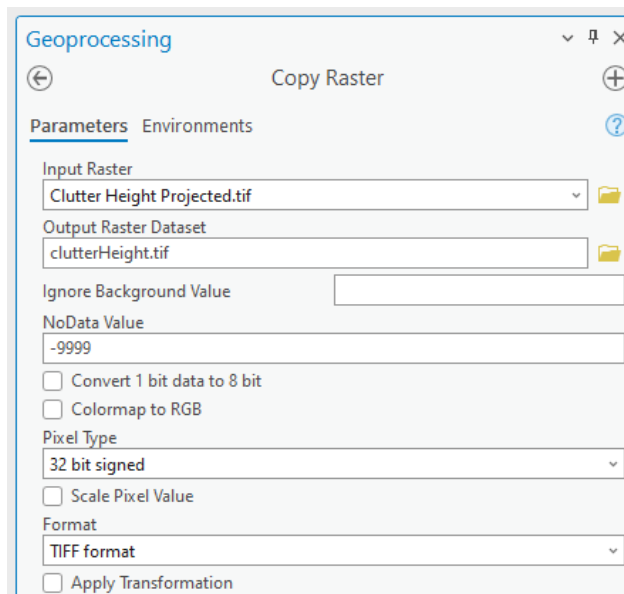
After setting the correct projection, assign the NoData attribute and specify the appropriate name for the Clutter Height raster. To do this, use the **Copy Raster** tool in Geoprocessing.



The screenshot shows the 'Geoprocessing' window for the 'Copy Raster' tool. The 'Parameters' tab is active. The 'Input Raster' field is empty. The 'Output Raster Dataset' field is empty. The 'Ignore Background Value' checkbox is unchecked. The 'NoData Value' field is empty. The 'Pixel Type' dropdown is set to '32 bit signed'. The 'Format' dropdown is set to 'TIFF format'. The 'Apply Transformation' checkbox is unchecked.

Configure the following settings:

- **Input Raster:** Select your newly projected Clutter Height raster.
- **Output Raster Dataset:** Specify the output location and set the raster name to clutterHeight.tif.
- **NoData Value:** Enter -9999.
- **Pixel Type:** Choose 32-bit signed or 32-bit float.
- **Format:** This will automatically be set to TIFF.

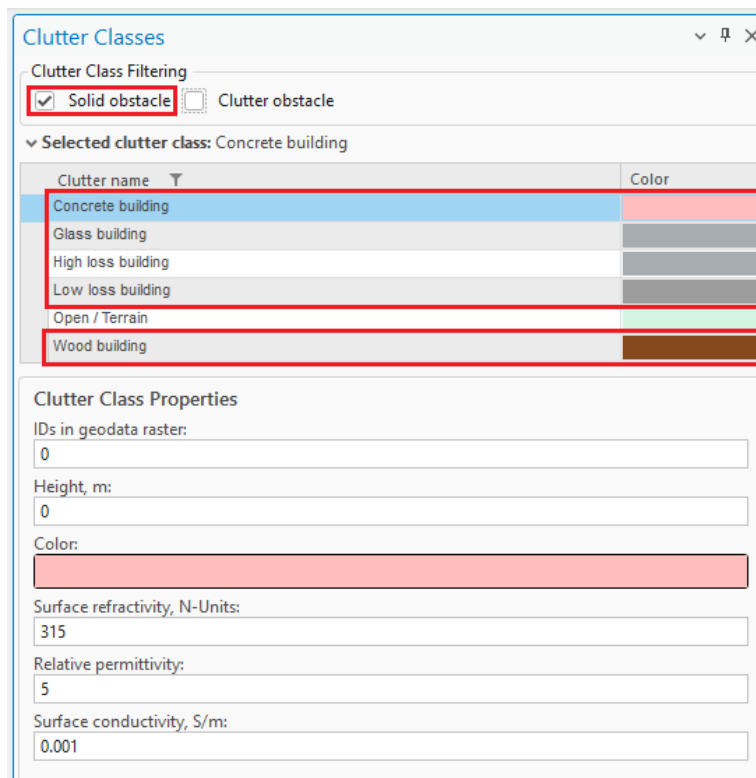


The screenshot shows the 'Geoprocessing' window for the 'Copy Raster' tool with the following settings configured: 'Input Raster' is 'Clutter Height Projected.tif', 'Output Raster Dataset' is 'clutterHeight.tif', 'NoData Value' is '-9999', 'Pixel Type' is '32 bit signed', and 'Format' is 'TIFF format'. The 'Apply Transformation' checkbox is still unchecked.

### 5.1.4 Buildings

Building features within the Clutter Classes raster are automatically identified and categorized using a range of dedicated **building-specific clutter types**. These clutter types are available within the **Clutter Classes tool** and are specifically designed to represent different architectural materials and structural characteristics.

Importantly, all building-related clutter classes are treated as **Solid Obstacles** – ensuring accurate modeling of signal attenuation and reflection in propagation calculations. This classification enhances the realism and precision of both indoor and outdoor network predictions by accounting for the physical impact of built environments.



| Clutter name       | Color       |
|--------------------|-------------|
| Concrete building  | Light blue  |
| Glass building     | Light grey  |
| High loss building | Dark grey   |
| Low loss building  | Medium grey |
| Open / Terrain     | Green       |
| Wood building      | Brown       |

**Clutter Class Properties**

IDs in geodata raster: 0

Height, m: 0

Color: [Red color swatch]

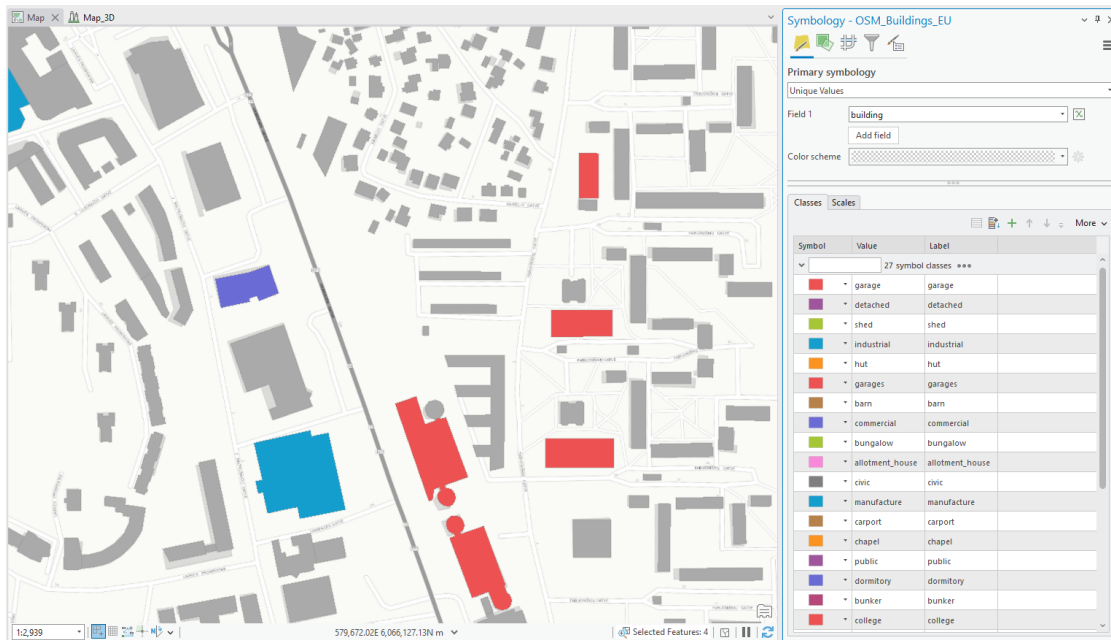
Surface refractivity, N-Units: 315

Relative permittivity: 5

Surface conductivity, S/m: 0.001

You can seamlessly **edit your Clutter Classes raster** using standard **ArcGIS Geoprocessing tools** to integrate detailed building data into your existing clutter layer.

The process supports both **vector and raster input formats**, giving you flexibility depending on your data source. Whether you're working with CAD-derived vectors or pre-classified raster layers, these tools allow you to enrich the clutter map with accurate building representations – essential for high-fidelity propagation modeling.



If your building data is in **vector format** (e.g., polygons), you'll first need to **convert it to raster** before incorporating it into the Clutter Classes layer.

1. **Convert Vector to Raster**

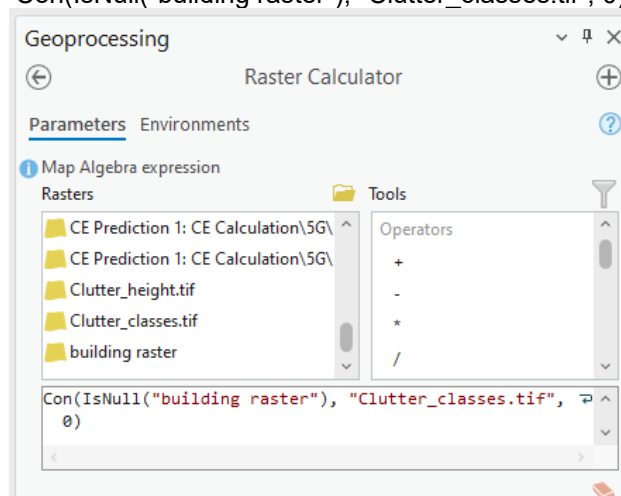
Use the **Polygon to Raster** tool available in the **Geoprocessing** pane. This will transform your vector-based building footprints into a raster format suitable for clutter classification.

2. **Update Clutter Classes Using Raster Calculator**

After conversion, open the **Raster Calculator** to integrate the new raster into your existing clutter layer:

- Navigate to **Geoprocessing**
- Go to **Spatial Analyst > Map Algebra > Raster Calculator**
- Use map algebra expressions to merge or replace values as needed, assigning appropriate clutter class codes to building areas.

Con(IsNull("building raster"), "Clutter\_classes.tif", 0)

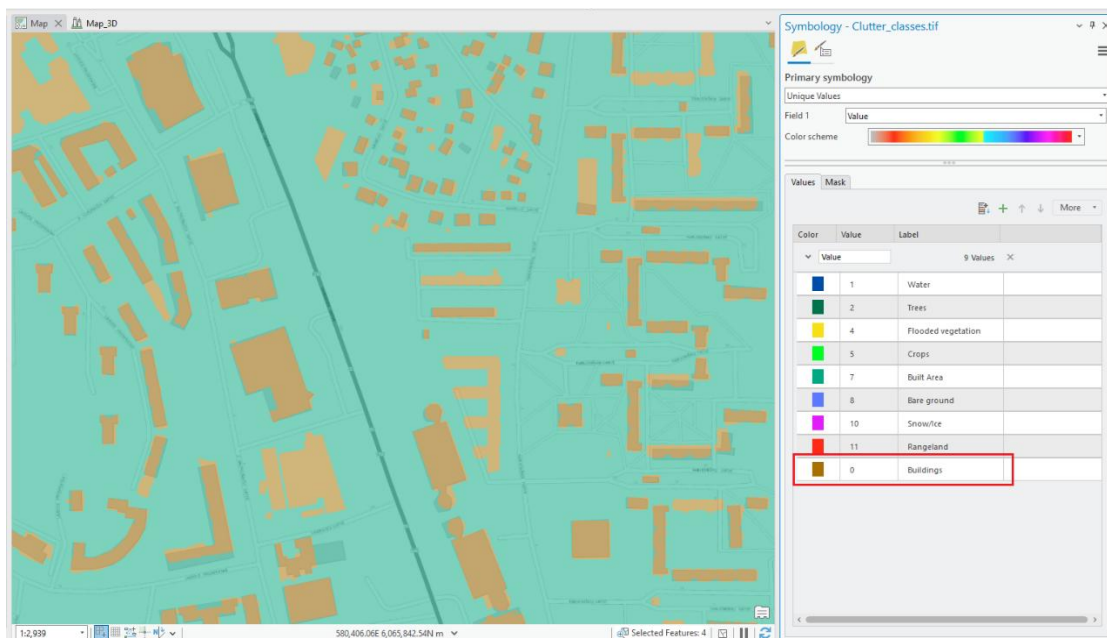


This workflow allows you to accurately reflect building data in your clutter map, improving prediction accuracy in urban and mixed-use environments.

When incorporating building data into the Clutter Classes raster, you have the flexibility to **assign any numeric value** to represent the **"Buildings"** clutter class. In the example provided, a value of **0** was used, but you may choose any value – **as long as it does not conflict with existing clutter class values** already present in your raster.

Once applied, the Clutter Classes raster will be updated to include a new entry for **Buildings**, enabling more precise signal propagation modeling that accounts for structural obstructions.

Be sure to document your chosen value and update your Clutter Class definitions accordingly within the Clutter Classes tool to maintain consistency across your modeling environment.



All propagation and prediction calculations reference **clutter types defined in the Clutter Classes table**. Each **building-related clutter class** is assigned a **unique ID value**, which is used during modeling to apply the appropriate path loss parameters for solid structures.

This ensures that **Buildings** are accurately represented in simulations, contributing to more realistic signal behavior in both indoor and outdoor environments.



**Clutter Classes**

Clutter Class Filtering

☒ Solid obstacle ☐ Clutter obstacle

Selected clutter class: Concrete building

| Clutter name       | Color |
|--------------------|-------|
| Concrete building  |       |
| Glass building     |       |
| High loss building |       |
| Low loss building  |       |
| Open / Terrain     |       |
| Wood building      |       |

Clutter Class Properties

IDs in geodata raster:  
0

Height, m:  
0

Color:

Surface refractivity, N-Units:  
315

Relative permittivity:  
5

Surface conductivity, S/m:  
0.001

### Building Height Determination in Clutter-Based Modeling

Pixels assigned to a **building clutter class ID** will be automatically recognized as **solid obstacle** during prediction calculations. Their heights are determined using the following priority:

- **Option 1:** From the associated **Clutter Height raster**, if available. This provides the most accurate, location-specific height information.
- **Option 2:** If no Clutter Height raster is present, the system will default to the **height value defined in the Clutter Classes table** for the corresponding clutter ID.

**Clutter Classes**

Clutter Class Filtering

☒ Solid obstacle ☐ Clutter obstacle

Selected clutter class: Concrete building

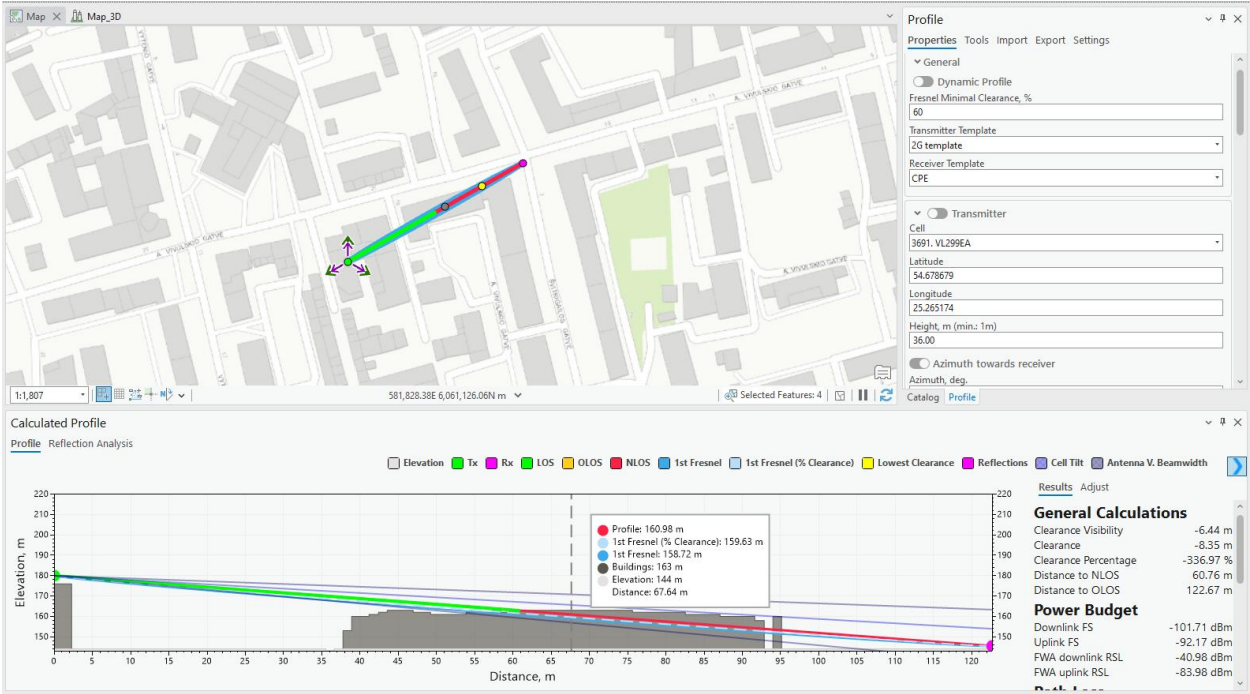
| Clutter name       | Color |
|--------------------|-------|
| Concrete building  |       |
| Glass building     |       |
| High loss building |       |
| Low loss building  |       |
| Open / Terrain     |       |
| Wood building      |       |

Clutter Class Properties

IDs in geodata raster:  
0

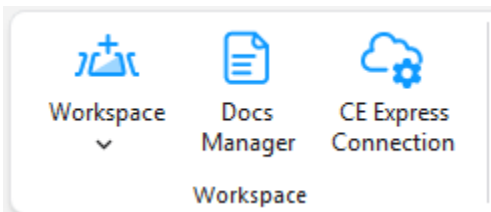
Height, m:  
10

The result:



## 6. Workspace

This chapter describes the Cellular Expert workspace functionality.



### 6.1 Workspace Tool

#### 6.1.1 Workspace Table

Cellular Expert workspace is a geodatabase containing data tables, feature datasets, and the workspace definition table. After creating a new workspace database, the workspace definition table will be named **CE\_WORKSPACE** and contain the information about the dataset.

|    | OBJECTID * | pName                     | pType  | pValue                   |
|----|------------|---------------------------|--------|--------------------------|
| 1  | 33         | Prediction Path           | STRING | C:\CE\Project\Workspace  |
| 2  | 34         | Calculation Path          | STRING | C:\CE\Project\Workspace  |
| 3  | 39         | Transmit power units      | STRING | dBm                      |
| 4  | 60         | Result Path               | STRING | C:\CE\Project\Workspace  |
| 5  | 66         | Geodata Folder Path       | STRING | C:\CE\Project\Workspace  |
| 6  | 67         | Calculation Tasks Data... | STRING | C:\CE\Project\Workspa... |
| 7  | 74         | CE Server URL             | STRING | < Null>                  |
| 8  | 75         | CE Server API             | STRING | < Null>                  |
| 9  | 76         | CE Server Environment     | STRING | < Null>                  |
| 10 | 77         | CE Server Workspace       | STRING | < Null>                  |
| 11 | 78         | CE Server Workspace Id    | STRING | < Null>                  |
| 12 | 79         | Use Clutter Layer         | STRING | 1                        |

Data field types and values:

- OBJECTID – long integer type
- pName – text type
- pType – text type
- pValue – text type

When moving your project's directory to another location (to another computer), it is necessary to update the workspace parameters by referencing the new paths to properly load the project:

- Prediction Path – path for storing prediction grids
- Calculation Path – path for temporary calculations
- Calculation Tasks Data Path – path for saving calculation tasks
- Geodata Folder Path – path for geodata (prediction models do not have geodata options, topographical data are taken from the Geodata Folder Path)
- Result Path – path for final results

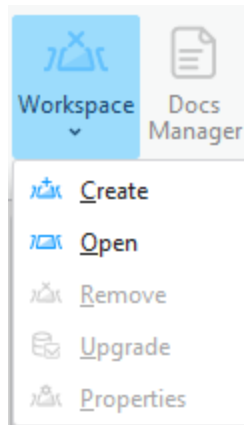
Workspace calculation paths and settings can be previewed in the dedicated tool, navigate to Workspace > Properties. It would show all parameters listed in CE\_WORKSPACE table.

| Project Paths               |                                                          |
|-----------------------------|----------------------------------------------------------|
| Calculation Path            | C:\CE\Project\Workspace\Temp                             |
| Calculation Tasks Data Path | C:\CE\Project\Workspace\SystemFiles\calculation_tasks.db |
| Geoclimatic Data Path       | C:\CE\Project\Workspace\Geoclimatic.gdb                  |
| Geodata Folder Path         | C:\CE\Project\Workspace\Geodata                          |
| Prediction Path             | C:\CE\Project\Workspace\Predictions                      |
| Result Path                 | C:\CE\Project\Workspace\Results                          |
| Volatile Calculation Path   | C:\CE\Project\Workspace\VolatileTemp                     |
| Volatile Result Path        | C:\CE\Project\Workspace\VolatileResults                  |
| Volatile Tasks Data Path    | C:\CE\Project\Workspace\SystemFiles\volatile_tasks.db    |

### 6.1.2 Create Workspace

Steps to create a new workspace:

1. Click the **Create** button in the **Cellular Expert Workspace** menu.



2. The **Create** dialogue will appear. Fill in the minimum required data:

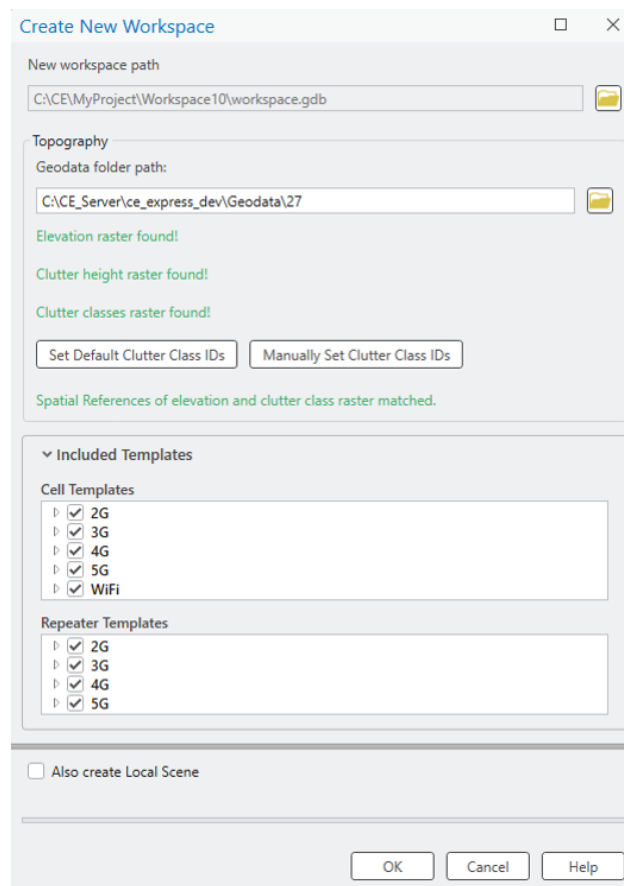
#### New workspace path

It is automatically filled based on the ArcGIS Pro project location. The recommendation is to first save your ArcGIS Pro project and then open the Workspace > Create tool. The project's workspace folder will be automatically created in the ArcGIS Pro project location catalog. This way, you will have both the ArcGIS Pro and CE workspace in the same location.

#### Geodata folder path

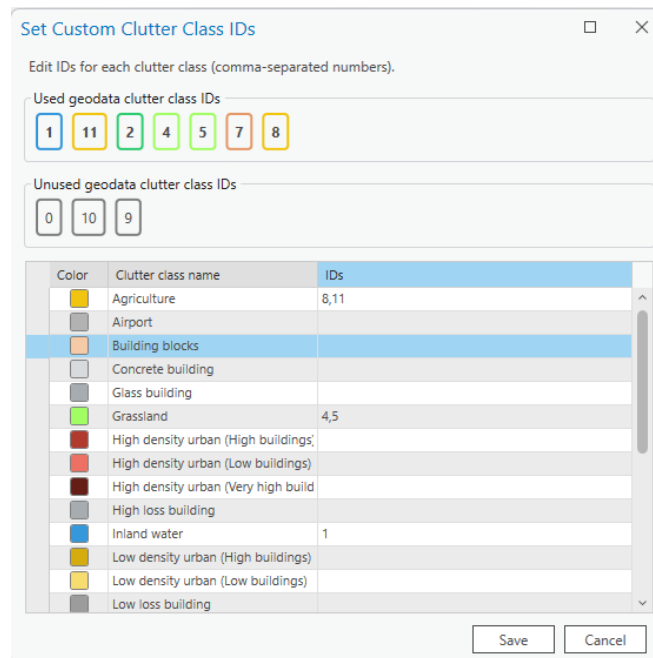
Catalog where geodata files are stored. More about Geodata requirements are listed in **5. Geographic data topic**. Click on  **Browse** button to define Geodata catalog. Geodata must have specific names:

- The Digital terrain model must be named *elevation.tif*
- The land use (or clutter) grid must be named *clutterClasses.tif*
- The clutter heights (typically building and vegetation height) grid must be named *clutterHeight.tif*



**Note:** The projected Coordinate System has been filled automatically and taken from the defined *Elevation grid*. This means that this coordinate system will be assigned to your project feature layers.

When creating a new workspace using geodata containing clutter classes raster, default class IDs can now be set by clicking *Set Default Clutter Class IDs* button. The default ID values are based on Living Atlas clutter. Clutter class IDs can also be set manually, either before or after the default values are set. To initiate the manual editing of clutter class IDs, click the *Manually Set Clutter Class IDs* button. Each clutter class can be edited to designate its used and unused types, also indicated by their distinct colors.



Set Custom Clutter Class IDs

Edit IDs for each clutter class (comma-separated numbers).

Used geodata clutter class IDs

1 11 2 4 5 7 8

Unused geodata clutter class IDs

0 10 9

| Color | Clutter class name                   | IDs  |
|-------|--------------------------------------|------|
|       | Agriculture                          | 8,11 |
|       | Airport                              |      |
|       | Building blocks                      |      |
|       | Concrete building                    |      |
|       | Glass building                       |      |
|       | Grassland                            | 4,5  |
|       | High density urban (High buildings)  |      |
|       | High density urban (Low buildings)   |      |
|       | High density urban (Very high build) |      |
|       | High loss building                   |      |
|       | Inland water                         | 1    |
|       | Low density urban (High buildings)   |      |
|       | Low density urban (Low buildings)    |      |
|       | Low loss building                    |      |

Save Cancel

Several messages related to Geodata:

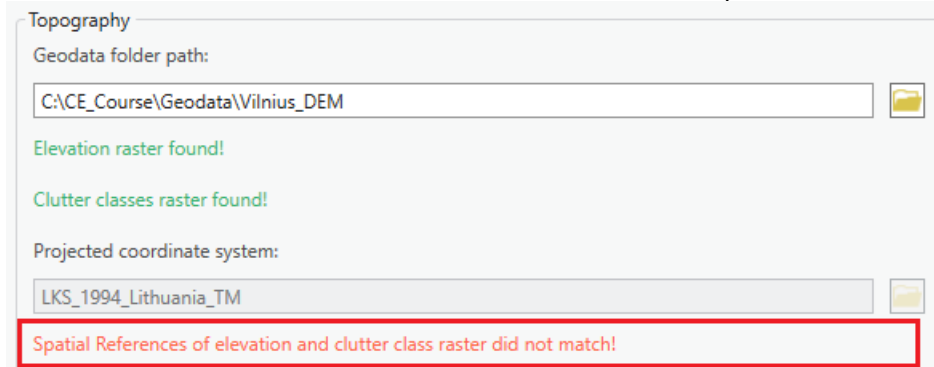
- If only Elevation exists in Geodata catalog, then tool will be filled with such information:

Elevation raster found!

(Optional) Clutter classes raster is missing.

Clutter Classes and Clutter Height are optional.

- If coordinates mismatch between elevation and clutter, then it will provide this message:



Topography

Geodata folder path:

C:\CE\_Course\Geodata\Vilnius\_DEM

Elevation raster found!

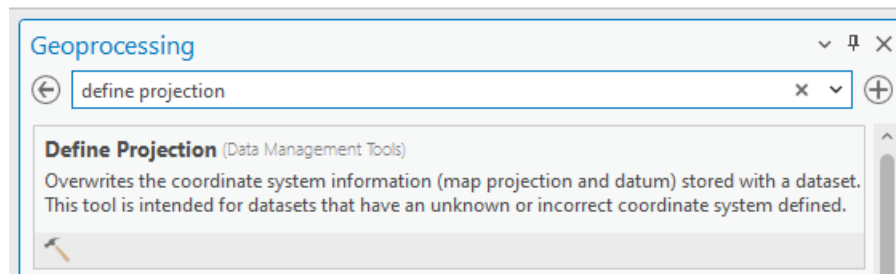
Clutter classes raster found!

Projected coordinate system:

LKS\_1994\_Lithuania\_TM

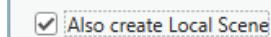
Spatial References of elevation and clutter class raster did not match!

Please close Workspace Create tool and fix your geodata. You can use Geoprocessing tools, Define Projection tool to fix this issue. Define the same coordinate as elevation.tif for your clutterClasses.tif and if available, clutterHeight.tif rasters.



### Also Create Local Scene

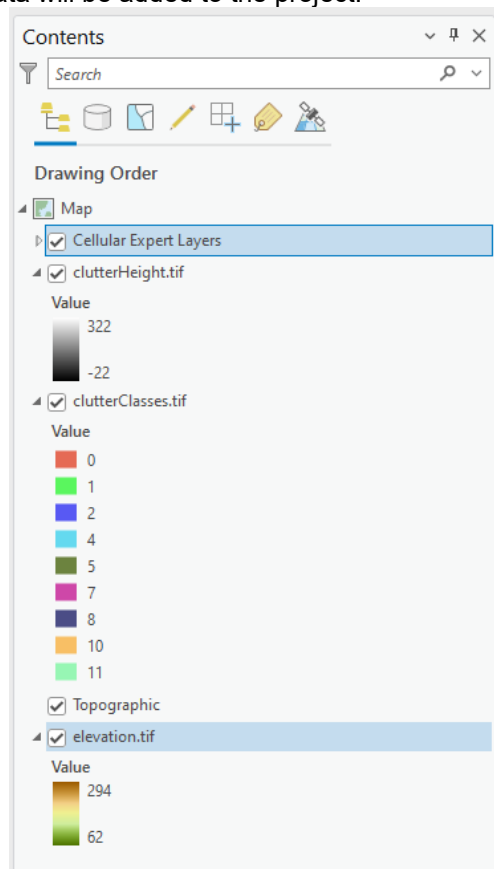
Use the option to create a second scene for 3D visualization.



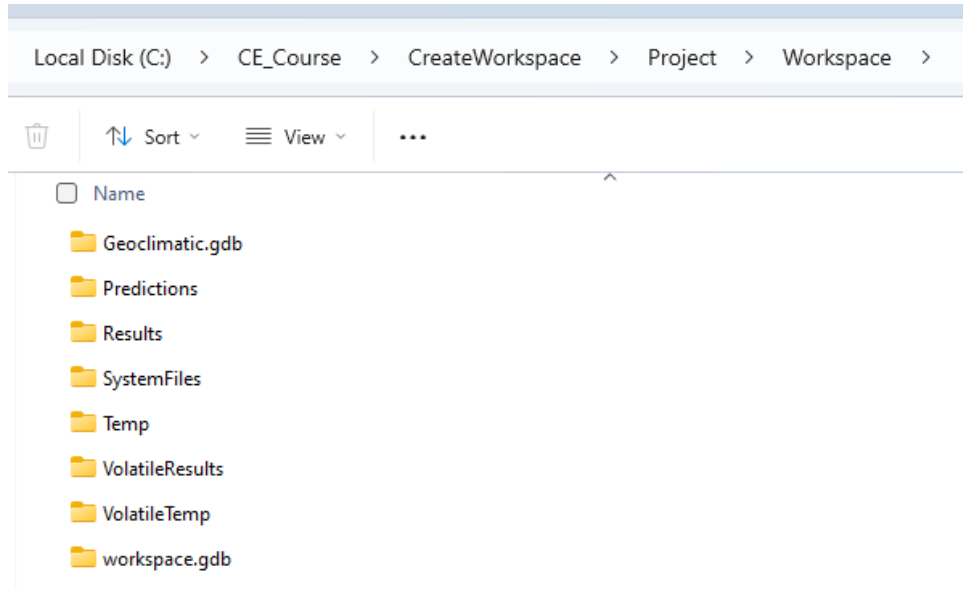
When creating a new workspace, there is also an option to choose which cell and repeater templates are provisioned into it based on technology (such as 2G, 3G, 4G, etc.). Templates are listed by technology group, and individual templates or entire groups can be selected or deselected. By default, all templates are enabled; deselecting unused technologies keeps the new workspace clean and uncluttered by objects from unrelated technology stacks. Previously created workspaces are not affected.

3. Press OK button to start workspace creation procedure.

Cellular Expert layer and geodata will be added to the project.



Workspace geodatabase and required folders will be created within successful workspace creation procedure.



The Project Paths will be filled in the **Workspace Properties** → **Properties** tab.

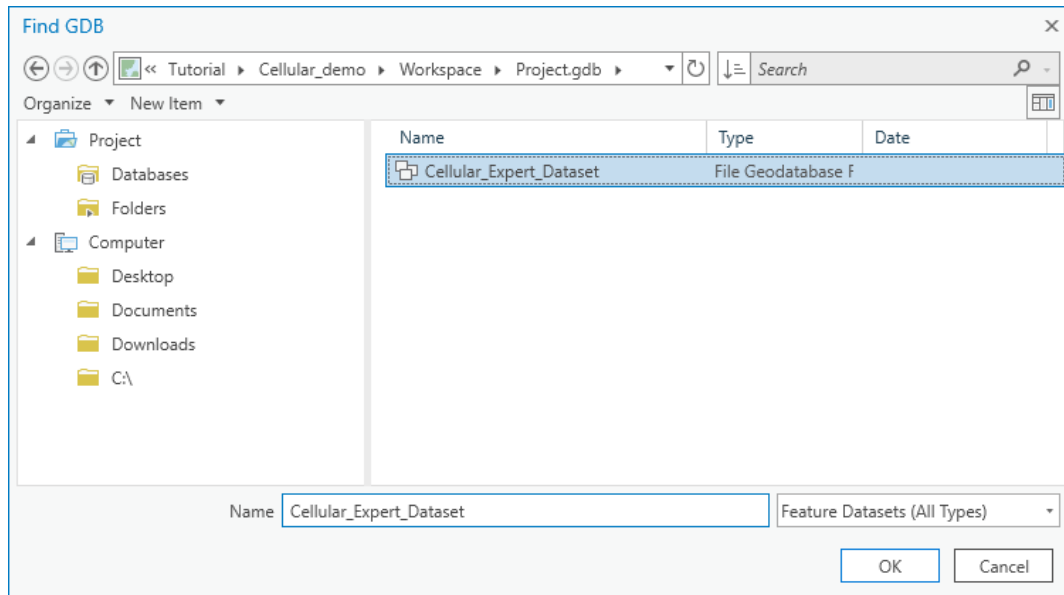
| Project Paths               |                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------|
| Calculation Path            | C:\CE_Course\CreateWorkspace\Project\Workspace\Temp                             |
| Calculation Tasks Data Path | C:\CE_Course\CreateWorkspace\Project\Workspace\SystemFiles\calculation_tasks.db |
| Geoclimatic Data Path       | C:\CE_Course\CreateWorkspace\Project\Workspace\Geoclimatic.gdb                  |
| Geodata Folder Path         | C:\CE_Course\Geodata\Vinivus                                                    |
| Prediction Path             | C:\CE_Course\CreateWorkspace\Project\Workspace\Predictions                      |
| Result Path                 | C:\CE_Course\CreateWorkspace\Project\Workspace\Results                          |
| Volatile Calculation Path   | C:\CE_Course\CreateWorkspace\Project\Workspace\VolatileTemp                     |
| Volatile Result Path        | C:\CE_Course\CreateWorkspace\Project\Workspace\VolatileResults                  |
| Volatile Tasks Data Path    | C:\CE_Course\CreateWorkspace\Project\Workspace\SystemFiles\volatile_tasks.db    |



### 6.1.3 Open Workspace

Steps to open a workspace:

1. Click the **Open** option in the **Workspace** menu of the **Cellular Expert** toolbar.
2. Specify the workspace name and click the OK button.



The specified workspace will be activated automatically.

### 6.1.4 Remove Workspace

Upon selecting **Remove** in the Workspace dropdown menu, the workspace will be removed from the currently open ArcGIS Pro project.

### 6.1.5 Workspace Upgrade

**Workspace Upgrade** is a tool that enables the user to update missing tables from the default database (*default.gdb*). If the table exists, Workspace Upgrade also checks if all default fields are present in that table. If the tables are toggled, they will be updated (upgraded) by the tool.

Usually, with new versions of *CE for ArcGIS Pro* come changes to default data tables. This tool is especially useful when checking if the newest version tables correspond to the current project tables.

Workspace Upgrade automatically checks all these parameters and notifies the user upon the start-up of the project.

**Workspace Upgrade**

▼ **Default Database Upgrades**

▼ Add Missing Tables

☒ Sirens

☒ ReflectionAnalysis

☒ model\_iso9613

☒ SirenTemplates

▼ Upgrade Existing Tables

▼ ☒ CE\_WORKSPACE

Add Missing Rows

|                              |
|------------------------------|
| Reference Value              |
| Receiver Height Reference    |
| Transmitter Height Reference |

▼ ☒ CE\_TABLEDATA

Add Missing Rows

|                 |
|-----------------|
| Reference Value |
| Sirens          |

▼ ☒ Radios

Add Missing Fields

| Field Name    | Field Type |
|---------------|------------|
| xpd           | Double     |
| cti_ratio_ber | Double     |

Upgrade Log

Analysis started.  
Please update your Workspace with the Upgrade tool  
Analysis complete.

☐ Refresh Cellular Expert Layers

☐ Reimport Antennas

☐ Generate Antenna Type

Run Analysis

Upgrade Database

**Refresh Cellular Expert Layers**

Imports missing layers from Cellular Expert Dataset.

**Reimport Antennas**

Imports the deleted default antennas back to the Antennas table.

**Generate Antenna Type**

Based on the Antenna data assign antennas a type if they are missing it.

**Run Analysis**

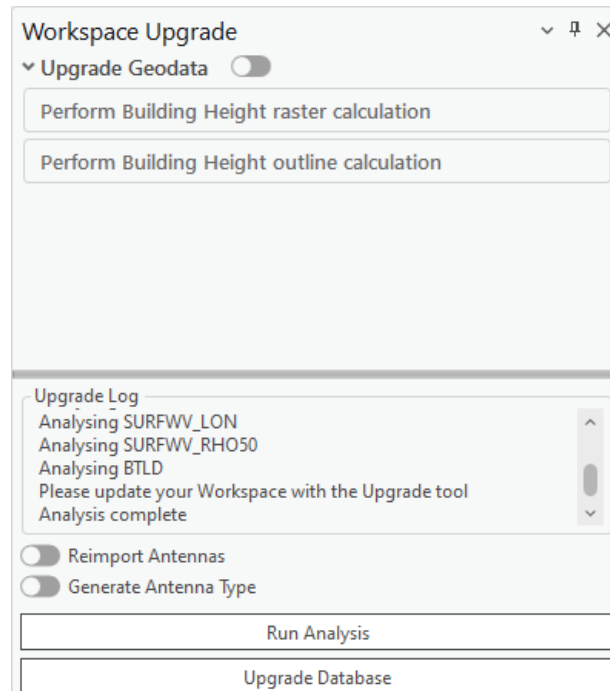
Manually checks if the default tables and their fields exist in the project.

**Upgrade Database**

Creates the missing tables and/or creates/adds back the missing fields from default tables.

If a project with incompatible geodata is loaded, the Workspace Upgrade tool analyzes the current geodata, as well as the owned user Esri Extension licenses to offer the optimal geodata upgrade path. This process is one-time only, and can be done by checking the Upgrade Geodata toggle.

- If only elevation and buildingHeight rasters exist, the buildingHeight can automatically be renamed to clutterHeight.
- **(Image Analyst and Spatial Analyst licenses only)** If elevation, buildingHeight and clutterHeight rasters exist, raster calculations are performed to merge the buildingHeight and clutterHeight rasters.
- **(Image Analyst and Spatial Analyst licenses only)** If elevation, buildingHeight, clutterHeight and clutterClasses rasters exist, raster calculations are performed to merge the buildingHeight and clutterHeight rasters, and modify the clutterClasses raster to add the building outlines with ID of 0.



### 6.1.6 Workspace Properties

The Workspace properties dialogue shows all the workspace information from the “CE\_WORKSPACE” data table. It also enables the user to customize the symbol visualization.

To open the **Workspace Properties** dialogue, click on the **Workspace** menu icon and choose **Properties**.



#### 6.1.6.1 Parameters

All information from the “CE\_WORKSPACE” table is represented in the **Parameters** tab of the Workspace Properties.

A screenshot of the 'Workspace Properties' dialog box, specifically the 'Parameters' tab. The dialog has a title bar with 'Workspace Properties' and standard window controls. Below the title bar, there are two tabs: 'Parameters' (selected) and 'Visualization'. The main area contains a table with two columns: 'Name' and 'Value'. The table is divided into two sections: 'CE Server' and 'Project Paths'. The 'CE Server' section has five rows: 'CE Server API', 'CE Server Environment', 'CE Server URL', 'CE Server Work Area', and 'CE Server Work Area Id'. The 'Project Paths' section has four rows: 'Batch Logs Path', 'Calculation Path', 'Calculation Tasks Data Path', and 'Geospatial Data Path'. The 'Calculation Path' and 'Calculation Tasks Data Path' rows have pre-filled values. At the bottom of the dialog, there are three buttons: 'OK', 'Cancel', and 'Help'.

#### OK

Click this button to save any changes and close the dialogue.

#### Cancel

Click this button to cancel any changes and close the dialogue.

**Help**

Get helpful information about the dialogue.

**CE Server Parameters****ArcGIS Portal URL**

Connection URL to the ArcGIS Portal website.

**CE Server API**

Connection URL to Cellular Expert Server API.

**CE Server Environment**

Cellular Expert Server environment in which it is run.

**CE Server URL**

Connection URL to Cellular Expert Server.

**CE Server Workspace**

The workspace used in the Cellular Expert Server.

**CE Server Workspace ID**

The ID of the workspace which is used in the Cellular Expert Server.

**Project Paths Parameters****Calculation Path**

Path for temporary calculation results.

**Calculation Tasks Data Path**

Path for the calculation results that will be displayed in the Calculation Task List.

**Geodata Folder Path**

Path to a folder in which the geodata from the Cellular Expert Workspace is stored.

**Prediction Path**

Path for storing prediction grids.

**Result Path**

Path for storing the final calculation results

**Volatile Calculation Path**

Path for temporary Quick Prediction calculation results.

**Volatile Result Path**

Path for storing the final Quick Prediction calculation results

**Volatile Tasks Data Path**

Path for the Quick Prediction calculation results that will be displayed in the Calculation Task List.

## **Project Settings Parameters**

### **Calculate EIRP**

Determines whether calculate EIRP or no in the prediction calculations.

- Value YES. EIRP will be calculated based on Power, Antenna Gain and Misc. Loss values.
- Value NO. EIRP will be taken as single value defined in Power field.

### **Enable GPU Acceleration**

Enables GPU Accelerations that optimizes the prediction calculations and makes them run faster. Possible values Yes/No.

### **Transmit Power Units**

Represents power value in dBm or Watts.

### **Receiver/Transmitter Height Reference**

The reference raster for calculating the receiver's and transmitter's height for these calculations: Profile, RF Predictions, Quick Predictions, Visibility and other prediction tools. Possible values:

- Elevation – reference layer will be used elevation.tif raster. This is a default parameter.
- Clutter height – reference layer will be used clutterHeight.tif raster. Receiver/transmitter height will be calculated over Clutter Height raster. clutterHeight.tif raster is a must to enable this parameter.
- Clutter height (buildings only) – Height reference is calculated using the Building class from the clutterClasses.tif raster. The Building class is defined in the Clutter Class dialog and linked to a specific clutter class ID in the clutterClasses.tif raster.
- Absolute – receiver and/or transmitter's absolute height (relative to sea level) will be used in relevant calculations.

### **Use Clutter Loss**

Determines whether clutterClasses.tif and clutterHeight.tif rasters are used in prediction calculations. Possible values Yes/No.

| Project Settings             |                |
|------------------------------|----------------|
| Calculate EIRP               | Yes            |
| Enable GPU Acceleration      | Yes            |
| Receiver Height Reference    | Clutter height |
| Transmitter Height Reference | Clutter height |
| Use Clutter Loss             | Yes            |

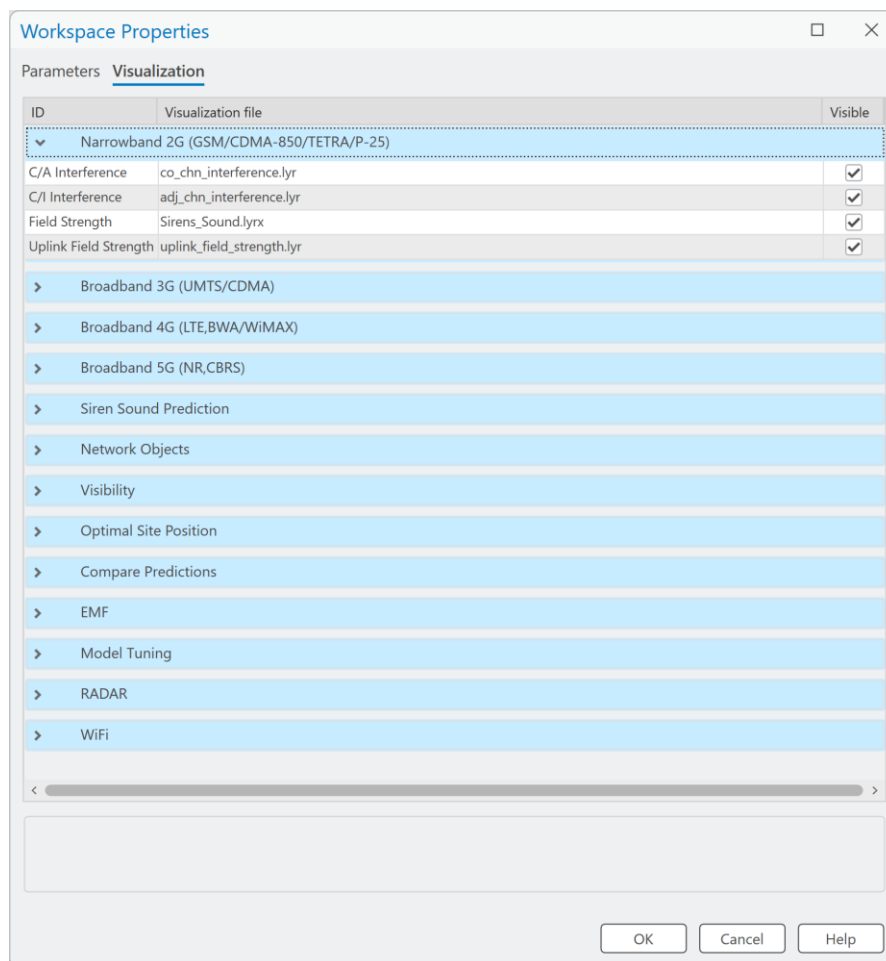
### **Rounding**

The rounding value for different parameters.

| Rounding                    |   |
|-----------------------------|---|
| Azimuth and Tilt, deg       | 2 |
| Default Rounding            | 2 |
| Frequency, MHz              | 2 |
| Geographic Coordinates, deg | 5 |
| Power, dBm                  | 2 |
| Projected Coordinates, m    | 3 |

### 6.1.6.2 Visualization

The Cellular Expert network objects (Sites, Cells, OMEN) and calculation result rasters are represented in ArcGIS with the symbology as defined in the .lyr files, located in the **Visualization** tab (“CE\_LAYERS” table).



**OK**

Click this button to save any changes and close the dialogue.

**Cancel**

Click this button to cancel any changes and close the dialogue.

**Help**

Get helpful information about the dialogue.

All visualization settings have the “Visible” option set to On by default. Suppose the Visible option is set to off, the next time a relevant calculation is performed. In that case, the rasters associated with the calculation are added to the map with the visibility disabled. This is done only for the calculations performed after the setting is changed, and does not impact already added rasters on the map.

The symbology is defined as a list of Layers and *.lyr* files:

- Narrowband 2G (GSM/CDMA-850/TETRA/P-25) – second generation network (like GSM) calculations or technology-independent calculations (for example, the symbology for antenna loss by tilt is defined by the same file for WiMAX, LTE, and other technology)
- Broadband 3G (UMTS/HSDPA) – results for UMTS, HSDPA and other 3 - 3.5 generation technologies
- Broadband 4G (LTE, BWA/WiMAX) – results of LTE technology calculations
- Broadband 5G (NR, CBRS) – results of 5G-NR technology calculations
- Siren Sound Prediction – results of sirens calculations
- Network objects – Sites, Cells, OMEN
- Visibility – results of Visibility Calculations
- Optimal Site Position – results of optimal site positioning.
- Compare predictions – the results of comparing several predictions.
- Model Tuning – results of model calibration
- Radar – results of radar coverage calculations
- Wi-Fi – results of Wi-Fi coverage calculations

For example, the layer *4G Downlink Throughput* with defined path *dl\_ul\_throughput.lyr* means that the 4G downlink bitrate prediction raster will be represented in ArcGIS using the symbology file *.../Cellular Expert/Layers/dl\_ul\_throughput.lyr*

Note: if you change the symbology with ArcGIS tools, it will be saved only in the current ArcGIS project. So, when you re-open the same Cellular Expert workspace in another ArcGIS project, the symbology will be defined in the **Visualization** tab of the **Workspace Properties** dialogue.

To create a new visualization and locate it in the **Workspace Properties**:

- Right-click on the layer with the modified symbology (for example *Sites*) in the ArcGIS **Table Of Contents**
- Choose **Sharing > Save As Layer File...** from the opened menu
- Save the file with a given name (for example *Sites\_my\_symbol.lyrx*)
- Copy/Paste it to C:\Program Files\Cellular Expert\Layers
- Open the Cellular Expert workspace in which you want to use your symbology
- Open the **Workspace Properties** dialogue and select the **Visualization** tab
- Select the row with the corresponding layer name (for example *Sites*)



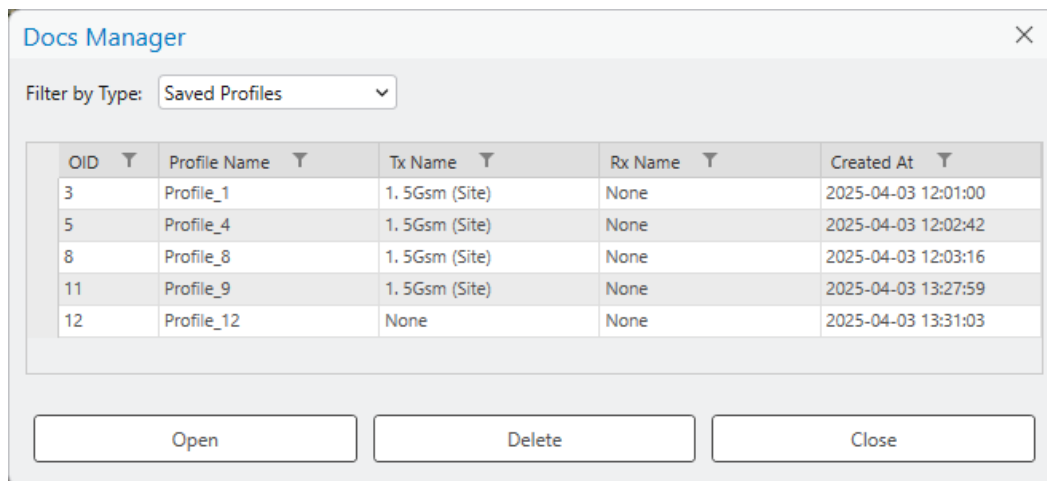
- Click the browse button  and locate your layer file (for example “C:\Program Files\Cellular Expert\Layers\Sites\_my\_symbol.lyrx”) which contains the modified symbols. Press the **OK** button.
- The symbology for the Cellular Expert network objects (Sites, Cells, OMEN) will be applied to the layers when you open the workspace next time. To see the changes immediately re-open the workspace (close it using the **Remove** option and open it again with the **Open** option from the workspace menu)

The new symbology will be uploaded and remain assigned to the workspace database independently of the ArcGIS project file.

When you use layer files for visualization, do not forget about them. Remember that when you move your workspace to another location, the location path settings can become incorrect. If Cellular Expert is not able to find your defined symbology file, it will use the default file from the location .../Cellular Expert/Layers.

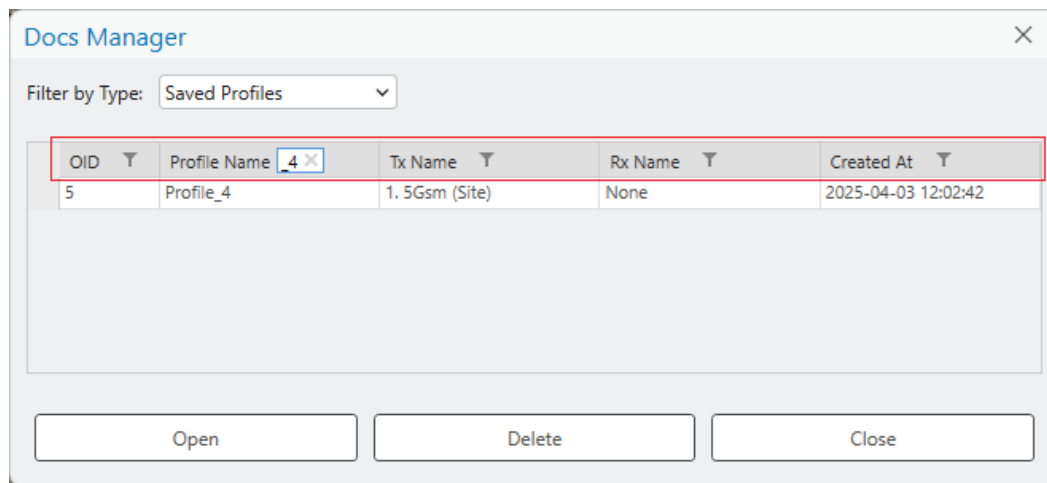
## 6.2 Docs Manager

Docs Manager is a tool for managing Saved Profiles between the transmitter (Tx) and receiver (Rx), which are generated in the Profile tool, as well as saved Link Prediction results, Profile Reports, and Link Prediction Reports. When a profile is saved in the Profile tool, it is automatically stored in Docs Manager, allowing users to reopen it at any time. This ensures that all parameters and calculations related to Tx and Rx are preserved for future reference, eliminating the need to reconfigure settings repeatedly. The same applies for Link Prediction results, and the Reports can be opened if the original exported document is, for example, deleted from Desktop or other saved location.



### How to Find a Saved Profile

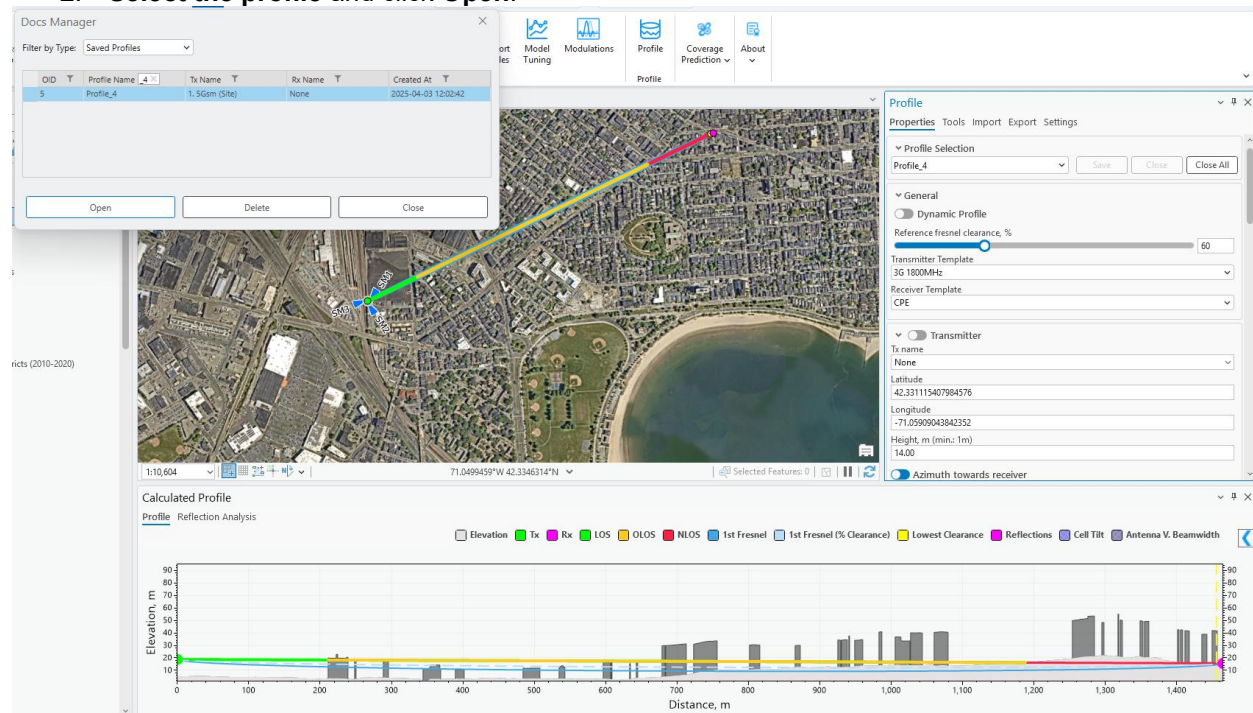
Use the filter option for each field to quickly locate the required profile from the list.



### How to Open a Profile

A saved profile can be accessed in two ways:

1. **Double-click** on the desired profile.
2. **Select the profile** and click **Open**.



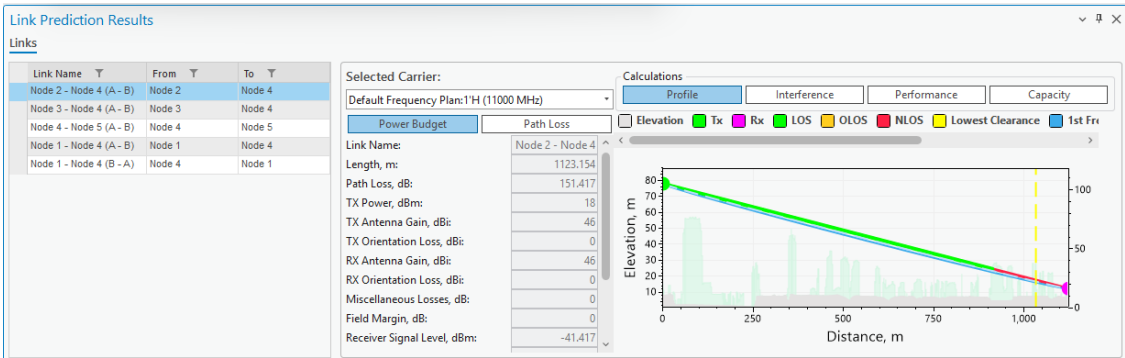
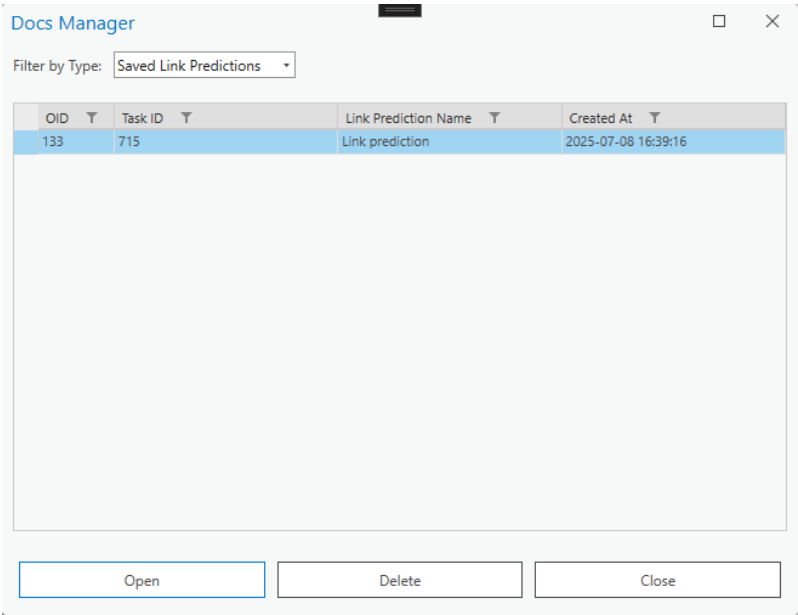
### Additional Functions

- **Delete:** Removes the selected profile.
- **Close:** Closes the Docs Manager dialog.

By using Docs Manager, users can efficiently store and retrieve profiles, ensuring consistency and accuracy in their Tx and Rx calculations.

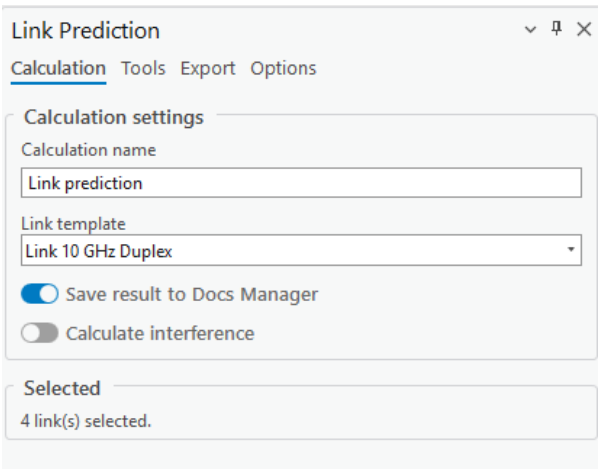
How to Open a Link Prediction

A saved Link Prediction, just like Profile, can be accessed either by double-clicking the desired Link Prediction result, or by select it and clicking Open.



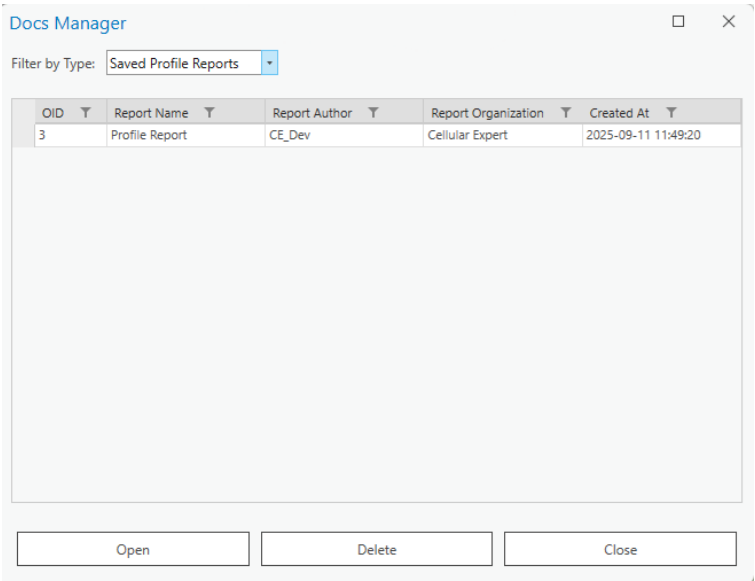
How to Save a Link Prediction

Link Prediction result can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Link Prediction tool.



How to Open a Profile Report

A Profile Report document can be accessed either by double-clicking the desired Profile Report, or by select it and clicking Open. The document will be opened using your default PDF document reader.



How to Save a Profile Report

Profile Report of the current drawn profile can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Export tab of the Profile tool.

The screenshot shows the 'Profile' dialog box with the 'Export' tab selected. The fields are as follows:

- Author:** CE\_Dev
- Organization:** Cellular Expert
- Report Name:** Profile Report
- Profile Report File Path:** C:\Users\jlekavicius\Desktop\Document
- Format:** PDF
- Save result to Docs Manager:** ☒
- Export Profile:** [Button]

### How to Open a Link Prediction Report

A Link Prediction Report document can be accessed either by double-clicking the desired Link Prediction Report, or by select it and clicking Open. The document will be opened using your default PDF document reader.

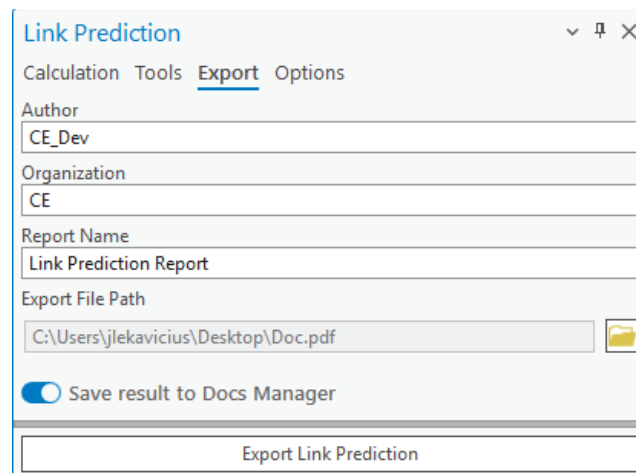
The screenshot shows the 'Docs Manager' dialog box. It has a filter dropdown set to 'Saved Link Prediction Re...' and a table with the following data:

| OID | Report Name            | Report Author | Report Organization | Created At          |
|-----|------------------------|---------------|---------------------|---------------------|
| 2   | Link Prediction Report | CE_Dev        | CE                  | 2025-09-11 11:47:53 |

At the bottom, there are three buttons: 'Open', 'Delete', and 'Close'.

### How to Save a Link Prediction Report

Link Prediction Report can be saved to Docs Manager by selecting **Save result to Docs Manager** in the Export tab of the Link Prediction tool.



The image shows a software dialog box titled "Link Prediction". It has a tabbed interface with "Calculation", "Tools", "Export" (selected), and "Options". The "Export" tab contains the following fields and controls:

- Author:** A text field containing "CE\_Dev".
- Organization:** A text field containing "CE".
- Report Name:** A text field containing "Link Prediction Report".
- Export File Path:** A text field containing "C:\Users\jlekavicius\Desktop\Doc.pdf" with a folder icon button to its right.
- Save result to Docs Manager:** A toggle switch that is currently turned on.
- Export Link Prediction:** A large button at the bottom of the dialog.

### 6.3 CE Express Connection

**CE Express Connection** is a tool that lets you establish a connection between the *CE Express* database and *CE for ArcGIS Pro*. When the connection is established, data can be retrieved from *CE Express* and uploaded to *CE for ArcGIS Pro* workspace.

For the connection to be established you must have access to **ArcGIS Portal** and have a valid **CE Express URL**.

#### 6.3.1 Properties



Click the  button and select the **Properties** tab to open the **CE Express Connection** dialogue.

To connect to *CE Server Express* and get the list of workspaces, insert the **Server URL** in the corresponding field. Then press the **Get Workspaces** button.

If you select one of the appearing workspaces, the properties of that workspace will be saved to your current ArcGIS Pro project automatically.

CE Express Connection Manager

Properties

Server URL:

Server ArcGIS Portal URL:

Server API URL:

Selected Workspace:

Selected Feature:

| cell_name | latitude    | height | azimuth | tilt | frequency |
|-----------|-------------|--------|---------|------|-----------|
| M2        | 42.30081856 | 47     | 90      | 0    | 3500      |
| SM3       | 42.33111541 | 10     | 251     | 0    | 3500      |
| SM2       | 42.33111541 | 10     | 146     | 0    | 3500      |
| SM1       | 42.33111541 | 10     | 33      | 0    | 3500      |
| M3        | 42.30081856 | 47     | 180     | 0    | 3500      |
| L3        | 42.30049188 | 45     | 180     | 0    | 3500      |
| L4        | 42.30049188 | 45     | 270     | 0    | 3500      |
| L2        | 42.30049188 | 45     | 90      | 0    | 3500      |
| L1        | 42.30049188 | 45     | 0       | 0    | 3500      |
| M4        | 42.30081856 | 47     | 270     | 0    | 3500      |

Complete in 00:00:00:046 seconds.

Get Workspaces Import Features

### Server URL

The URL to the *CE Express* database where the workspaces will be located. This parameter must be defined to connect to the database.

### Server ArcGIS Portal URL

The URL to your organization's ArcGIS Portal (filled in automatically).

### Server API URL

The API is used in the connection process (filled in automatically).

### Selected Workspace

The list of all workspaces was retrieved from the *CE Express Database*.

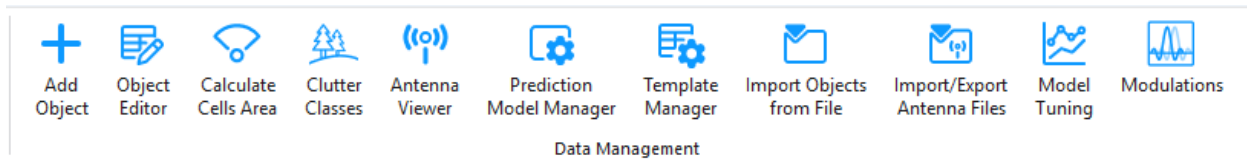
### Get Workspaces

Establishes the connection between the provided *Server URL* and *CE for ArcGIS Pro*. Upon clicking the button, the user may be redirected to a browser window in which he will have to log in to the **ArcGIS Portal**. After doing so, the workspaces will be retrieved.

### Import Features

Imports the retrieved objects to the currently opened CE workspace.

## 7. Data Management



### 7.1 Network Objects

Cellular Expert network objects are:

**Sites** – represent the geographical location of a radio station. They are identified by the unique site ID and contain information about the geographical coordinates, ground altitude, and base height.

**Cells** – primary network object for prediction calculations. They denote the carrier list, equipment information, height, and other common radio channel parameters.

**Sirens** – represent siren objects and their geographical locations on the map. They contain essential parameters required for siren prediction.

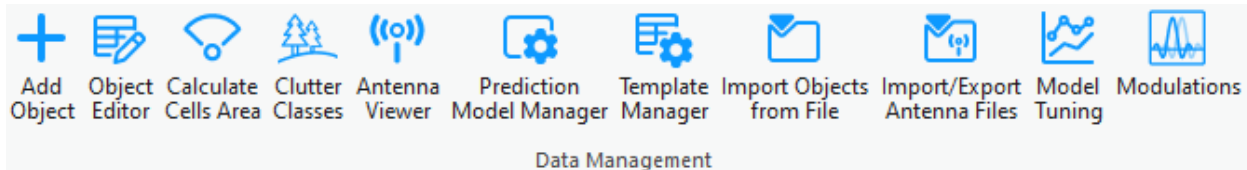
**Links** – represents a radio connection between a transmitter and a receiver. It can be established between the selected transmitter and receiver. The link operates at a fixed frequency from the transmitter's carriers list. Available within CE RLP license.

**Repeaters** – represents the geographical location of a signal repeater. Only used for microwave point-to-point network planning using Links.

**CPE** – stands for Customer Premises Equipment and represents the geographic location of a client.

**Radar** – represents the geographical location of a radiolocation system that uses radio waves to determine the distance, angle, and radial velocity of objects.

With the data management tools located in the **Data Management** section, you can view and edit the Cellular Expert network objects.



Note: most of these tools work only in editing sessions on the currently active workspace.

### 7.2 Add Object

New network objects can be created in several ways. They can be:

- Imported using the CE for ArcGIS Pro functionality
- Created with Cellular Expert tools from zero (define all parameters in the process)

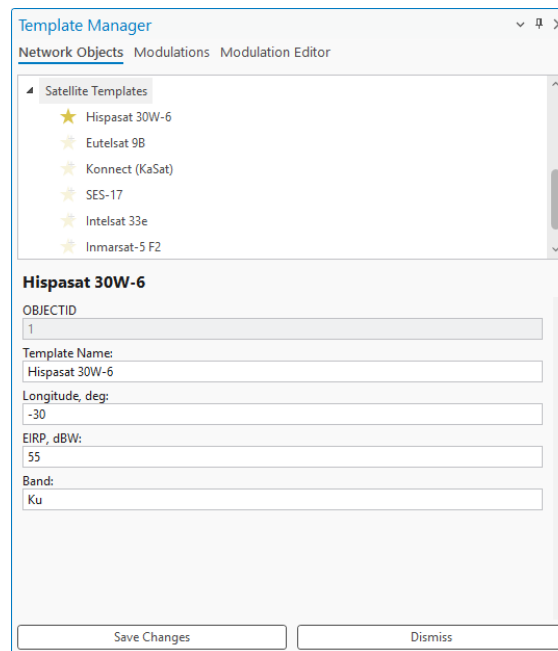


- Created from templates

There are three tabs with different functionalities for adding objects: Network Objects, Feature Set Templates, and Batch Creation.

### 7.2.1 Add Satellite

Currently, satellites can be used as templates in Template Manager, under the Satellite Templates group in the Network Objects tab. These can be edited or duplicated to create new templates for use in Quick SAT Link Budget and Sky Clearance tools (when GEO satellite type is selected).

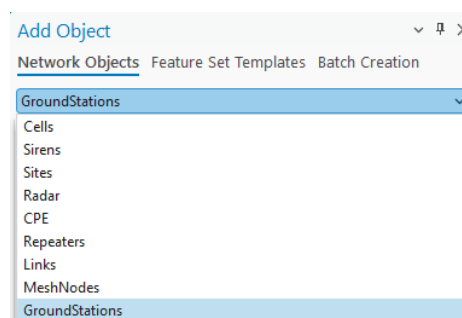


### 7.2.2 Add Ground Station

The object represents a fixed earth-based satellite terminal. It carries the station's location, the antenna it uses, and the terminal and link parameters needed for satellite calculations. It is used in the Sky Clearance and Quick SAT Link Budget calculations.



1. Choose the  button from the toolbar and select **GroundStations** from the dropdown list.



- Define the location of the new Ground Station by pressing the mouse left button on the map. The new Ground Station will be placed right in that location.

**Add Object** ⌵ ⌵ ✕

Network Objects Feature Set Templates Batch Creation

GroundStations ⌵

---

**Ground Station Properties**

Template:  
★ Starlink Standard ⌵

Name:

Latitude (degrees):  
54 ° 41 ' 12.99 ''

Longitude (degrees):  
25 ° 17 ' 0.22 ''

Latitude:

Longitude:

X:

Y:

Z:

Height above ground, m:

Ground altitude, m:

The Ground Station object can be created by entering exact coordinates in:

- Latitude (degrees) and Longitude (degrees) section.

Latitude (degrees):  
54 ° 41 ' 49.51 ''

Longitude (degrees):  
25 ° 14 ' 51.69 ''

- Latitude and Longitude

Longitude:

Latitude:

- X and Y (projected coordinate system)

|    |              |
|----|--------------|
| X: | 580427.2251  |
| Y: | 6063011.8366 |

3. Define Ground Station name and press Save Changes to save object to the database.

**Save Changes**

Creates the object with the given parameters.

**Dismiss**

Cancels object creation and closes the dialogue.

**Ground Station Properties****Name**

Ground Station identification.

**Latitude (degrees)**

Latitude degrees, minutes, and seconds Y type coordinate in the WGS 1984 geographical coordinate system.

**Longitude (degrees)**

Longitude degrees, minutes, and seconds X type coordinate in the WGS 1984 geographical coordinate system.

**Latitude**

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

**Longitude**

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

**X**

Coordinate in the projected coordinate system.

**Y**

Coordinate in the projected coordinate system.

**Z**

3D dimensions representing an object's height above sea level, used for visualizing objects in a 3D scene.

**Height Above Ground**

Object's height above the terrain.

**Ground Altitude**

Ground elevation above sea level at the network object's location.

**Antenna**

Define antenna patterns for the Ground Station object.

**Terminal G/T**

Ratio of antenna gain to system noise temperature. Determines the terminal's receive sensitivity. Value in dB/K.

**Minimum elevation**

Lowest elevation angle at which the terminal can use a satellite. Value in degrees.

**Polarization**

Signal polarization, options: Horizontal, Vertical, or Circular.

**Band**

Frequency band of the terminal, such as Ka, Ku, or C.

**EIRP**

Effective isotropic radiated power of the terminal. Value in dBW.

**Downlink frequency**

Receive (downlink) frequency. Value in MHz.

**Uplink frequency**

Transmit (uplink) frequency. Value in MHz.

**Bandwidth**

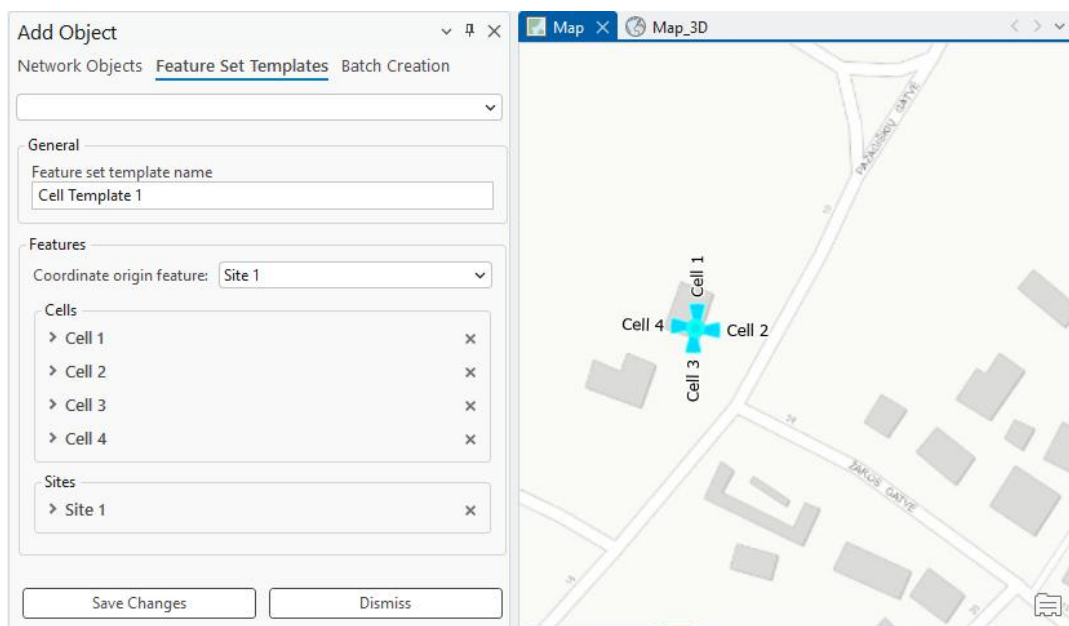
Channel bandwidth Value in MHz.

**Required C/N**

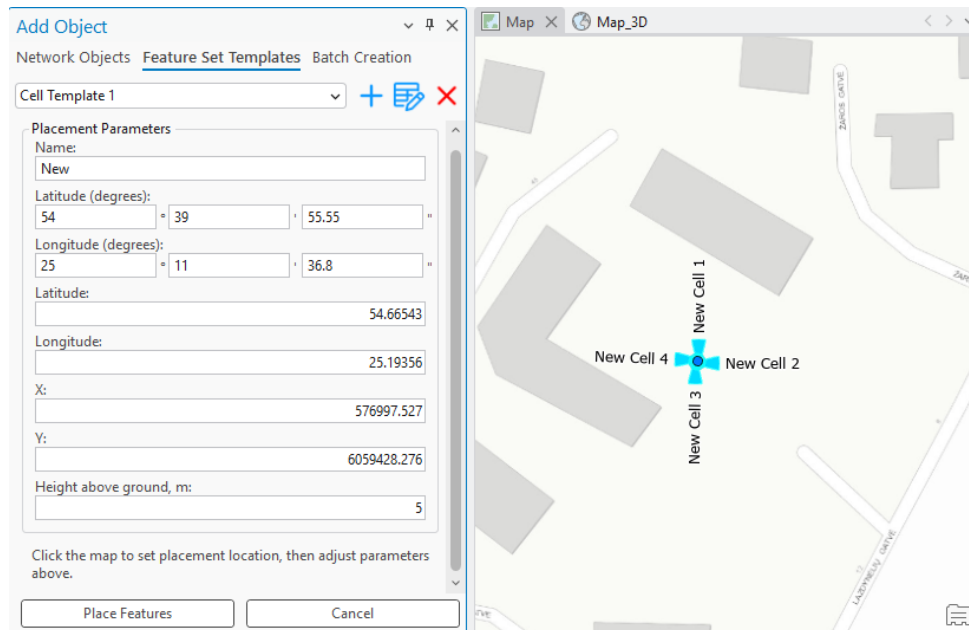
Minimum carrier-to-noise ratio, needed for the link to close. Value in dB.

### 7.2.3 Feature Set Templates

The Feature Set Templates tab in Add Object supports composite templates that describe a group of multiple objects, preserving the relative geometry between them. A composite template is created by selecting two or more existing objects and saving them as a template.



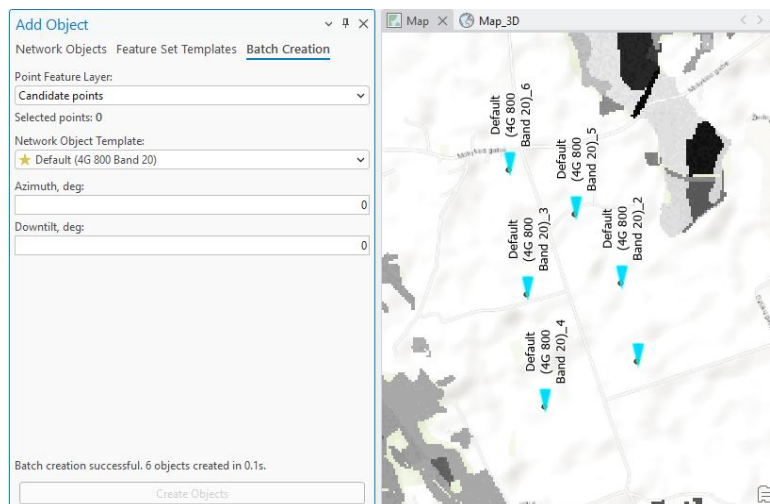
When a composite template is selected in the Add Object tool, the whole group can be placed on the map in a single action, either by clicking a location on the map or by entering explicit coordinates (X/Y or latitude/longitude). All objects in the group are created at their correct positions relative to the placement point and remain individually editable after placement.



### 7.2.4 Batch Creation

Objects can also be created in bulk from an existing point feature layer. The user can select points from any point feature layer available in the ArcGIS Pro project, pick a network object template, and create an object or an object group at each selected location using the template default values.

Progress is shown for larger batches, and a completion summary reports the total number of points selected, the objects successfully created, and any failures with their reasons. Objects created in a batch are individually editable afterwards, and the source feature layer is not modified.



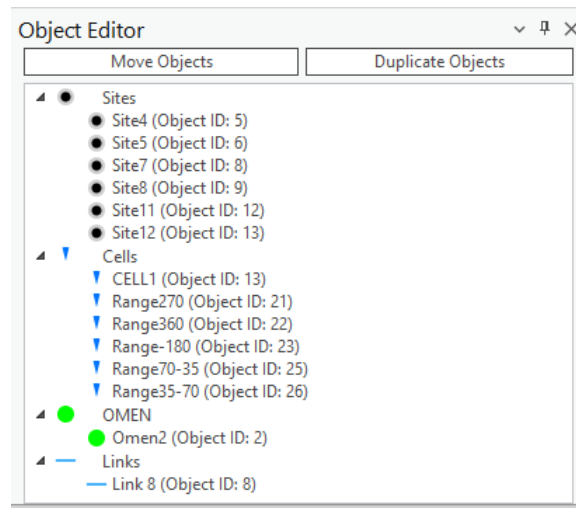
### 7.3 Object Editor

The Object Editor enables the user to make changes to a network object after it is created and placed on the map.

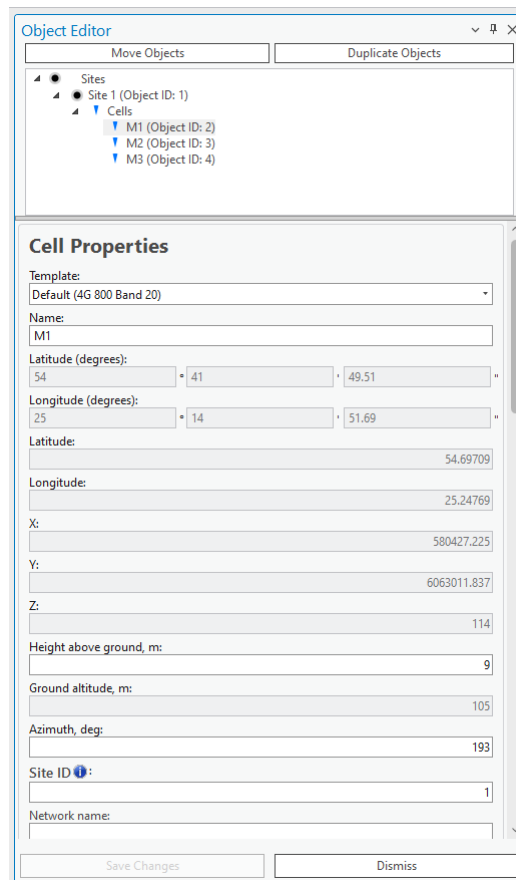


Choose the  button to open the **Object Editor** dialogue.

Select objects by navigating to the ArcGIS Pro **Edit** → **Selection** section and choosing the **Select** tool. The selected objects will appear in the Object Editor in a tree hierarchy.

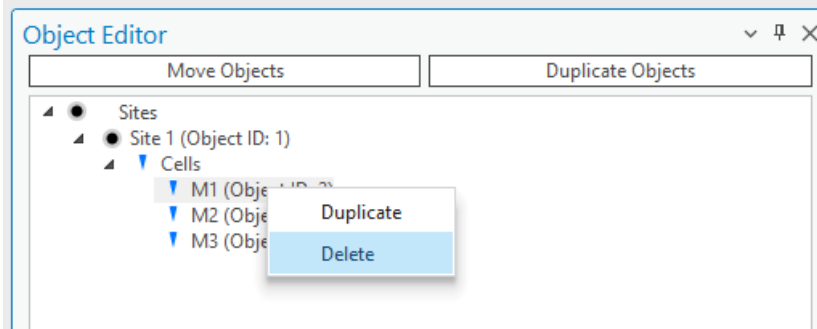


To edit one of the selected objects, left-click on that object and the corresponding editing menu will open below the list.



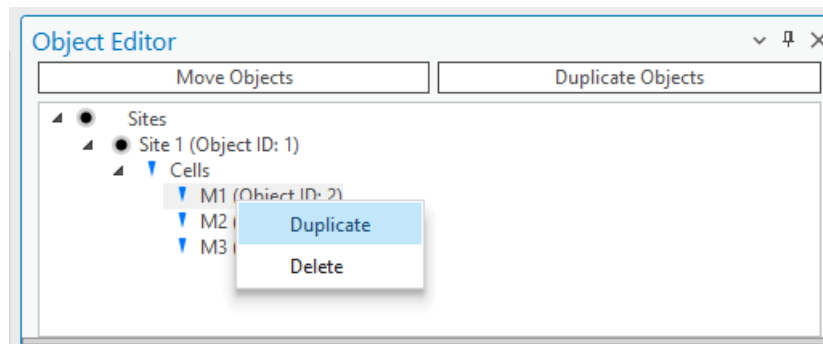
### Delete Object

Select and right-click any network object from the selection, then choose the **Delete** option from the popup.



### Duplicate Object

Select and right-click any object from the selection, then choose the **Duplicate** option from the popup. The duplicated object will retain all information from the original object including coordinates/meridians. If you want to duplicate objects and change their coordinates/meridians at the same time, use the separate **Duplicate Objects** button from above the selection tree.



### 7.3.1 Move Objects



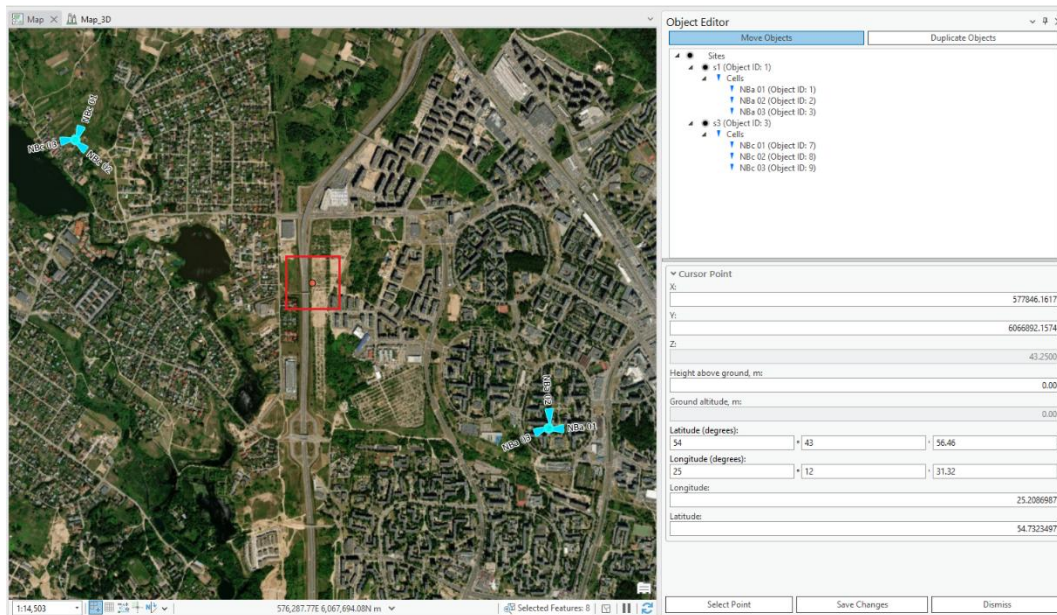
Choose the **Object Editor** button to open the **Object Editor** dialogue. Select the desirable objects with the **Select** tool and press the **Move Objects** button in the Object Editor.

The screenshot shows the 'Object Editor' window with the 'Move Objects' tab selected. The tree view on the left shows a hierarchy: 'Sites' (expanded) -> 'H1 (Object ID: 2)' (expanded) -> 'Cells' (expanded) -> 'D1 (Object ID: 6)' and 'D2 (Object ID: 7)'. The 'Cursor Point' section is expanded, showing fields for X, Y, Z, Height above ground, Ground altitude, Latitude (degrees), Longitude (degrees), and Longitude/Latitude coordinates. The 'Select Point', 'Save Changes', and 'Dismiss' buttons are at the bottom.

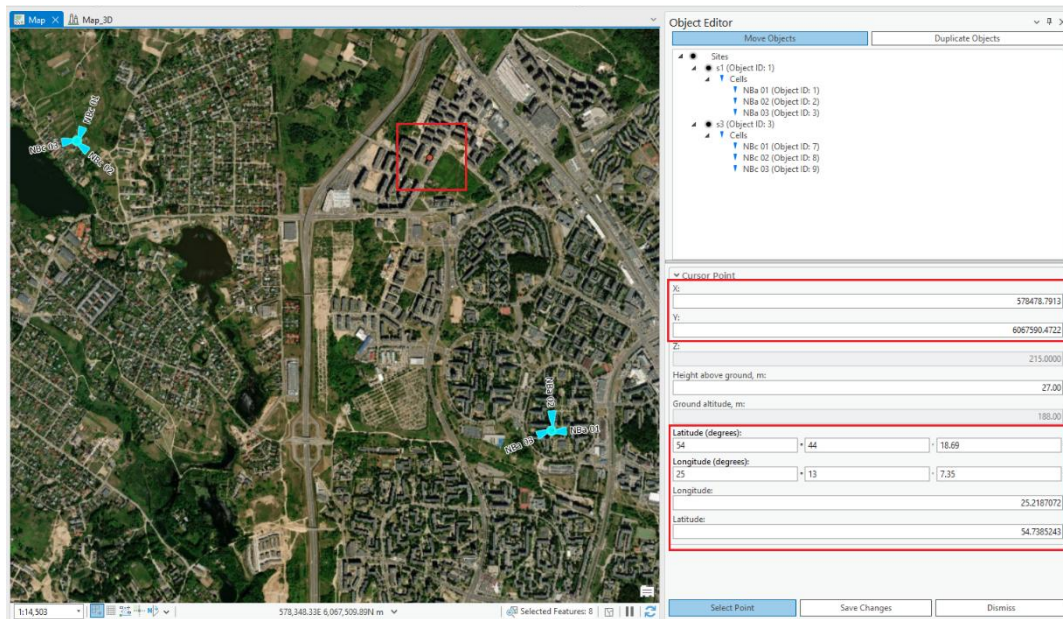


There are multiple ways to move objects:

- If multiple objects are selected, the move objects display will show the geospatial properties of the center point between the objects denoted as the “Cursor point”.



The "Cursor Point" functions as a reference marker depicted by a red dot on the map. This marker is centrally located among the objects on the map and shows where the cursor was placed. You can adjust the position of the Cursor Point either by entering new coordinates directly or by clicking “Select Point” and choosing a different location on the map. When you move the Cursor Point, all the objects on the map will shift their position to maintain their relative distances from this central point.



- If a single point object is selected or several objects at the same location, the move objects display will show the geospatial properties of these objects denoted as “Cursor Point”.

Object Editor

Move Objects Duplicate Objects

- Sites
  - s3 (Object ID: 3)
    - Cells
      - NBc 01 (Object ID: 7)
      - NBc 02 (Object ID: 8)
      - NBc 03 (Object ID: 9)

Cursor Point

X: 576548.6890

Y: 6067679.9148

Z: 55.0000

Height above ground, m: 0.00

Ground altitude, m: 0.00

Latitude (degrees): 54 ° 44 ' 22.65 "

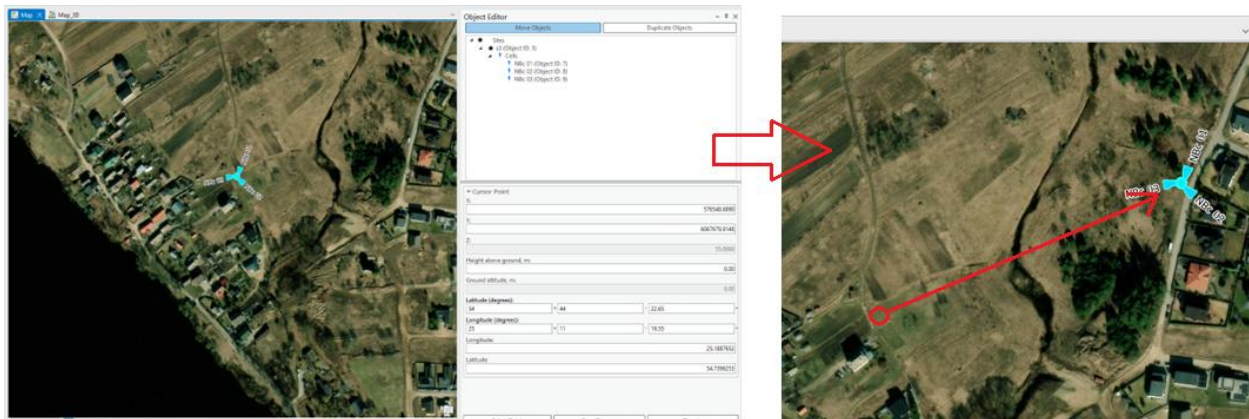
Longitude (degrees): 25 ° 11 ' 19.55 "

Longitude: 25.1887652

Latitude: 54.7396253

Select Point Save Changes Dismiss

If a single point object is selected on the map, the "Cursor Point" serves as its positional anchor. Positioned as a red dot, the Cursor Point represents the current coordinates of the selected object and shows where the cursor was last placed. You can alter the location of the selected object by manually updating the Cursor Point's coordinates or by clicking "Select Point" and choosing a new location on the map. Any movement of the Cursor Point will directly translate the selected object to the corresponding new location while keeping its spatial relationship with the Cursor Point consistent.



### Save Changes

Saves the changes to the objects.

### Dismiss

Cancels the changes to the objects and closes the dialogue.

### 7.3.2 Duplicate Objects



Choose the **Object Editor** button to open the **Object Editor** dialogue. Select the object with the **Select** tool and press the **Duplicate Objects** button in the Object Editor dialogue.

**Object Editor**

Move Objects | Duplicate Objects

- Sites
  - s1 (Object ID: 1)
    - Cells
      - NBa 01 (Object ID: 1)
      - NBa 02 (Object ID: 2)
      - NBa 03 (Object ID: 3)

▼ Cursor Point

X: 579143.6345

Y: 6066104.4000

Z: 31.5000

Height above ground, m: 0.00

Ground altitude, m: 0.00

Latitude (degrees): 54 43 30.25

Longitude (degrees): 25 13 43.05

Longitude: 25.2286250

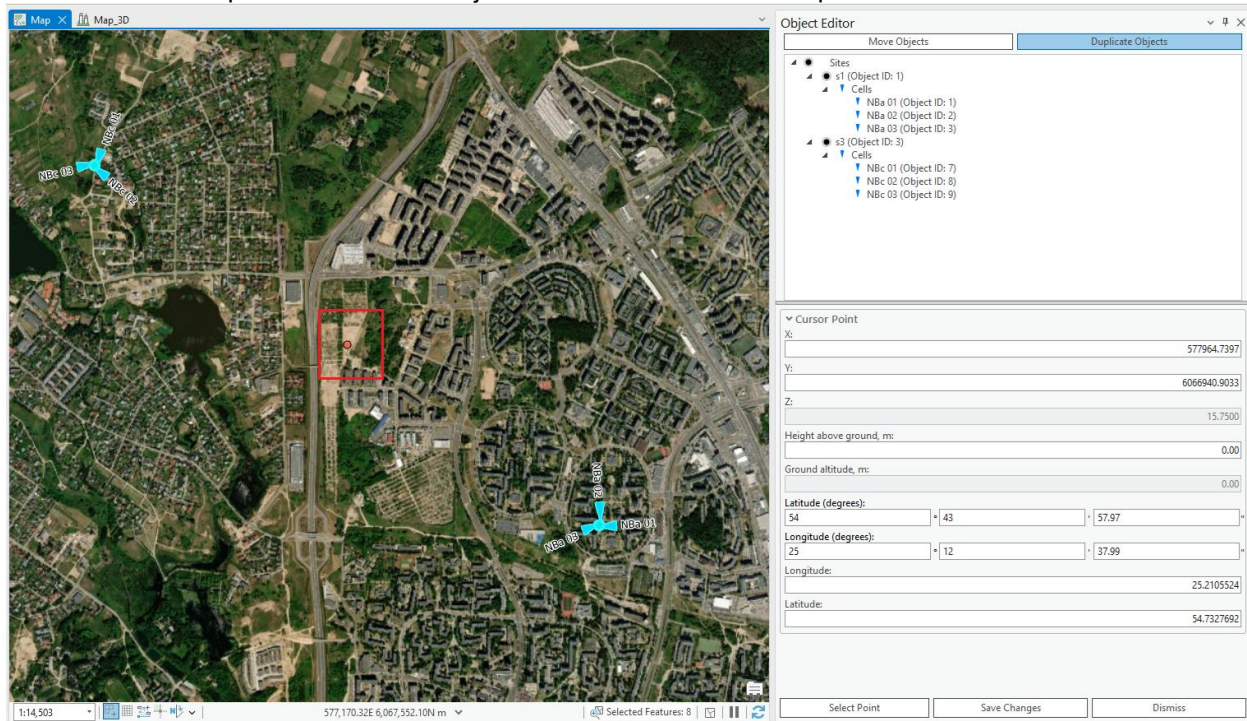
Latitude: 54.7250708

Select Point | Save Changes | Dismiss

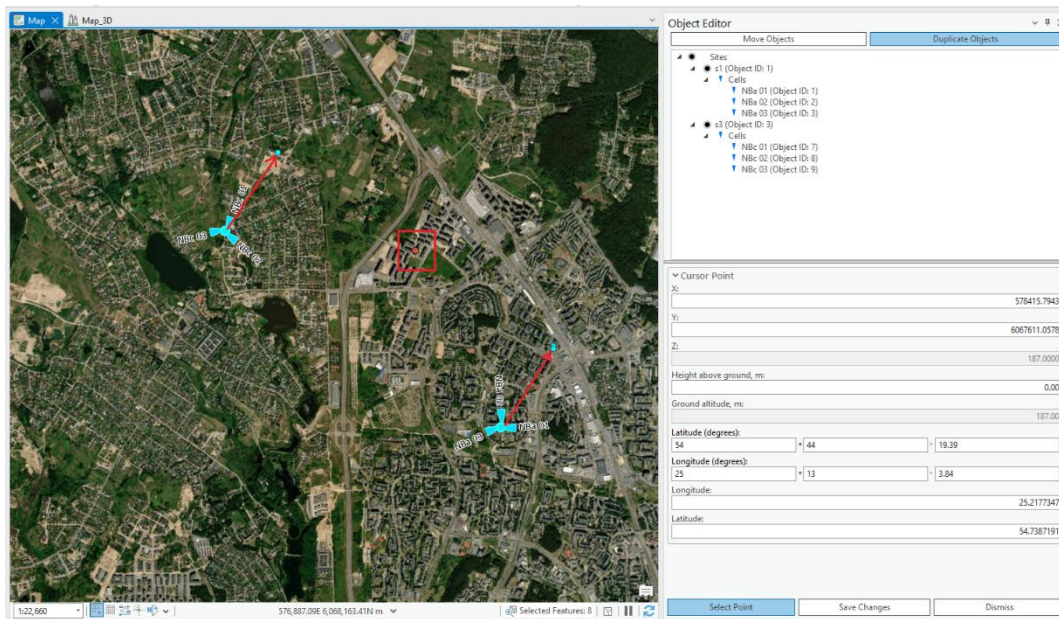


There are multiple ways to duplicate objects:

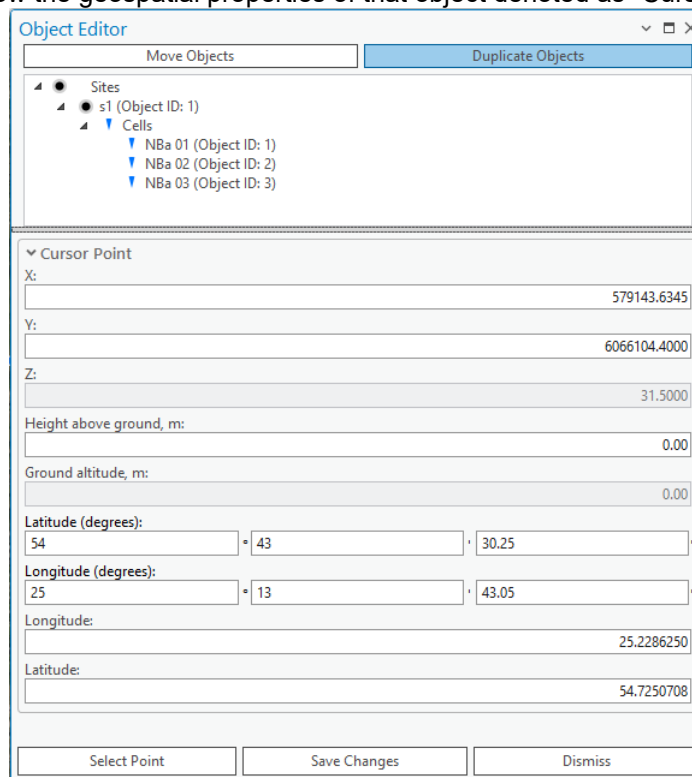
- If multiple objects are selected, the duplicate objects display will show the geospatial properties of the center point between the objects denoted as the “Cursor point”.



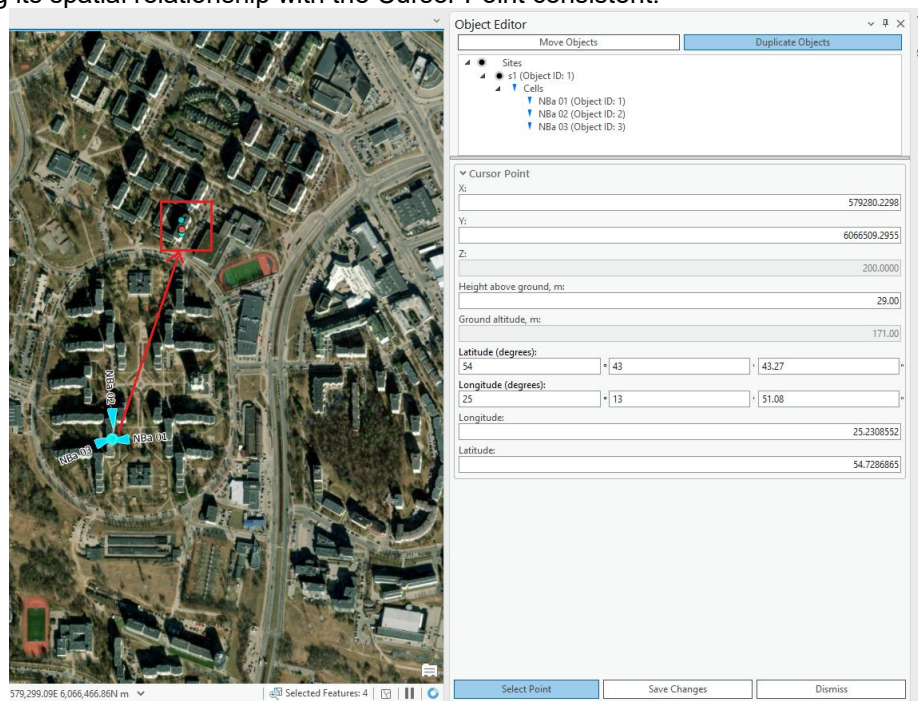
The "Cursor Point" functions as a reference marker depicted by a red dot on the map. This marker is centrally located among the objects on the map and shows where the cursor was placed. You can adjust the position of the Cursor Point either by entering new coordinates directly or by clicking “Select Point” and choosing a different location on the map. When you move the Cursor Point, all the objects on the map will shift their position to maintain their relative distances from this central point.



- If a single point object or several objects on the same location are selected, the duplicate objects display will show the geospatial properties of that object denoted as “Cursor Point”.



If a single point object is selected on the map, the "Cursor Point" serves as its positional anchor. Positioned as a red dot, the Cursor Point represents the current coordinates of the selected object and also shows where the cursor was last placed. You can alter the location of the selected object by manually updating the Cursor Point's coordinates or by clicking "Select Point" and choosing a new location on the map. Any movement of the Cursor Point will directly translate the selected object to the corresponding new location while keeping its spatial relationship with the Cursor Point consistent.



### Save Changes

Duplicates the objects.

### Dismiss

Cancels the changes to the objects and closes the dialogue.

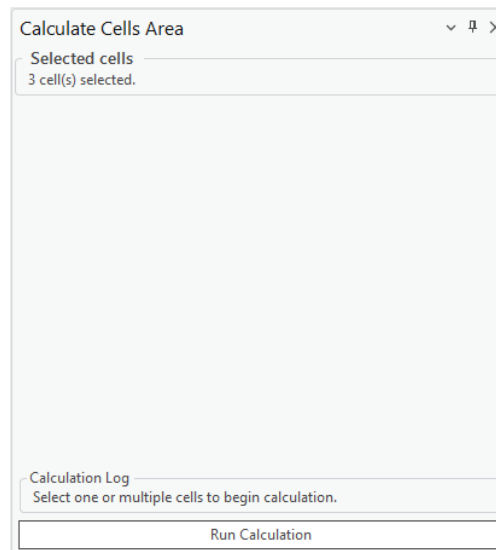
## 7.4 Calculate Cells Area

The **Calculate Cells Area** tool is designed to compute the service areas of selected cell sites within a given map and display them as polygons in a new layer. By selecting one or more cells, users can initiate a calculation that considers various parameters like cell azimuth, horizontal beamwidth, and antenna patterns to generate precise coverage polygons.

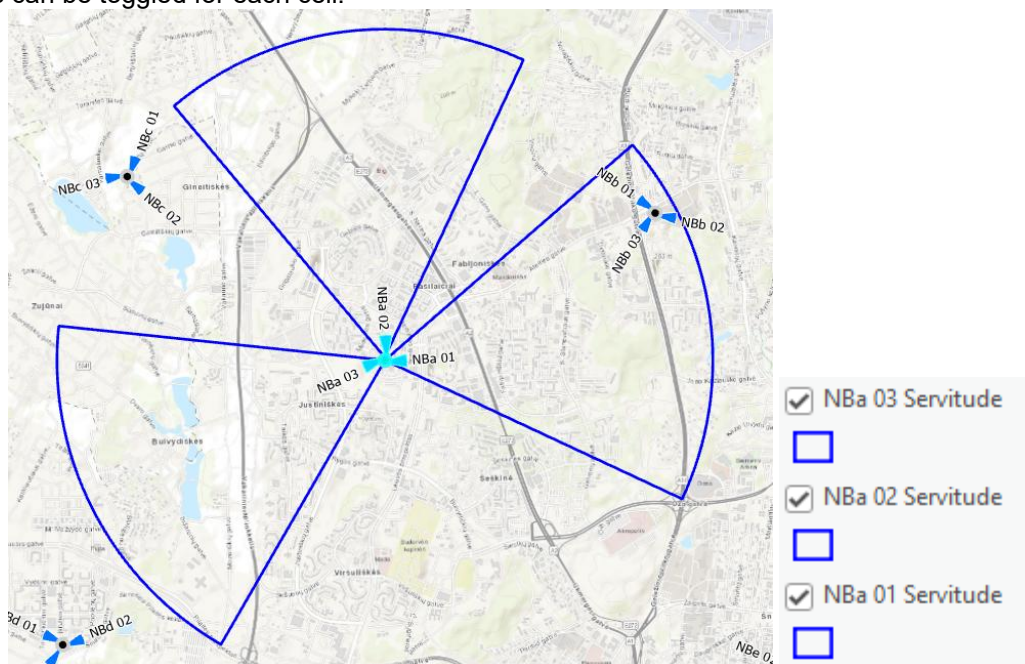


Choose the **Calculate Cells Area** button to open the **Calculate Cells Area** tool.



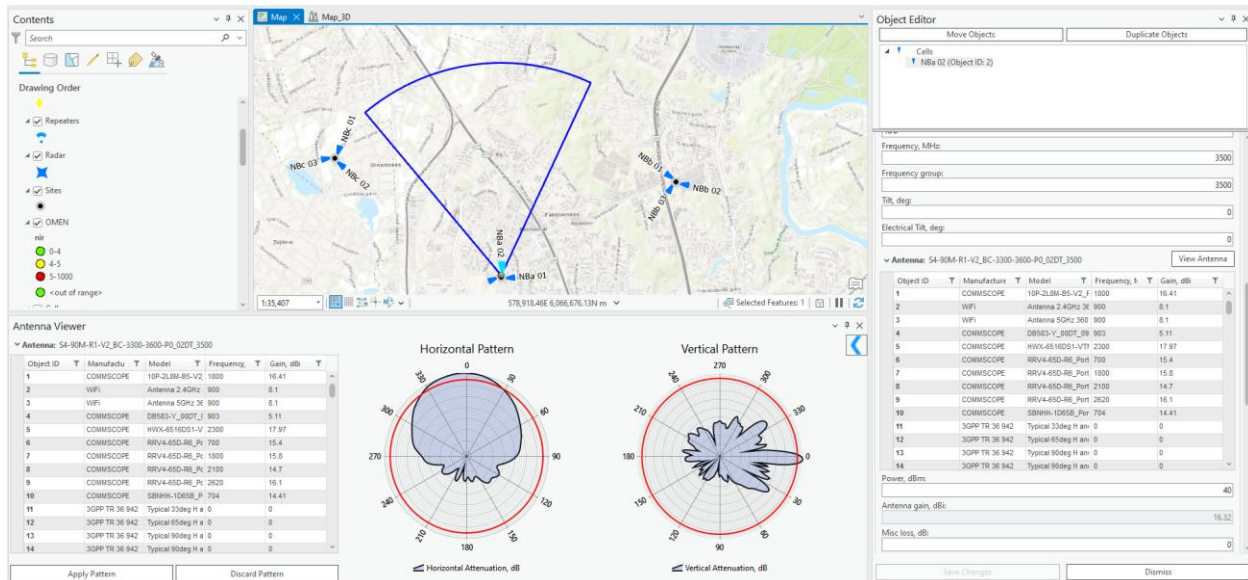


Select the cells which the calculations will be performed for, and press **Run Calculation**. The cell areas will be drawn on the map, in a separate layer **CE Cell Area Results**. In the results group layer, the servitude polygons can be toggled for each cell.



The polygons represent the cell area where the antenna's horizontal pattern's attenuation at each angle is lower than the threshold of 3.

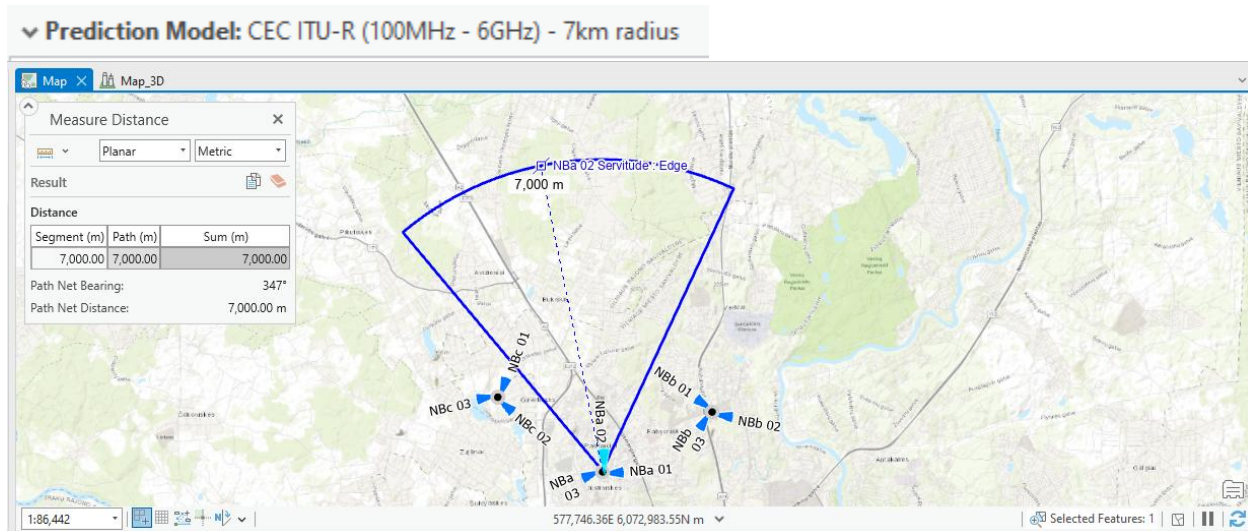




Maximum radius depends on defined Prediction Model, Radius value for the Cell object.

▼ Prediction Model: CEC ITU-R (100MHz - 6GHz) - 3km radius

Using bigger radius:



## 7.5 Clutter Classes

The **Clutter Classes** tool is designed to manage categories describing different types of environments in telecommunication networks. It relies on the Clutter Classes raster used in the project. The raster values must match the ID values in the Clutter Classes dialog. For example, if "Trees" has a value of 2 in the Clutter Classes raster, this ID must be defined in the Clutter Classes tool. By default, ESRI's Sentinel-2 Land Use values are used after workspace creation: [ESRI Sentinel-2 Land Cover](#)

Here is the table view of Clutter table, which comes with default database.

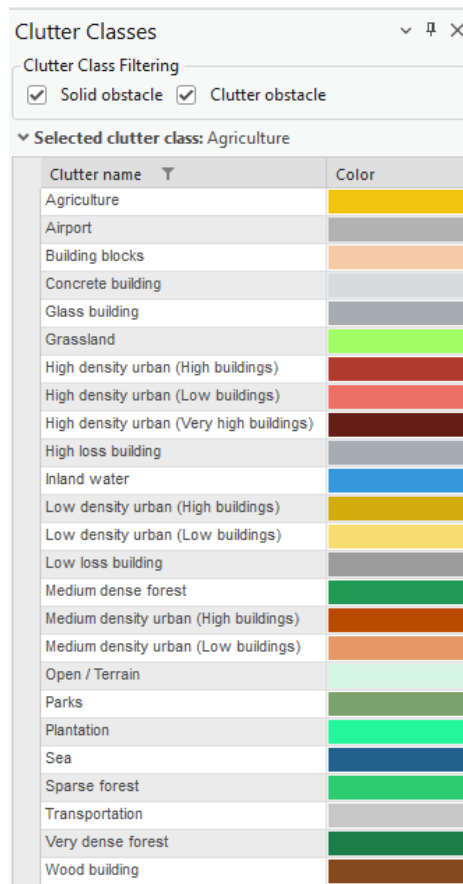
| clutter                                                                                    |                         |        |                  |                      |                       |                      |            |         |        |                                          |
|--------------------------------------------------------------------------------------------|-------------------------|--------|------------------|----------------------|-----------------------|----------------------|------------|---------|--------|------------------------------------------|
| Field: Add Calculate Selection: Select By Attributes Switch Clear Delete Copy Rows: Insert |                         |        |                  |                      |                       |                      |            |         |        |                                          |
| OBJECTID *                                                                                 | clutter_class_name      | height | nominal_distance | surface_refractivity | relative_permittivity | surface_conductivity | express_id | color   | ids    | clutter_class_alias                      |
| 1                                                                                          | open                    | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #d5f5e3 | <Null> | Open / Terrain                           |
| 2                                                                                          | grassland               | 0      | 27               | 315                  | 20                    | 0.03                 | <Null>     | #A0FE62 | <Null> | Grassland                                |
| 3                                                                                          | sparseForest            | 0      | 27               | 315                  | 13                    | 0.004                | <Null>     | #2ecc71 | <Null> | Sparse forest                            |
| 4                                                                                          | mediumDenseForest       | 0      | 27               | 315                  | 13                    | 0.004                | <Null>     | #229954 | <Null> | Medium dense forest                      |
| 5                                                                                          | veryDenseForest         | 0      | 15               | 315                  | 13                    | 0.004                | <Null>     | #1C7D46 | <Null> | Very dense forest                        |
| 6                                                                                          | lowDensityUrbanLowB...  | 0      | 27               | 315                  | 5                     | 0.001                | <Null>     | #f7dc6f | <Null> | Low density urban (Low buildings)        |
| 7                                                                                          | lowDensityUrbanHigh...  | 0      | 27               | 315                  | 5                     | 0.001                | <Null>     | #d4ac0d | <Null> | Low density urban (High buildings)       |
| 8                                                                                          | mediumDensityUrbanL...  | 0      | 15               | 365                  | 5                     | 0.001                | <Null>     | #e59866 | <Null> | Medium density urban (Low buildings)     |
| 9                                                                                          | mediumDensityUrban...   | 0      | 15               | 315                  | 5                     | 0.001                | <Null>     | #ba4a00 | <Null> | Medium density urban (High buildings)    |
| 10                                                                                         | highDensityUrbanLow...  | 0      | 10               | 315                  | 5                     | 0.001                | <Null>     | #ec7063 | <Null> | High density urban (Low buildings)       |
| 11                                                                                         | highDensityUrbanHig...  | 0      | 10               | 315                  | 5                     | 0.001                | <Null>     | #b03a2e | <Null> | High density urban (High buildings)      |
| 12                                                                                         | highDensityUrbanVery... | 0      | 10               | 315                  | 5                     | 0.001                | <Null>     | #641e16 | <Null> | High density urban (Very high buildings) |
| 13                                                                                         | buildingBlocks          | 0      | 15               | 315                  | 5                     | 0.001                | <Null>     | #f5cba7 | <Null> | Building blocks                          |
| 14                                                                                         | transportation          | 0      | 27               | 315                  | 10                    | 0.002                | <Null>     | #c7c7c7 | <Null> | Transportation                           |
| 15                                                                                         | agriculture             | 0      | 27               | 315                  | 20                    | 0.03                 | <Null>     | #f1c40f | <Null> | Agriculture                              |
| 16                                                                                         | plantation              | 0      | 27               | 315                  | 20                    | 0.03                 | <Null>     | #25f69c | <Null> | Plantation                               |
| 17                                                                                         | parks                   | 0      | 27               | 315                  | 20                    | 0.03                 | <Null>     | #7ba16d | <Null> | Parks                                    |
| 18                                                                                         | airport                 | 0      | 27               | 315                  | 5                     | 0.001                | <Null>     | #b2b2b2 | <Null> | Airport                                  |
| 19                                                                                         | sea                     | 0      | 27               | 315                  | 80                    | 1                    | <Null>     | #21618c | <Null> | Sea                                      |
| 20                                                                                         | inlandWater             | 0      | 27               | 315                  | 80                    | 1                    | <Null>     | #3498db | <Null> | Inland water                             |
| 21                                                                                         | concreteBuilding        | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #d7dbdd | <Null> | Concrete building                        |
| 22                                                                                         | glassBuilding           | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #a6acaf | <Null> | Glass building                           |
| 23                                                                                         | woodBuilding            | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #86491E | <Null> | Wood building                            |
| 24                                                                                         | lowLossBuilding         | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #9C9C9C | <Null> | Low loss building                        |
| 25                                                                                         | highLossBuilding        | 0      | 1                | 315                  | 5                     | 0.001                | <Null>     | #a6acaf | <Null> | High loss building                       |
| Click to add new row.                                                                      |                         |        |                  |                      |                       |                      |            |         |        |                                          |

Sentinel-2 clutter classes raster provides information about these classes:

|    |            |
|----|------------|
| 1  | Water      |
| 2  | Trees      |
| 4  | Flooded... |
| 5  | Crops      |
| 7  | Built Area |
| 8  | Bare...    |
| 10 | Snow/Ice   |
| 11 | Rangeland  |



Choose the **Clutter Classes** button to open the **Clutter Classes** dialogue.



Select one of the clutter classes to open their properties. Clutter class list can also be filtered by solid obstacle and/or clutter obstacle categories.

Clutter Classes

Clutter Class Filtering

☒ Solid obstacle

☒ Clutter obstacle

Selected clutter class: Sparse forest

| Clutter name                             | Color |
|------------------------------------------|-------|
| Agriculture                              |       |
| Airport                                  |       |
| Building blocks                          |       |
| Concrete building                        |       |
| Glass building                           |       |
| Grassland                                |       |
| High density urban (High buildings)      |       |
| High density urban (Low buildings)       |       |
| High density urban (Very high buildings) |       |
| High loss building                       |       |
| Inland water                             |       |
| Low density urban (High buildings)       |       |
| Low density urban (Low buildings)        |       |
| Low loss building                        |       |
| Medium dense forest                      |       |
| Medium density urban (High buildings)    |       |
| Medium density urban (Low buildings)     |       |
| Open / Terrain                           |       |
| Parks                                    |       |
| Plantation                               |       |
| Sea                                      |       |
| Sparse forest                            |       |
| Transportation                           |       |
| Very dense forest                        |       |
| Wood building                            |       |

Clutter Class Properties

IDs in geodata raster:

Height, m:

0

Color:

Surface refractivity, N-Units:

315

Relative permittivity:

13

Surface conductivity, S/m:

0.004

New mapping between land use raster and clutter class data should be done in Clutter Classes dialog.

**Apply**

Saves changes made to clutter classes.

**OK**

Saves changes made to clutter classes and closes the dialogue.

**Dismiss**

Cancels clutter class changes and closes the dialogue.

## 7.6 Antenna Viewer

The Antenna Viewer enables the user to preview the default antennae, compare their vertical and horizontal patterns as well as view the values of these patterns in a table.



Click the **Antenna Viewer** button to open the **Antenna Viewer**.

You can filter the entries of the Antenna Table by pressing the **Filter** symbol near one of the field names and entering the desired filter value.

▼ Antenna: 10P-2L8M-B5-V2\_Port 3 +45\_01DT\_1800

| Object ID | Manufi: WiFi | Model            | Frequency, | Gain, dBi |
|-----------|--------------|------------------|------------|-----------|
| 2         | WiFi         | Antenna 2.4GHz 3 | 900        | 8,1       |
| 3         | WiFi         | Antenna 5GHz 36  | 900        | 8,1       |

### 7.6.1 Preview Antenna Patterns

The antenna Type dropdown list contains two antenna types: **Sector** and **Parabolic**. Sector antennas have 2 patterns and pertain to a single network object. Parabolic antennas have 4 patterns and pertain to point-to-point network objects. In Cellular Expert these objects are Links.

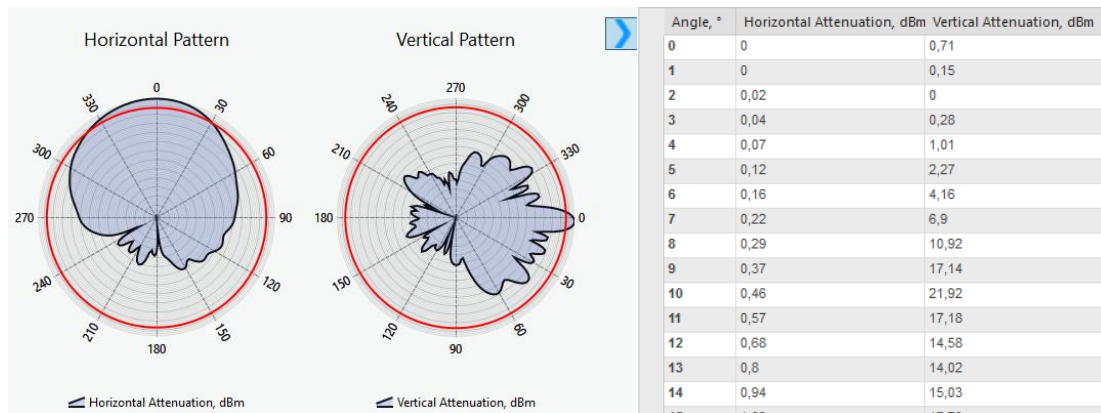
To compare the horizontal and vertical patterns of an antenna in the Antenna Viewer, double-click one of the antennas from the antenna list and the patterns as well as their corresponding values will appear.

Antenna Viewer

Antenna Type: Sector

▼ Antenna: RRV4-65D-R6\_Port 5 +45\_02DT\_2100

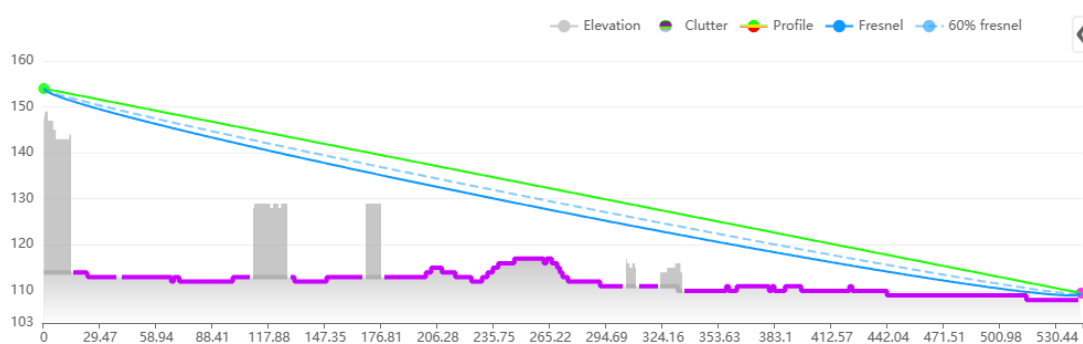
| Object ID | Manufatu       | Model             | Frequency, | Gain, dBi |
|-----------|----------------|-------------------|------------|-----------|
| 1         | COMMSCOPE      | 10P-2L8M-B5-V2    | 1800       | 16,41     |
| 2         | WiFi           | Antenna 2.4GHz    | 900        | 8,1       |
| 3         | WiFi           | Antenna 5GHz 36   | 900        | 8,1       |
| 4         | COMMSCOPE      | DB583-Y_00DT_     | 903        | 5,11      |
| 5         | COMMSCOPE      | HWX-6516DS1-V     | 2300       | 17,97     |
| 6         | COMMSCOPE      | RRV4-65D-R6_Pc    | 700        | 15,4      |
| 7         | COMMSCOPE      | RRV4-65D-R6_Pc    | 1800       | 15,8      |
| 8         | COMMSCOPE      | RRV4-65D-R6_Pc    | 2100       | 14,7      |
| 9         | COMMSCOPE      | RRV4-65D-R6_Pc    | 2620       | 16,1      |
| 10        | COMMSCOPE      | SBNHH-1D65B_P     | 704        | 14,41     |
| 11        | 3GPP TR 36 942 | Typical 33deg H a | 0          | 0         |
| 12        | 3GPP TR 36 942 | Typical 65deg H a | 0          | 0         |
| 13        | 3GPP TR 36 942 | Typical 90deg H a | 0          | 0         |
| 14        | 3GPP TR 36 942 | Typical 90deg H a | 0          | 0         |



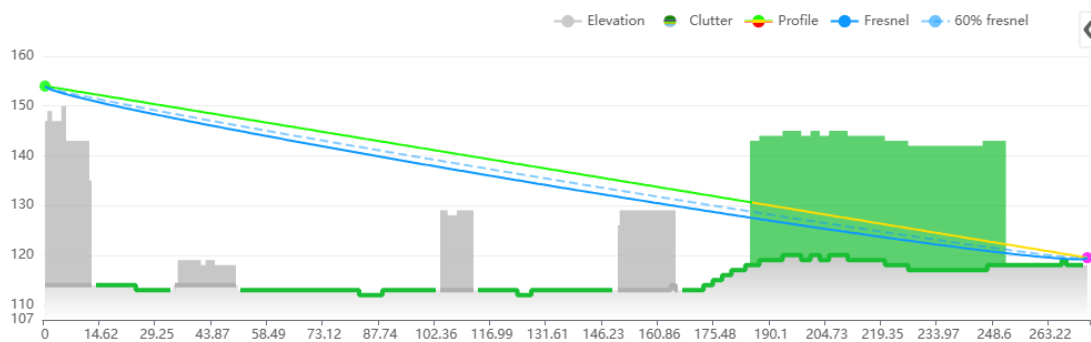
## 7.7 Prediction Model Manager

The CE Path Loss Modelling aims to perform near-deterministic calculation of received signal levels at each specific point (pixel) in the network's target coverage area by applying selective path loss model depending on the radio visibility condition between the transmitter antenna vis-à-vis a receiver antenna located at a given point in coverage area. The radio visibility is evaluated based on the DTM, Obstacles and Clutter path profile information, as described in previous section. This verification of radio visibility will result in the receiver antenna point assigned into one of three possible radio visibility conditions:

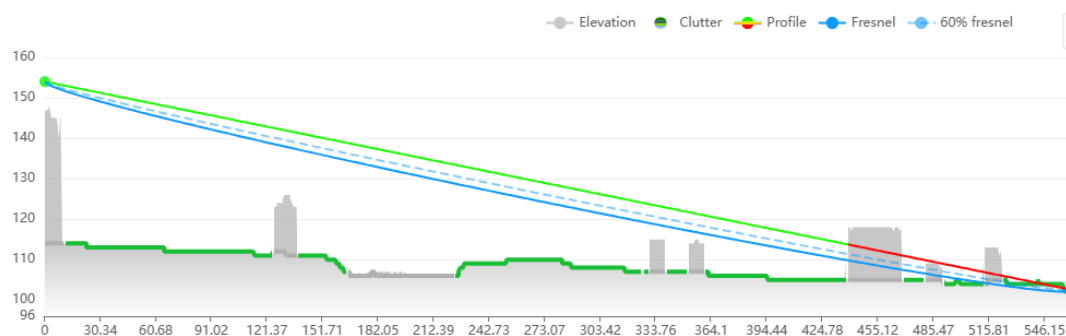
- **Line-of-Sight (LOS)** – occurs when there are neither terrain irregularities, obstacles or clutter interposing the direct radio path between the transmitter and receiver antennas. The radio path is understood to include the 1<sup>st</sup> Fresnel zone around the direct line and account for Spherical Earth effect. The LOS condition is illustrated by the path profile depicted in Fig. 3(a).
- **Obstructed LOS (OLOS)** – occurs when the direct radio propagation line is interposed by clutter, see illustration in Fig. 3(b).
- **Non-LOS (NLOS)** – occurs when the direct radio propagation line is interposed by terrain bulges or obstacles, see illustration in Fig. 3(c).



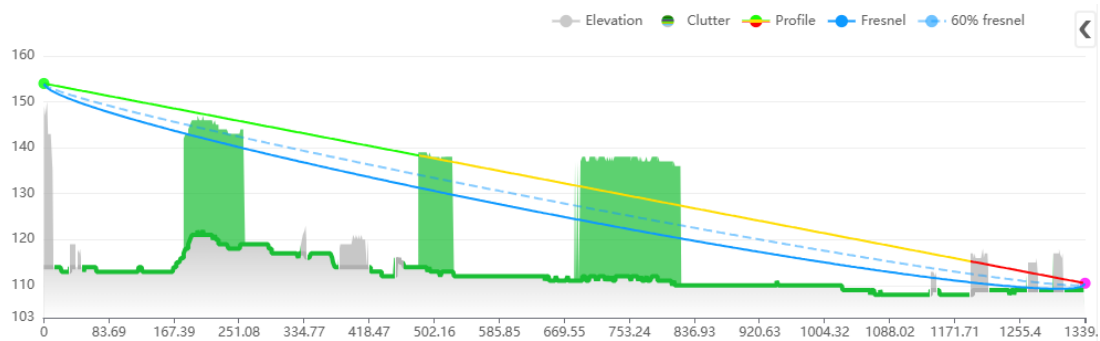
(a) Example of path profile with LOS condition (green line of direct radio link)



(b) Example of path profile with OLOS condition (yellow segment of radio link path)



(c) Example of path profile with NLOS condition (red segment of radio link path)



(d) Example of path profile with OLOS+NLOS condition (yellow+red segment of radio link path)

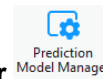
**Fig. 3. Illustration of different LOS conditions**

Depending on the LOS condition for the receive antenna at specific location (area map pixel), the CE tools will apply the specific sub-set of path loss prediction model, as explained in the following section. Note that when the receiver is located indoors, the special Outdoor-to-Indoor propagation function will be applied in addition to basic path loss, as explained in the separate section at the end of this chapter.

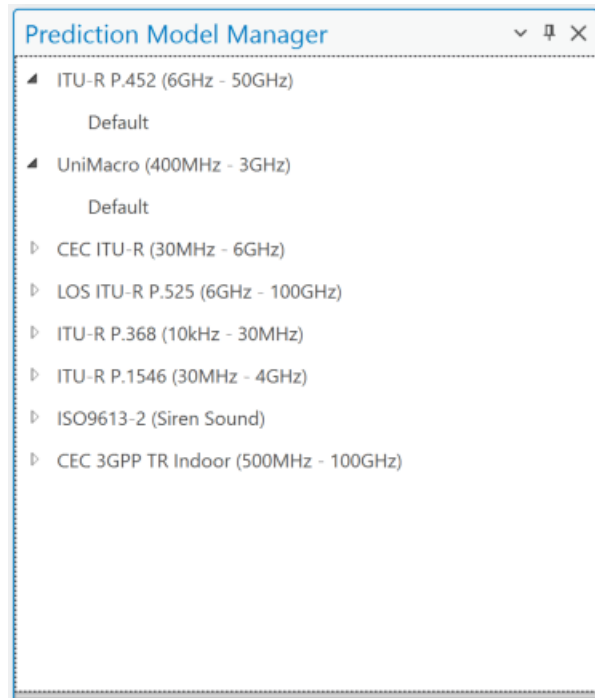
### 7.7.1 Models

Prediction models available in Cellular Expert support frequencies from 10kHz to 350 GHz.





To open the **Prediction Model Manager** dialogue, click on the **Prediction Model Manager** tool in the **Data Management** section.



**CEC ITU-R 3GPP Model (100MHz – 6GHz)** is a combination model intended for use in a variety of different radiocommunication systems which is derived explicitly from ITU-R path loss modelling methods as follows:

- Receive antenna in LOS condition – path loss calculated as FSL based on Recommendation ITU-R P.525 ([ref URL](#)).
- Receive antenna in OLOS condition – total path loss modelled as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option and clutter loss modelling based on Recommendation ITU-R P.2108 ([ref URL](#)).
- Receive antenna in NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)).
- Receive antenna in OLOS+NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) with included dual slope option, additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)), and clutter loss modelling based on Recommendation ITU-R P.2108 ([ref URL](#)).
- Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation (if receiver is in vegetation type clutter) or based on 3GPP TR 38.901 ([ref URL](#)) (if receiver is in a building).

**ITU-R P.452 Model (6GHz – 50GHz)** is provided as a universally applicable model with a very wide frequency range from 0.1-50 GHz. Its implementation is based on the methodology described in the Recommendation ITU-R P.452 ([ref URL](#)). This model does not provide for definition of OLOS visibility condition; instead, it considers clutter as part of the general obstacles category and accordingly distinguishes only two radio visibility cases:



- a. Receive antenna in LOS condition – path loss model based on FSL principle.
- b. Receive antenna in NLOS condition – total path loss modelled using a combination of basic transmission losses and losses due to diffraction.

**ITU-R P.1546 Model (30MHz – 4GHz)** ([ref URL](#)) is a widely recognized radio propagation prediction method developed by the **International Telecommunication Union (ITU)**. It is primarily used for estimating **point-to-area radio signal coverage** in the frequency range from **30 MHz to 4000 MHz** over **terrestrial paths**. This model is especially suitable for **broadcasting, land mobile, and fixed services**.

#### Key Features

- **Versatile Application:** Supports predictions over **land, sea, and mixed paths**, making it adaptable to various geographic conditions.
- **Input Parameters:** Takes into account factors such as **transmitter and receiver heights, terrain profile, clutter (buildings, vegetation), climate, and time/location variability**.
- **Time and Location Variability:** Predictions can be tailored for different statistical reliability levels (e.g., 50% or 10% time availability).
- **Clutter and Terrain Handling:** The model can incorporate detailed **digital elevation models (DEM)** and **clutter data** for more accurate predictions, reflecting the influence of buildings, forests, and other surface features.

#### Use in Cellular Expert

In the **Cellular Expert** software, the ITU-R P.1546 model is implemented to support **real-world coverage planning** and **regulatory studies**. Users can configure environmental parameters and resolution settings to match local conditions and improve prediction accuracy.

**LOS ITU-R P.525 Model (6GHz – 100GHz)** is the FSL path loss calculated based on the method in Recommendation ITU-R P.525 ([ref URL](#)). As such it could be used for modelling radio links where LOS is considered a necessary condition, e.g., for Fixed (Point-to-Point) Links or Mobile Systems in mmWave bands.

**UniMacro Model (400MHz – 3GHz)** is the CE's proprietary combination model developed over the years of practical experience with the operational planning of cellular mobile networks in the frequency ranges from 400-3000 MHz. It had been fine-tuned to produce coverage predictions that are most closely aligned with what could be expected to be experienced by the actual mobile network users in the field. The model will model different path losses depending on radio visibility conditions as follows:

- a. Receive antenna in LOS condition – path loss model based on FSL principle and dual slope based on breakpoint distance.
- b. Receive antenna in OLOS, or NLOS condition – path loss modelled using Extended Hata (Open Area) model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)).
- c. Receive antenna in the clutter (building, vegetation, etc) – path loss is calculated as described above based on LOS, OLOS and NLOS conditions, and additional penetration loss is added to simulate Outdoor-to-Indoor scenario which is based on ITU-R P.833 recommendation (if receiver is in vegetation type clutter) or based on 3GPP TR 38.901 ([ref URL](#)) (if receiver is in a building).

**ITU-R P.368 (10kHz – 30MHz)** provides a standardized prediction method for assessing the ground-wave field strength of radio waves in the 10 kHz to 30 MHz frequency range. This frequency band is primarily associated with long-range communication systems using amplitude modulation (AM) and shortwave bands, often for maritime, aeronautical, military, and broadcasting services.

This model offers guidance for engineers, planners, and researchers working on system design and analysis in the MF (Medium Frequency) and HF (High Frequency) bands.

The ITU-R P.368 model calculates signal strength based on several key environmental and system parameters:

**Frequency (f)**

Higher frequencies tend to attenuate more rapidly over ground. The attenuation rate increases significantly above 3 MHz.

**Distance (d)**

Field strength diminishes with increasing distance due to geometrical spreading and absorption by the ground and atmosphere.

**Surface refractivity, Surface Conductivity ( $\sigma$ ) and Relative Permittivity ( $\epsilon_r$ )**

The surface over which the wave propagates critically affects signal strength:

- Sea water: High conductivity, minimal loss
- Dry land or desert: Low conductivity, high loss
- Typical values range from:
  - Conductivity:  $10^{-4}$  to 5 S/m
  - Relative permittivity: 4 to 81

**ISO 9613** standard provides a validated, practical method for predicting the outdoor propagation of sound, and is increasingly applied in siren sound modeling within public security, emergency warning systems, and defense operations. This modeling ensures that acoustic alert systems (e.g. civil defense sirens, disaster warnings, military alert signals) achieve their intended coverage, intelligibility, and effectiveness across various terrain and urban environments.

**Purpose in the Public Security Context**

In emergency and defense scenarios, reliable audibility of sirens is critical for:

- Civil alert and evacuation systems
- Military base perimeter alarms
- Air raid or missile defense warning networks
- Disaster alert systems (e.g. earthquakes, tsunamis, nuclear incidents)

By applying ISO 9613-2, engineers can model how far a siren can be heard under specific environmental conditions, optimizing:

- Placement and spacing of sirens
- Sound power selection
- Minimization of acoustic shadow zones
- Compliance with national safety and civil defense regulations

The model assumes standard favorable propagation:

- Downwind or moderate inversion conditions
- Ambient temperature  $\sim 10^\circ\text{C}$
- Relative humidity  $\sim 70\%$

These are conservative conditions ensuring that siren reach is never overestimated, supporting public safety margin planning.

**CEC 3GPP TR Indoor (500MHz – 100GHz)** Propagation Model is a high-frequency path loss model designed for indoor radiocommunication systems operating within the 500 MHz to 100 GHz range. It builds upon the CEC ITU-R 3GPP model (100 MHz – 6 GHz) by adapting it to complex indoor environments, such as:

- Office buildings
- Residential units
- Shopping centers
- Industrial halls

This model integrates core **ITU-R recommendations** for free-space loss and penetration effects, while leveraging 3GPP-specific methods for accurate simulation of indoor multipath, wall attenuation, and frequency-dependent fading.

### **Purpose and Use Cases**

The model is intended for:

- Indoor wireless access network design (e.g., Wi-Fi, 5G NR, mmWave)
- System-level simulations for indoor coverage planning
- Performance evaluation of in-building penetration for outdoor base stations
- Integration with dual-slope and multi-scenario path loss modeling frameworks

#### **7.7.1.1 CEC ITU-R 3GPP Model**

##### **Model application**

This deterministic model is designed for precise tracking of the main, strongest radio ray, while also empirically modeling the scattering of other rays around the receiver. It applies to all ranges of cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within the 30 MHz to 6 GHz frequency range.

The model is recommended for accurate wide-area propagation and coverage modeling, especially when precise and up-to-date topographic data are available. This includes Digital Terrain Models (DTM), building data (including height information), and vegetation data (with height details) derived from a Digital Surface Model (DSM). Ideally, these data should be created with LiDAR or similar methods, at a resolution of at least 10 meters, though 5, 2, or even 1 meter or higher is preferable for optimal accuracy.

When building data and their heights are not available, and only DTM and clutter data at a resolution of 10 meters or lower are accessible, the UniMacro Model should be considered for wide-area propagation and coverage modeling in a slightly narrower frequency range of 400 MHz to 3 GHz.

##### **Default settings**

#### ***General settings to calculate Model loss***

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Distance coefficient obstructed – represents the slope based on the obstructed distance between

the cell and the receiver location. The default value is 40.

- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

|                                  |    |
|----------------------------------|----|
| Offset coefficient, dB:          | 32 |
| Distance coefficient:            | 20 |
| Distance coefficient obstructed: | 40 |
| Frequency coefficient:           | 20 |

***Clutter class to calculate diffraction, clutter loss, penetration loss and receiver loss***

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients.

☒ Solid obstacle ☒ Clutter obstacle

▼ Selected clutter class:

| Clutter name                             | Color |
|------------------------------------------|-------|
| Agriculture                              |       |
| Airport                                  |       |
| Building blocks                          |       |
| Concrete building                        |       |
| Glass building                           |       |
| Grassland                                |       |
| High density urban (High buildings)      |       |
| High density urban (Low buildings)       |       |
| High density urban (Very high buildings) |       |
| High loss building                       |       |
| Inland water                             |       |
| Low density urban (High buildings)       |       |
| Low density urban (Low buildings)        |       |
| Low loss building                        |       |
| Medium dense forest                      |       |
| Medium density urban (High buildings)    |       |
| Medium density urban (Low buildings)     |       |
| Open / Terrain                           |       |
| Parks                                    |       |
| Plantation                               |       |
| Sea                                      |       |
| Sparse forest                            |       |
| Transportation                           |       |
| Very dense forest                        |       |
| Wood building                            |       |

These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Nominal distance, m:

1

Diffraction loss coefficient:

1

Enclosed receiver loss offset, dB:

12.7

Enclosed receiver loss scaling coefficient:

0.5

Enclosed receiver loss frequency exponent coefficient:

0.58

Receiver point loss offset, dB:

0

#### Key Parameters:

- Nominal distance, m – the average distance between objects within the clutter class, ranging from 1 to 100 meters.

- Diffraction loss coefficient – a multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter types. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher than forest or other clutter classes.
- Enclosed receiver loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.
- Enclosed receiver loss scaling coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Enclosed receiver loss frequency exponent coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.
- Receiver point loss offset, dB – an additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

#### Clutter Classes default values

|                                          | Penetration receiver loss offset | Penetration receiver loss scaling coefficient | Penetration receiver loss frequency exponent coefficient |
|------------------------------------------|----------------------------------|-----------------------------------------------|----------------------------------------------------------|
| Open / Terrain                           | 17                               | 0.25                                          | 1                                                        |
| Grassland                                | 0                                | 0.82                                          | 0.65                                                     |
| Sparse forest                            | 0                                | 0.82                                          | 0.65                                                     |
| Medium dense forest                      | 0                                | 0.89                                          | 0.65                                                     |
| Very dense forest                        | 0                                | 0.95                                          | 0.65                                                     |
| Low density urban (Low buildings)        | 0                                | 0.89                                          | 0.65                                                     |
| Low density urban (High buildings)       | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (Low buildings)     | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (High buildings)    | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (Low buildings)       | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (High buildings)      | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (Very high buildings) | 0                                | 0.89                                          | 0.65                                                     |
| Building blocks                          | 0                                | 0.89                                          | 0.65                                                     |
| Transportation                           | 0                                | 0.89                                          | 0.65                                                     |
| Agriculture                              | 0                                | 0.89                                          | 0.65                                                     |
| Plantation                               | 0                                | 0.89                                          | 0.65                                                     |
| Parks                                    | 0                                | 0.89                                          | 0.65                                                     |
| Airport                                  | 0                                | 0.89                                          | 0.65                                                     |
| Sea                                      | 0                                | 0.89                                          | 0.65                                                     |
| Inland water                             | 0                                | 0.89                                          | 0.65                                                     |
| Concrete building                        | 5                                | 0.25                                          | 1                                                        |

|                    |     |      |   |
|--------------------|-----|------|---|
| Glass building     | 2   | 0.25 | 1 |
| Wood building      | 2   | 0.25 | 1 |
| Low loss building  | 8.5 | 0.25 | 1 |
| High loss building | 17  | 0.25 | 1 |

### 7.7.1.2 ITU-R P.452 Model

#### Model application

This model is designed to estimate radio signal propagation over long distances, including terrestrial paths. It is specifically designed to predict signal attenuation caused by various mechanisms, such as diffraction, tropospheric scatter, ducting, and reflections from the Earth's surface, across frequencies ranging from 0.1 GHz to 50 GHz. While other prediction model covers frequencies from 100MHz to 6GHz, we recommend to use it from 6GHz to 50GHz frequencies.

This model is particularly well-suited for microwave links and is widely used for planning and interference analysis in fixed and mobile radio communication systems. By accounting for the effects of terrain, atmospheric conditions, and other factors, it helps engineers assess link reliability and optimize network performance in a variety of environmental scenarios.

#### Default settings

##### *General settings to calculate Model loss*

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

|                         |    |
|-------------------------|----|
| Offset coefficient, dB: | 32 |
| Distance coefficient:   | 20 |
| Frequency coefficient:  | 20 |

#### *Multipath and focusing*

In ITU-R P.452, the correction for multipath and focusing effects accounts for signal enhancements caused by constructive interference and atmospheric focusing. This adjustment reduces the total path loss under

favorable conditions, such as over-water paths or specific atmospheric gradients, ensuring more accurate signal predictions.

Possible values Yes or No.

Use Multipath Focusing:

No

Yes

No

### ***Clutter class to calculate penetration loss***

Clutter class option describes several fixed Clutter names with penetration loss coefficients.

These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Penetration loss offset, dB:

20

Penetration loss distance coefficient:

1.1

Penetration loss frequency coefficient:

0.58

Key Parameters:

- Penetration loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.
- Penetration loss distance coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Penetration loss frequency coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.

### **Clutter Classes default values**

|                                    | Penetration receiver loss offset | Penetration receiver loss scaling coefficient | Penetration receiver loss frequency exponent coefficient |
|------------------------------------|----------------------------------|-----------------------------------------------|----------------------------------------------------------|
| Open / Terrain                     | 17                               | 0.25                                          | 1                                                        |
| Grassland                          | 0                                | 0.82                                          | 0.65                                                     |
| Sparse forest                      | 0                                | 0.82                                          | 0.65                                                     |
| Medium dense forest                | 0                                | 0.89                                          | 0.65                                                     |
| Very dense forest                  | 0                                | 0.95                                          | 0.65                                                     |
| Low density urban (Low buildings)  | 0                                | 0.89                                          | 0.65                                                     |
| Low density urban (High buildings) | 0                                | 0.89                                          | 0.65                                                     |



|                                          |     |      |      |
|------------------------------------------|-----|------|------|
| Medium density urban (Low buildings)     | 0   | 0.89 | 0.65 |
| Medium density urban (High buildings)    | 0   | 0.89 | 0.65 |
| High density urban (Low buildings)       | 0   | 0.89 | 0.65 |
| High density urban (High buildings)      | 0   | 0.89 | 0.65 |
| High density urban (Very high buildings) | 0   | 0.89 | 0.65 |
| Building blocks                          | 0   | 0.89 | 0.65 |
| Transportation                           | 0   | 0.89 | 0.65 |
| Agriculture                              | 0   | 0.89 | 0.65 |
| Plantation                               | 0   | 0.89 | 0.65 |
| Parks                                    | 0   | 0.89 | 0.65 |
| Airport                                  | 0   | 0.89 | 0.65 |
| Sea                                      | 0   | 0.89 | 0.65 |
| Inland water                             | 0   | 0.89 | 0.65 |
| Concrete building                        | 5   | 0.25 | 1    |
| Glass building                           | 2   | 0.25 | 1    |
| Wood building                            | 2   | 0.25 | 1    |
| Low loss building                        | 8.5 | 0.25 | 1    |
| High loss building                       | 17  | 0.25 | 1    |

### 7.7.1.3 UniMacro Model

#### Model application

This model is designed for deterministic tracking of the main, strongest radio ray in Line of Sight (LOS) areas, while propagation modeling in Obstructed Line of Sight (OLOS) and Non-Line of Sight (NLOS) areas uses empirically determined parameters defined in ITU-R and 3GPP recommendations. It also models the scattering of other rays around the receiver. The model applies empirically validated values for the 400 MHz to 3 GHz frequency range and is suitable for modeling all cellular mobile and public safety networks, including 2G, 3G, 4G, and 5G, within that frequency range.

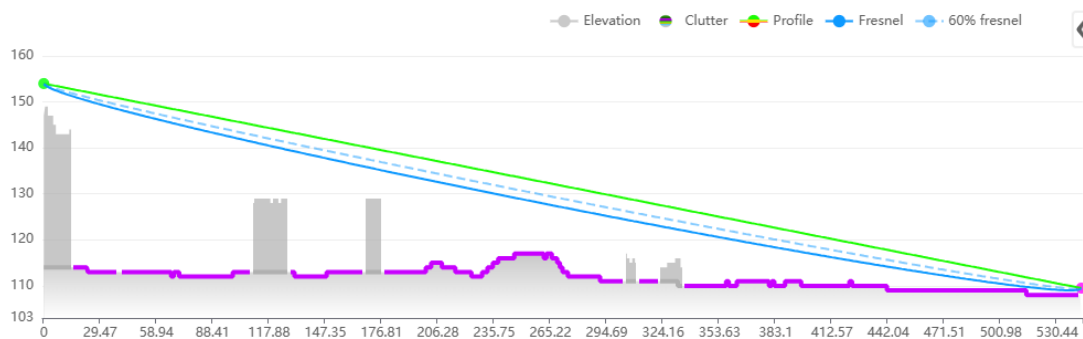
This model is recommended for wide-area propagation and coverage modeling when building data and their heights are unavailable, and only DTM and clutter data at a resolution of 10 meters or lower are accessible.

For accurate wide-area propagation and coverage modeling where building data and their heights are available, the CEC ITU-R 3GPP Model is recommended, offering a broader application frequency range of 100 MHz to 6 GHz.

#### Default settings

##### *General settings to calculate Model loss*

If Tx and Rx is in LOS condition:



**Fig. 4. Illustration of LOS conditions**

Line of Sight coefficients are used to calculate general model loss.

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient near – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Distance coefficient far – represents the slope based on breakpoint distance between the cell and the receiver location. The default value is 40.
- Use custom break distance – if value Yes, enables custom Fresnel breakpoint distance value. The path loss dependence on the distance is split into near and far zones by the breakpoint distance and the effect is only applied for Line of Sight condition.
- Custom break distance, km – Fresnel breakpoint distance then path loss is calculated using Distance coefficient far parameter.
- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

|                            |     |
|----------------------------|-----|
| Offset coefficient, dB:    | 32  |
| Distance coefficient near: | 20  |
| Distance coefficient far:  | 40  |
| Use custom break distance: | Yes |
| Custom break distance, km: | 2   |
| Frequency coefficient:     | 20  |

If Tx and Rx is in OLOS or NLOS condition, then Hata 9999 equation is used. 9999 Model is the Ericsson's implementation of Hata Model. Ericsson provides the steering parameters of 9999 Model for different environments; therefore it's very convenient just to apply in this form as the default parameters.

- Hata Loss: A0 - constant offset in dB this value simply added to loss grid. Adjusting this value, you can minimize mean error. It regulates the absolute level of the loss curve. Default value 36.2.
- Hata Loss: A1 - distance influence coefficient. Physically it represents loss dependant on distance such as atmospheric (dust, hydrometeors, etc...) losses. It regulates slope of the curve. Default value 30.2.
- Hata Loss: A2 - transmitter height influence coefficient. It is related to errors in DTM, real Earth curvature, etc. It regulates loss curve vertical position like the A0, but with respect to antenna height. Default value -12.
- Hata Loss: A3 - Okumura-Hata type of multiplying factor for  $\log(h_M)\log(d)$ . Default value 0.1.

|               |      |
|---------------|------|
| Hata Loss: A0 | 36.2 |
| Hata Loss: A1 | 30.2 |
| Hata Loss: A2 | -12  |
| Hata Loss: A3 | 0.1  |

***Clutter class to calculate diffraction, clutter loss, penetration loss and receiver loss***

The Clutter Class option defines several predefined clutter categories, each with unique values for diffraction loss, clutter loss, penetration loss, and receiver loss coefficients.

☒ Solid obstacle ☒ Clutter obstacle

▼ Selected clutter class:

| Clutter name                             | Color |
|------------------------------------------|-------|
| Agriculture                              |       |
| Airport                                  |       |
| Building blocks                          |       |
| Concrete building                        |       |
| Glass building                           |       |
| Grassland                                |       |
| High density urban (High buildings)      |       |
| High density urban (Low buildings)       |       |
| High density urban (Very high buildings) |       |
| High loss building                       |       |
| Inland water                             |       |
| Low density urban (High buildings)       |       |
| Low density urban (Low buildings)        |       |
| Low loss building                        |       |
| Medium dense forest                      |       |
| Medium density urban (High buildings)    |       |
| Medium density urban (Low buildings)     |       |
| Open / Terrain                           |       |
| Parks                                    |       |
| Plantation                               |       |
| Sea                                      |       |
| Sparse forest                            |       |
| Transportation                           |       |
| Very dense forest                        |       |
| Wood building                            |       |

These parameters describe how a signal is impacted when it passes through or terminates in a specific clutter class.

Nominal distance, m:

1

Diffraction loss coefficient:

1

Penetration loss offset, dB:

12.7

Penetration loss distance coefficient:

0.5

Penetration loss frequency coefficient:

0.58

Receiver point loss offset, dB:

0

Key Parameters:

- Nominal distance, m – the average distance between objects within the clutter class, ranging from 1 to 100 meters.
- Diffraction loss coefficient – a multiplier used in diffraction calculations. Lower values result in reduced diffraction loss, while higher values increase it. Typically, this coefficient is higher for buildings compared to forests or other clutter types. If value is lower, diffraction will be lower, if higher – then diffraction will be higher. Usually, for buildings clutter class this parameter is higher than forest or other clutter classes.
- Penetration loss offset, dB – the initial entry loss into the clutter class, expressed as an offset in dB, which is added to the path loss grid.
- Penetration loss distance coefficient – represents additional signal loss as a function of the distance traveled within the clutter class. Higher values increase path loss.
- Penetration loss frequency coefficient – reflects additional loss inside the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.
- Receiver point loss offset, dB – an additional loss offset in dB applied to the path loss grid, representing user equipment (UE) losses.

#### Clutter Classes default values

|                                          | Penetration receiver loss offset | Penetration receiver loss scaling coefficient | Penetration receiver loss frequency exponent coefficient |
|------------------------------------------|----------------------------------|-----------------------------------------------|----------------------------------------------------------|
| Open / Terrain                           | 17                               | 0.25                                          | 1                                                        |
| Grassland                                | 0                                | 0.82                                          | 0.65                                                     |
| Sparse forest                            | 0                                | 0.82                                          | 0.65                                                     |
| Medium dense forest                      | 0                                | 0.89                                          | 0.65                                                     |
| Very dense forest                        | 0                                | 0.95                                          | 0.65                                                     |
| Low density urban (Low buildings)        | 0                                | 0.89                                          | 0.65                                                     |
| Low density urban (High buildings)       | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (Low buildings)     | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (High buildings)    | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (Low buildings)       | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (High buildings)      | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (Very high buildings) | 0                                | 0.89                                          | 0.65                                                     |
| Building blocks                          | 0                                | 0.89                                          | 0.65                                                     |
| Transportation                           | 0                                | 0.89                                          | 0.65                                                     |
| Agriculture                              | 0                                | 0.89                                          | 0.65                                                     |
| Plantation                               | 0                                | 0.89                                          | 0.65                                                     |
| Parks                                    | 0                                | 0.89                                          | 0.65                                                     |
| Airport                                  | 0                                | 0.89                                          | 0.65                                                     |
| Sea                                      | 0                                | 0.89                                          | 0.65                                                     |

|                    |     |      |      |
|--------------------|-----|------|------|
| Inland water       | 0   | 0.89 | 0.65 |
| Concrete building  | 5   | 0.25 | 1    |
| Glass building     | 2   | 0.25 | 1    |
| Wood building      | 2   | 0.25 | 1    |
| Low loss building  | 8.5 | 0.25 | 1    |
| High loss building | 17  | 0.25 | 1    |

#### 7.7.1.4 LOS ITU-R Model

##### Model application

Line of Sight model is typically used for mmWave band frequencies within the 6 GHz – 100 GHz frequency range and provides results only for line-of-sight areas.

##### Default settings

##### *General settings to calculate Model loss*

- Offset coefficient (dB) – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- Distance coefficient – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- Frequency coefficient – indicates the slope determined by the frequency value, with a default value of 20.

|                         |    |
|-------------------------|----|
| Offset coefficient, dB: | 32 |
| Distance coefficient:   | 20 |
| Frequency coefficient:  | 20 |

#### 7.7.1.5 ITU-R P.368 Model

##### Model application

ITU-R P.368 is ground-wave propagation of radio signals model. It is specifically designed to estimate the field strength and attenuation of radio waves over the Earth's surface, particularly for frequencies below 30 MHz.

This model is widely used in planning and designing long-distance communication systems, such as

maritime, broadcasting, and low-frequency navigation systems, where ground-wave propagation plays a critical role. It accounts for factors such as terrain conductivity, dielectric properties, and surface roughness to deliver accurate predictions of signal behavior.

### Default settings

The general model parameters include the radius and receiver height. Additional parameters used for path loss calculations are derived from the clutter classes. Each clutter class has its own unique set of parameters:

- **Surface refractivity** - a measure of the refractive index's influence on electromagnetic wave propagation, particularly in the lower atmosphere close to the Earth's surface. It is expressed as a dimensionless value, typically dependent on atmospheric pressure, temperature, and humidity. High surface refractivity can significantly affect radio wave bending and propagation, such as ducting or anomalous refraction.
- **Relative permittivity** - quantifies a material's ability to permit electric field propagation relative to vacuum. It is a complex quantity with the real part representing energy storage capability and the imaginary part representing energy dissipation within the material. ITU-R P.368 uses relative permittivity to model how radio waves interact with various surface materials, such as soil, water, or vegetation. These interactions influence reflection, refraction, and absorption phenomena at the surface.
- **Surface conductivity** - refers to a material's ability to conduct electrical currents across its surface. It is measured in siemens per meter (S/m). Higher conductivity indicates that a surface can easily allow current flow, affecting the reflection and absorption of radio waves. According to ITU-R P.368, surface conductivity is a critical factor in determining the reflective properties of surfaces, such as dry versus wet soil, or metallic versus dielectric surfaces.

|                                      | <b>Buildings</b> | <b>Forest</b> | <b>Dense forest</b> | <b>Urban</b> | <b>Dense urban</b> | <b>Bare ground</b> | <b>Crops</b> | <b>Water</b> | <b>Road</b> |
|--------------------------------------|------------------|---------------|---------------------|--------------|--------------------|--------------------|--------------|--------------|-------------|
| <b>Surface refractivity, N-Units</b> | 315              | 315           | 315                 | 315          | 315                | 315                | 315          | 365          | 315         |
| <b>Relative permittivity</b>         | 5                | 13            | 13                  | 5            | 5                  | 10                 | 20           | 80           | 10          |
| <b>Surface conductivity, S/m</b>     | 0.001            | 0.004         | 0.004               | 0.001        | 0.001              | 0.002              | 0.03         | 1            | 0.002       |

#### 7.7.1.6 ITU-R P.1546 Model

### Model application

This model is designed for wide-area radio propagation prediction based on empirical data and statistical analysis of measured field strengths. It provides path loss estimations over land, sea, and mixed terrain for frequencies ranging from 30 MHz to 3 GHz, using parameters defined in ITU-R Recommendation P.1546. The model is applicable to terrestrial broadcasting, mobile, and public safety networks, including technologies such as 2G, 3G, and 4G, within its frequency range.

The ITU-R P.1546 model accounts for antenna heights, terrain elevation (DTM), land cover types (clutter), and environmental conditions, incorporating corrections for time variability and location-specific effects. It is particularly suitable for modeling line-of-sight (LOS) and non-line-of-sight (NLOS) propagation over long distances where detailed building data is not available.

This model is recommended for national or regional coverage planning, especially in cases where high-resolution DTM (e.g., 30m) and clutter data (e.g., 10m resolution) are available, but building heights and detailed 3D structures are not. Its statistical approach allows for reliable estimations of field strength in both urban and rural environments, without requiring ray-tracing or detailed geometry-based modeling.

For scenarios where accurate building geometry and heights are available and a higher modeling frequency range (up to 6 GHz) is required, the CEC ITU-R 3GPP Model is recommended instead, offering greater precision in dense urban and high-frequency applications.

### Default settings

The following key parameters are used to configure the ITU-R P.1546 radio propagation predictions within the software. These settings directly influence the coverage calculation and modeling accuracy:

#### 1. Radius

- Description: This setting defines the maximum distance from the transmitter (or prediction center) over which the propagation prediction will be performed.
- Purpose: It limits the spatial extent of the coverage area to optimize calculation time and resource usage.
- Typical Values: Often set between 10 km to 100 km, depending on the transmitter's power, terrain, and target coverage region.

#### 2. Receiver Height (m)

- Description: Specifies the height of the receiving antenna above ground level, in meters.
- Purpose: Receiver height impacts the predicted signal strength, especially in terrain with elevation changes or obstacles.
- Guidelines:
  - 1.5 to 2 m for handheld/mobile users (e.g., mobile phones, public safety devices).
  - 10 m or higher for fixed installations (e.g., rooftop or vehicular antennas).
- Note: Accurate setting of receiver height is essential for meaningful signal level predictions.

#### 3. Time Percentage (%)

- Description: Indicates the percentage of time during which the predicted field strength is expected to be met or exceeded.
- Purpose: Reflects the statistical variability of signal propagation due to atmospheric and environmental effects.
- Common Use Cases:



- 50% time: Typical for general service coverage maps (median conditions).
- 10% time: Used for high-reliability or interference studies, ensuring signal presence under less favorable conditions.
- Interpretation: A 10% time prediction means the signal level is met or exceeded during 10% of the time, capturing worst-case propagation conditions (i.e., stronger signal occurrence during favorable propagation).

|                     |         |
|---------------------|---------|
| Object ID:          | 1       |
| Configuration name: | Default |
| Radius, km:         | 3       |
| Receiver height, m: | 1.5     |
| Time percentage, %: | 10      |

#### 7.7.1.7 ISO9613 Model

##### Model application

ISO 9613 is an international standard used for predicting outdoor sound propagation, including the assessment of siren noise levels. It provides a structured method for calculating sound attenuation over distance, considering factors such as geometric spreading, atmospheric absorption, ground effects, reflections, and obstacles.

In the context of siren sound prediction, ISO 9613 helps determine the effective coverage area, ensuring that warning signals reach the intended audience with sufficient audibility. This standard is essential for optimizing siren placement, regulatory compliance, and designing effective emergency alert systems.

The primary factors included in the standard are:

1. Geometric Spreading
  - This refers to how sound spreads out as it moves away from its source. Sound intensity decreases as the distance from the source increases, following the inverse square law (with spherical spreading) or other forms depending on terrain.
  - As the distance from the sound source increases, the intensity of sound diminishes, which is considered in the calculation of sound levels at various receiver points.

$$A_{div} = 20 * LOG10(d) + 11$$

2. Atmospheric Absorption

- Sound is absorbed by the atmosphere as it travels, particularly at higher frequencies. The

absorption depends on factors like temperature, humidity, and air pressure, and is often more significant over longer distances.

- Atmospheric absorption reduces the intensity of sound as it propagates, especially at higher frequencies. The standard takes into account these effects to calculate the reduction in sound level.

### 3. Obstacles

- Physical barriers such as walls, hills, and buildings block or scatter sound waves, reducing the sound level that reaches certain areas.
- Obstacles can cause significant sound reduction, especially if they are large or located between the sound source and the receiver. The standard accounts for the shadow zones created by these barriers.

### 4. Directivity of the Source

- This parameter accounts for the directionality of the sound source, which may not emit sound equally in all directions. Sirens, for example, may have directional characteristics that focus their sound output in certain directions.
- The directivity of the sound source influences how the sound energy is distributed, and therefore, how far and in what pattern the sound propagates.

### 5. Meteorological Conditions

- Weather conditions such as wind speed, temperature gradients, and humidity can significantly affect sound propagation. For example, sound may travel farther downwind or be absorbed more by humid air.
- These conditions are integrated into the model to adjust the calculations of sound attenuation, ensuring more accurate predictions for different weather scenarios.

## Default settings

- **Distance coefficient: 20.**

Describes slope coefficient based on distance. Value is included in Geometric spreading calculations.

- **Temperature: 20 C°**
- **Humidity: 50%**
- **Meteorological conditions: 3 dB.** It is a factor, in decibels, which depends on local meteorological statistics for wind speed and direction, and temperature gradients. Experience indicates that values of Meteorological conditions (C0) in practice are limited to the range from zero to approximately + 5 dB.

**Ground factor** parameter for the ISO9613-2 model's clutter classes allows users to define the acoustic reflectivity (from 0 to 1, indicating hard-soft respectively) of the ground surface for more accurate sound level predictions.

**Clutter Classes default values**

|                                          | <b>Ground factor</b> |
|------------------------------------------|----------------------|
| Open / Terrain                           | 1                    |
| Grassland                                | 1                    |
| Sparse forest                            | 1                    |
| Medium dense forest                      | 1                    |
| Very dense forest                        | 1                    |
| Low density urban (Low buildings)        | 0.5                  |
| Low density urban (High buildings)       | 0.3                  |
| Medium density urban (Low buildings)     | 0.3                  |
| Medium density urban (High buildings)    | 0.1                  |
| High density urban (Low buildings)       | 0.1                  |
| High density urban (High buildings)      | 0                    |
| High density urban (Very high buildings) | 0                    |
| Building blocks                          | 0                    |
| Transportation                           | 0                    |
| Agriculture                              | 1                    |
| Plantation                               | 1                    |
| Parks                                    | 0.7                  |
| Airport                                  | 0                    |
| Sea                                      | 0                    |
| Inland water                             | 0                    |
| Concrete building                        | 0                    |
| Glass building                           | 0                    |
| Wood building                            | 0                    |
| Low loss building                        | 0                    |
| High loss building                       | 0                    |

**7.7.1.8 CEC 3GPP TR Indoor (500MHz – 100GHz)****Model application**

The **CEC 3GPP TR Indoor Model** (500 MHz – 100 GHz) is a robust, scalable prediction method based on ITU-R principles and 3GPP extensions, specifically engineered for indoor radiowave propagation. By

supporting a wide range of scenarios — from clean LOS corridors to deeply obstructed NLOS paths — it enables highly accurate modeling of next-generation wireless systems, ensuring reliable, secure, and efficient communication in the most demanding indoor environments.

### Key Enhancements for Indoor Use

- **Frequency scaling up to 100 GHz** supports mmWave and terahertz
- **Wall material database** from 3GPP TR 38.901:
  - Standard drywall: ~2–8 dB per wall
  - Concrete: 5–15 dB
  - Glass: 2–10 dB
- **Multi-floor attenuation** (floor penetration factor)
- **Path loss floors**: ensures minimum attenuation beyond near field
- **LOS probability models**: stochastic treatment of LOS in large buildings

### Default settings

|                                  |         |
|----------------------------------|---------|
| Object ID:                       | 1       |
| Configuration name:              | Default |
| Radius, km:                      | 0.1     |
| Effective Earth radius, km:      | 8500    |
| Receiver height, m:              | 1.5     |
| Offset coefficient, dB:          | 37      |
| Distance coefficient:            | 20      |
| Distance coefficient obstructed: | 30      |
| Frequency coefficient:           | 20      |

#### Configuration name

Name of the prediction configuration name.

#### Radius

Maximum prediction radius in kilometers to calculate path loss.

#### Receiver height

Receiver height above the receiver reference height selected in the workspace settings.

#### Effective earth radius

Earth radius in kilometers, used for the calculations.

**Offset coefficient**

Represents the offset in decibels added to the path loss grid. The default value is 37 dB.

**Distance coefficient**

Defines the slope based on the distance between the cell and the receiver location, with a default value of 20.

**Distance coefficient obstructed**

Represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 30.

**Frequency coefficient**

Indicates the slope determined by the frequency value, with a default value of 20.

**Penetration loss offset**

The initial entry loss applied when crossing an obstacle within the clutter class, expressed as an offset in dB, which is added to the path loss grid.

**Penetration loss scaling coefficient**

Represents additional signal loss as a function of the distance travelled in an obstacle within the clutter class. Higher values increase path loss.

**Penetration loss frequency exponent coefficient**

Reflects additional loss inside an obstacle within the clutter class based on frequency. Higher values increase path loss, particularly at higher frequencies.

Note: penetration losses stack when multiple obstacles are crossed.

**Clutter Classes default values**

|                                       | Penetration receiver loss offset | Penetration receiver loss scaling coefficient | Penetration receiver loss frequency exponent coefficient |
|---------------------------------------|----------------------------------|-----------------------------------------------|----------------------------------------------------------|
| Open / Terrain                        | 17                               | 0.25                                          | 1                                                        |
| Grassland                             | 0                                | 0.82                                          | 0.65                                                     |
| Sparse forest                         | 0                                | 0.82                                          | 0.65                                                     |
| Medium dense forest                   | 0                                | 0.89                                          | 0.65                                                     |
| Very dense forest                     | 0                                | 0.95                                          | 0.65                                                     |
| Low density urban (Low buildings)     | 0                                | 0.89                                          | 0.65                                                     |
| Low density urban (High buildings)    | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (Low buildings)  | 0                                | 0.89                                          | 0.65                                                     |
| Medium density urban (High buildings) | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (Low buildings)    | 0                                | 0.89                                          | 0.65                                                     |
| High density urban (High buildings)   | 0                                | 0.89                                          | 0.65                                                     |

|                                          |     |      |      |
|------------------------------------------|-----|------|------|
| High density urban (Very high buildings) | 0   | 0.89 | 0.65 |
| Building blocks                          | 0   | 0.89 | 0.65 |
| Transportation                           | 0   | 0.89 | 0.65 |
| Agriculture                              | 0   | 0.89 | 0.65 |
| Plantation                               | 0   | 0.89 | 0.65 |
| Parks                                    | 0   | 0.89 | 0.65 |
| Airport                                  | 0   | 0.89 | 0.65 |
| Sea                                      | 0   | 0.89 | 0.65 |
| Inland water                             | 0   | 0.89 | 0.65 |
| Concrete building                        | 5   | 0.25 | 1    |
| Glass building                           | 2   | 0.25 | 1    |
| Wood building                            | 2   | 0.25 | 1    |
| Low loss building                        | 8.5 | 0.25 | 1    |
| High loss building                       | 17  | 0.25 | 1    |

#### 7.7.1.9 CNOSSOS-EU Model

##### Model application

Developed by the European Commission's Joint Research Centre as the Common Noise Assessment Methods in Europe, CNOSSOS-EU is the reference methodology adopted under the Environmental Noise Directive (2002/49/EC) for strategic noise mapping across EU Member States, making it the preferred choice for projects that must align with European regulatory requirements.

Unlike ISO 9613-2, which assumes downwind or moderate temperature inversion conditions throughout, CNOSSOS-EU evaluates sound propagation under two distinct meteorological scenarios — favourable conditions (downwind or temperature inversion, which enhance propagation toward the receiver) and homogeneous conditions (neutral atmosphere) — and combines the two results using the long-term occurrence probability of favourable conditions for the given direction. This produces long-term average sound pressure levels that more realistically reflect the variability of outdoor propagation over extended periods.

The model also features more refined calculations of geometrical divergence, atmospheric absorption, ground effect, diffraction over obstacles and around vertical edges, and reflections from building façades and other vertical surfaces. Ground effect is handled separately for the source-side, middle, and receiver-side regions of the propagation path, with acoustic reflectivity defined per clutter class via the Ground factor parameter shared with the ISO 9613-2 model. Diffraction is computed using path-difference geometry consistent with the CNOSSOS-EU specification, and terrain profiles, clutter classification, and building data from the workspace are used directly in the calculation. The primary factors included in the standard are:

##### Default settings

- **Distance coefficient: 20.**

Describes slope coefficient based on distance. Value is included in Geometric spreading calculations.

- **Temperature: 15 C°**
- **Humidity: 70%**
- **Favourable conditions occurrence: 0.25:** Probability or fraction of time during which meteorological conditions are "favourable" for sound propagation from a source to a receiver.

**Ground factor** parameter for the CNOSSOS-EU model's clutter classes allows users to define the acoustic reflectivity (from 0 to 1, indicating hard-soft respectively) of the ground surface for more accurate sound level predictions. When the clutter classes raster is absent from the geodata, a ground factor value of 0.5 is used.

#### Clutter Classes default values

|                                          | Ground factor |
|------------------------------------------|---------------|
| Open / Terrain                           | 1             |
| Grassland                                | 1             |
| Sparse forest                            | 1             |
| Medium dense forest                      | 1             |
| Very dense forest                        | 1             |
| Low density urban (Low buildings)        | 0.5           |
| Low density urban (High buildings)       | 0.3           |
| Medium density urban (Low buildings)     | 0.3           |
| Medium density urban (High buildings)    | 0.1           |
| High density urban (Low buildings)       | 0.1           |
| High density urban (High buildings)      | 0             |
| High density urban (Very high buildings) | 0             |
| Building blocks                          | 0             |
| Transportation                           | 0             |
| Agriculture                              | 1             |
| Plantation                               | 1             |
| Parks                                    | 0.7           |
| Airport                                  | 0             |
| Sea                                      | 0             |
| Inland water                             | 0             |
| Concrete building                        | 0             |
| Glass building                           | 0             |

|                    |   |
|--------------------|---|
| Wood building      | 0 |
| Low loss building  | 0 |
| High loss building | 0 |

## 7.8 Template Manager

**Template Manager** allows the user to edit current project templates residing in the various Template tables of the *default.gdb*. The user may change the field values of the templates, create new or delete existing templates.

Templates are essential tools designed to streamline and simplify the configuration process by predefining parameters automatically. Their primary purpose is to save time and reduce the risk of human error during setup, particularly in complex scenarios like wireless network configuration.

Here's why templates are beneficial:

1. **Automatic Parameter Filling:** Templates automatically populate required parameters, eliminating the need for users to input them one by one. This ensures consistency and speeds up the setup process, especially when working with large datasets or multiple network layers, such as cells, sites, or CPEs.
2. **Error Prevention:** In cases where a parameter might be inadvertently missed during manual configuration, templates act as a safeguard. They ensure all necessary parameters are accounted for, reducing the likelihood of incomplete or incorrect setups.
3. **Coverage Prediction Assurance:** Templates are particularly valuable when performing tasks like coverage predictions. If a parameter in the wireless network layer is forgotten or overlooked (e.g., for a cell, site, or CPE), the template provides a fallback, ensuring that the prediction can still be performed accurately and without interruption.

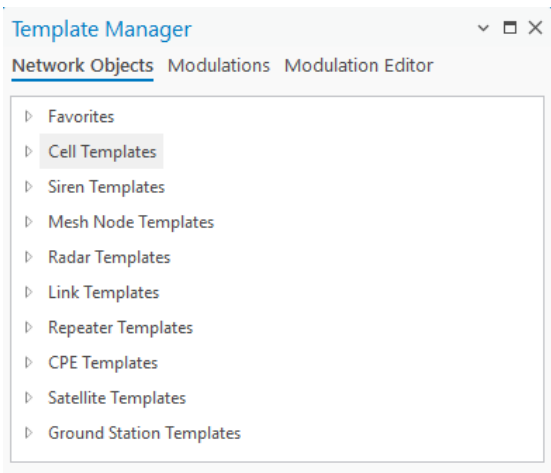
By using templates, users can maintain efficiency, accuracy, and reliability in network planning and analysis, even in "just-in-case" scenarios where manual inputs might fall short.

Templates can be marked as favorites in the Template Manager with a single click. Favorite templates are visually distinguished in the Template Manager and appear at the top of every template selection list in the application, including the Add Object and RF Prediction tools, separated from the remaining templates. The ordering always reflects the current favorite status, so newly starred templates move to the top the next time a picker is opened.

The templates are divided into categories:

- Network Objects – edit and manage network objects (cells, sites, links, etc) templates. Templates are divided into different network layers, each network have unique template structure based on available parameters in network layer.
- Modulations – create modulation configurations that can be used for MW links > Radios.
- Modulation Editor – create single modulations, which can be used in the Modulations tab.

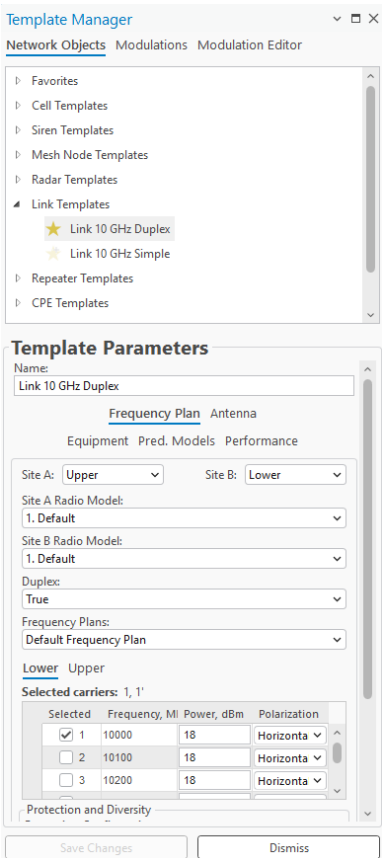




7.8.1 Edit Network Objects template



Click the  button to open the **Template Manager** dialogue. Select one of the opened templates to edit them.



**Save Changes**

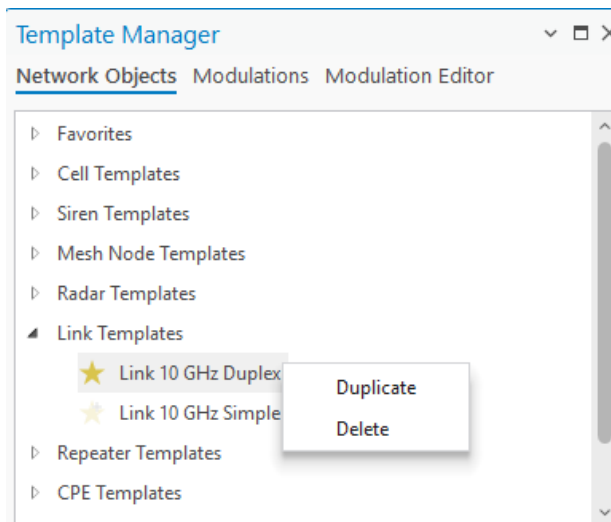
Saves the changes to the objects.

**Dismiss**

Cancels the changes to the objects and closes the dialogue.

### 7.8.2 Manage Network Object Template

Right-click on a selected template to open the context menu with Duplicate and Delete options.



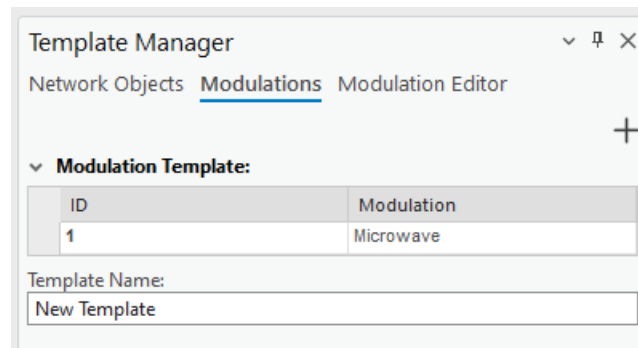
**Duplicate** will create a new template with the same parameters.

**Delete** will remove this selected template.

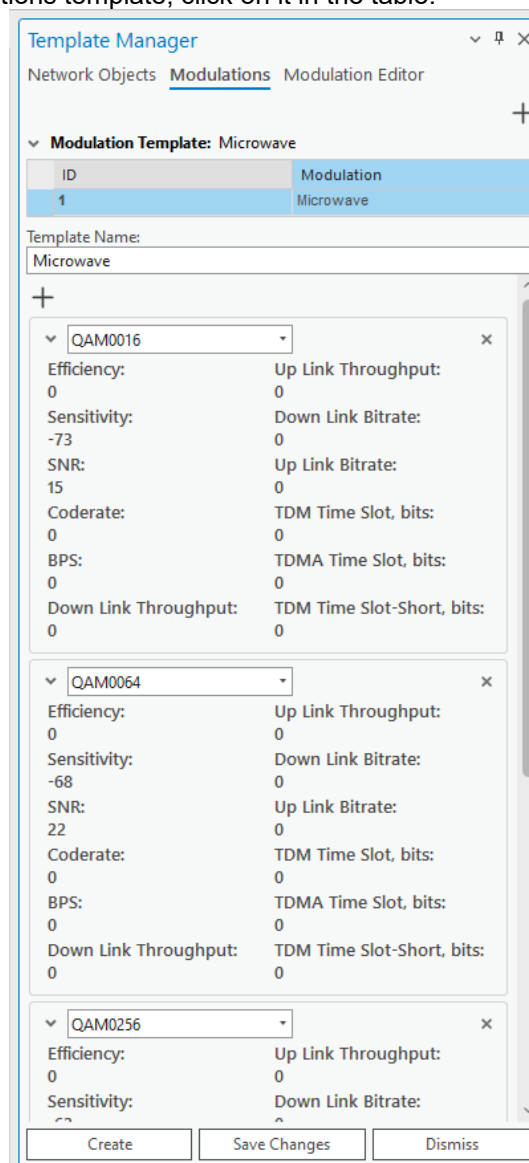
Click the star icon on the left of the template to mark it as favorite. Solid star symbol indicates that the template is marked as favorite and will appear in Favorites category at the top of Network Objects list in Template Manager. To unmark the template as favorite, click the star icon again. Transparent star icon indicates that the template is not marked as favorite.

### 7.8.3 Modulations

The modulations are used in MW link calculations and specifically can be defined for Radios. Instead of specifying Modulations one by one, the customer can create a set of them, and it is available in this tab.



To preview and edit the Modulations template, click on it in the table.



 (top right)

A new modulation with default values can be initialized.



Modulation can be removed.

### Save changes

All changes will be saved.

### Dismiss

Remove changes.

To create a new Modulation template, press  button in the top right corner of the dialog to initialize a new template.

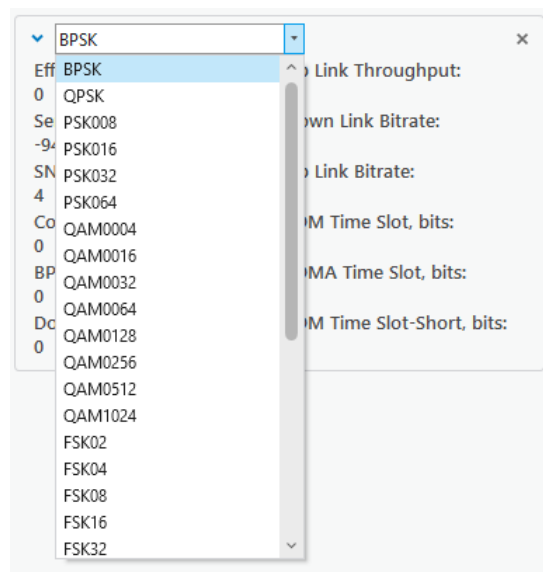
### Template name

The name of the modulation template.



(under the template name field)  
Assign modulation to the template.

Use the modulation drop-down list menu to apply and include the modulations.



Once the changes are made, click the Create button at the bottom of the window to create a new modulation template.

#### 7.8.4 Modulation Editor

The list of available modulations is taken from the Modulation Editor tab.

Template Manager

Network ObjectsModulationsModulation Editor

Modulation: QAM0004

| ID  | Modulation | Sensitivity |
|-----|------------|-------------|
| 103 | QAM0004    | -80         |
| 104 | ASK64      | -88         |
| 105 | BPSK       | -94         |
| 106 | ASK64      | -88         |
| 107 | BPSK       | -94         |
| 108 | QPSK       | -81         |
| 109 | PSK008     | -82         |
| 110 | PSK016     | -82         |
| 111 | PSK032     | -82         |
| 112 | PSK064     | -82         |
| 113 | QAM0004    | -80         |
| 114 | QAM0032    | -70         |
| 115 | QAM0128    | -65         |
| 116 | QAM1024    | -55         |

To add a new modulation, click the  at the top right of the Modulation Editor window to initialize a new modulation with default values, and press the Create button at the bottom of the window. Alternatively, click on an existing one to edit its parameters.

Technology:  
New Technology

Modulation:  
Modulation 1

Efficiency:  
0

Sensitivity:  
0

SNR:  
0

Code rate:  
0

BPS:  
0

DL Throughput:  
0

UL Throughput:  
0

DL Bitrate:  
0

UL Bitrate:  
0

TDM Time Slot:  
0

TDMA Time Slot:  
0

TDMA Time Slot Short:  
0

Radio ID:  
0

CreateSave ChangesCancel

**Create**  
Create a new modulation with the specified parameters.

**Save Changes**

Apply changes to a currently selected modulation. The button is disabled if the modulation is not created.

**Cancel**

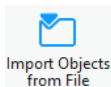
Dismiss changes made to the modulation.

## 7.9 Import Objects

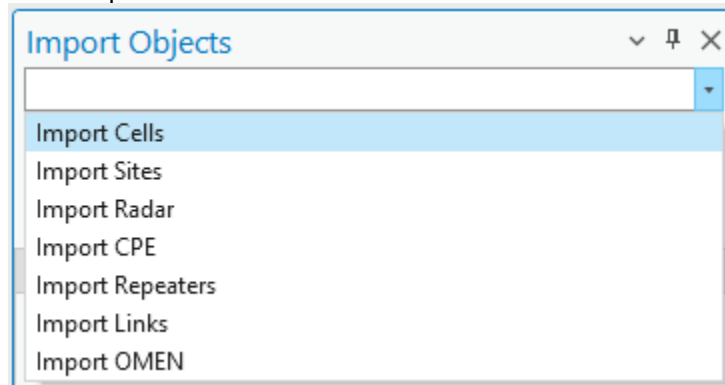
The Import Objects feature further enhances efficiency by allowing users to bring in network objects from multiple external documents without manually creating them. This capability is invaluable when dealing with extensive or complex datasets. Cellular Expert for ArcGIS Pro supports the import of three widely used file formats: .xls, .xlsx, and .csv. Additionally, for mapping files, it supports the .json format.

Key benefits of the Import Objects feature include:

- **Time-Saving:** Instead of manually entering or recreating network objects, users can quickly import data from existing documents, significantly reducing setup time.
- **Data Consistency:** Importing ensures that all network objects maintain consistency with the original data source, minimizing discrepancies and errors.
- **Support for Multiple Formats:** With support for commonly used file formats like .xls, .xlsx, and .csv, the tool is versatile and can integrate seamlessly with various data workflows.



Click the **Import Objects from File** button to open the **Import Objects** dialogue. Expand Import drop-down menu and select the object, which should be imported.



The dialog will be filled with the options to define data and mapping files.

### 7.9.1 Import Cells

The option enables possibility to import Cells in the Cellular Expert workspace. It has additional parameters compared to other import options.

#### Template

Take necessary parameters from the template during the import. Template values are taken if some parameters are missing in text file and mapping file.

#### Import HCM Patterns

Antenna patterns would be created and imported based on specific antenna name values provided in HCM agreement. HCM antenna pattern import requires the data files to have an *antenna\_type* field.

#### Generate Cell Name

Generate a cell name for all selected cells based on these parameters in this exact order: "longitude", "latitude", "height", "azimuth", "power", "antenna\_gain", and "frequency". If any of the fields are missing, that field will be skipped.

#### Use Default Mapping File

A default mapping file will be applied to the imported data.

#### Select Data Files

Opens a dialogue window where the user can select the files that define the network objects to be imported. The supported formats: *.xls*, *.xlsx*, *.csv*, and tables from *.sde* connection. Upon successful selection, the button will light up green.

### Select Mapping File (Optional)

Opens a dialogue window where the user can select a file that defines the data to be imported and the conditions under which said data is processed. The supported format is *.json*. Upon successful selection, the button will light up green. More on mapping files are below.

Steps.

1. Click on Select Data Files, and define your text file. The file will be loaded into the dialog.

**Import Objects**

Import Cells

3G 1800MHz

The template's default values will fill all fields (that were not specified) for predictions

☐ Import HCM Patterns  
☐ Generate Cell Name  
☐ Use Default Mapping File

Select Data Files (Cells.csv)

Select Mapping File (Optional)

Select Territory Polygon (Optional)

| Cell Name | Height Above Ground | azimuth | tilt | frequency | Cell Power | bandwidth | TxMIMO | RxMIMO | Tech | Tower ID |
|-----------|---------------------|---------|------|-----------|------------|-----------|--------|--------|------|----------|
| NBa 01    | 33                  | 86      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1       |
| NBa 02    | 30                  | 357     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1       |
| NBa 03    | 30                  | 247     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1       |
| NBb 01    | 30                  | 305     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2       |
| NBb 02    | 30                  | 99      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2       |
| NBb 03    | 30                  | 212     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2       |
| NBc 01    | 55                  | 19      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3       |
| NBc 02    | 55                  | 129     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3       |
| NBc 03    | 55                  | 259     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3       |
| NBd 01    | 25                  | 294     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4       |
| NBd 02    | 25                  | 69      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4       |
| NBd 03    | 22                  | 214     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4       |
| NBe 01    | 16                  | 31      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5       |
| NBe 02    | 16                  | 292     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5       |
| NBe 03    | 16                  | 210     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5       |

2. If your data structure is different compared to Cellular Expert database, then Mapping file should be used to map your data structure and Cellular Expert workspace, Cells layer structure. More information about mapping file is available below.



Select Data Files (Cells.csv)

Select Mapping File (cellsmapping\_small.json)

Select Territory Polygon (Optional)

| Cell Name | Height Above Ground | azimuth | tilt | frequency | Cell Power | bandwidth | TxMIMO | RxMIMO | Tech | Tower N |
|-----------|---------------------|---------|------|-----------|------------|-----------|--------|--------|------|---------|
| NBa 01    | 33                  | 86      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1      |
| NBa 02    | 30                  | 357     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1      |
| NBa 03    | 30                  | 247     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s1      |
| NBb 01    | 30                  | 305     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2      |
| NBb 02    | 30                  | 99      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2      |
| NBb 03    | 30                  | 212     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s2      |
| NBc 01    | 55                  | 19      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3      |
| NBc 02    | 55                  | 129     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3      |
| NBc 03    | 55                  | 259     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s3      |
| NBd 01    | 25                  | 294     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4      |
| NBd 02    | 25                  | 69      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4      |
| NBd 03    | 22                  | 214     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s4      |
| NBe 01    | 16                  | 31      | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5      |
| NBe 02    | 16                  | 292     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5      |
| NBe 03    | 16                  | 210     | 0    | 3500      | 40         | 100       | 4      | 4      | 5G   | s5      |

3. Click on Import Cells button to start importing procedure.

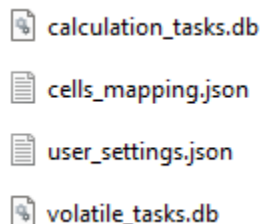
Sites can be imported together with Cells, if:

- The mapping file is not used, and text file has site\_id field, which contains information about Site name. It must be text data.
- The mapping file is used, then the data should be mapped with site\_id field. It must be text data.

#### 7.9.1.1 Mapping file

The data in the import files may have names, values, and units that do not match the data in the Cellular Expert database. To resolve such issues an additional Mapping file should be imported in which these data conflicts ought to be addressed.

The empty mapping file can be found in the project's workspace catalog, SystemFiles folder.



The mapping files are not necessary if the import file data corresponds to the Cellular Expert database data, otherwise mapping files are a must for a successful import.

Values that are not mentioned in the mapping file will not be affected.

The structure of a mapping file:

```
"table_name": "Cells",  
  
"fields": [  
  {  
    "current_name": "lat",  
    "destination_name": "latitude",  
    "default_value": "41.268"  
  },  
]
```

**“table\_name”** - defines which network object’s values are being mapped.

**“current\_name”** - the name of the value that is written in the data file. In this example “lat” is a column name in the data file and will be changed to “latitude” when the mapping file is applied, and objects are imported.

**“destination\_name”** - the proper name of the property (table column name) in the Cellular Expert database. This property will be checked when the mapping file is imported.

**“default\_value”** – value that will be used when an object in the data file has no value for a particular property. In this example, if “latitude” is not set, then it will by default be assigned the value of 41.258. Leaving the default to empty quotation marks (“”) means that no default value will be applied.

### 7.9.2 Import Sites, Radar, CPE, Sirens, or Repeaters

Other network objects can be imported to Cellular Expert workspace. The dialog contains information about.

#### Template

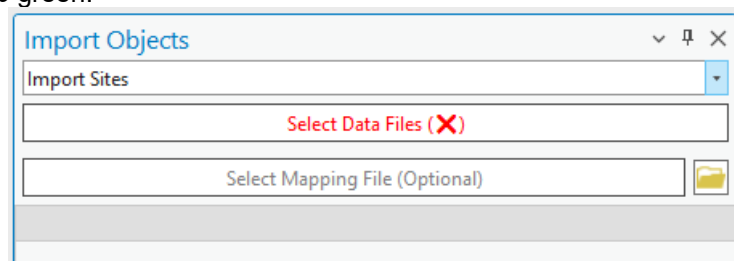
Take necessary parameters from the template during the import. Template values are taken if some parameters are missing in text file and mapping file.

#### Select Data Files

Opens a dialogue window where the user can select the files that define the network objects to be imported. The supported formats: .xls, .xlsx, .csv, and tables from .sde connection. Upon successful selection, the button will light up green.

#### Select Mapping File (Optional)

Opens a dialogue window where the user can select a file that defines the data to be imported and the conditions under which said data is processed. The supported format is .json. Upon successful selection, the button will light up green.

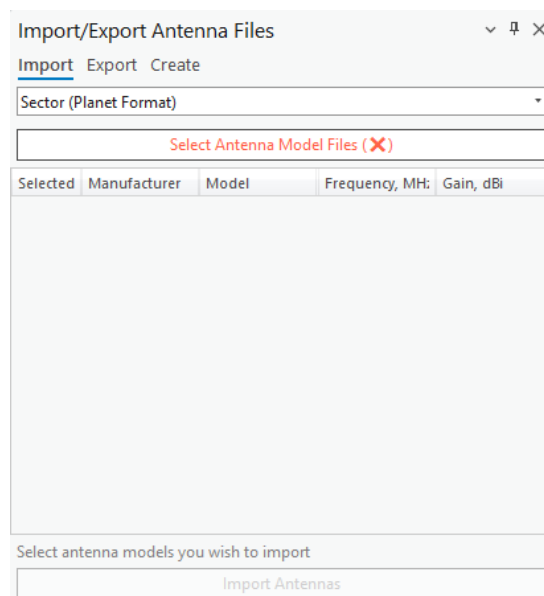


## 7.10 Import/Export Antenna Files

### 7.10.1 Import Antennas



Click the toolbar button  and select **Import** to import antenna patterns. The command opens a dialogue window where the user can select the antenna pattern files to be imported into the Cellular Expert database. Select the antenna type in the dropdown list and proceed.



#### Select Files

This button opens a dialogue in which you can select one or more antenna pattern files to be imported. The supported type formats are *Planet*, *Andrew*, and *NSMA*.

#### Import Antennas

Imports the selected Antenna pattern files to the Cellular Expert database.

#### Import Antennas

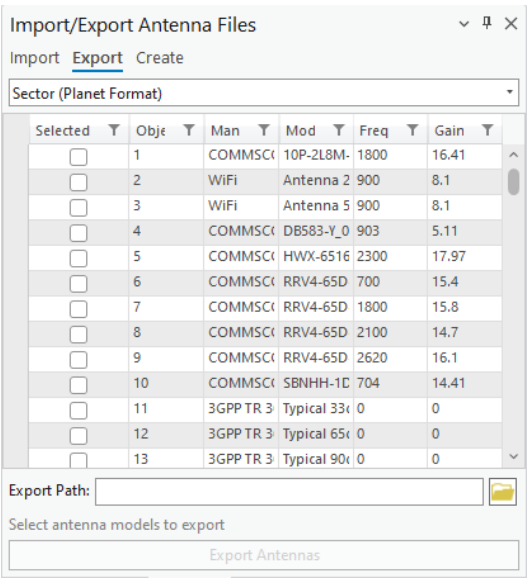
Refreshes the data table if changes have been made to it.

### 7.10.2 Export Antennas

Exports selected antennas to a desired format.



Click on the  button and select **Export** to export antenna patterns. Select the **Export path** on your local hard drive, check the desired antennas, and click the *Export* button.



Export Path

The destination folder on your local hard drive to which the selected antennas will be exported. Can be entered manually or by clicking the **Open** button and selecting it in the dialog. After defining the antenna format, all available antennas of that format will be displayed in the Antennas table. The supported type formats are *Planet*, *Andrew*, and *NSMA*.

Export Antennas

Exports the chosen antennas to the selected destination folder.

7.10.3 Create Antennas

Create new antennas based on input parameters.



Click on the **Import/Export Antenna Files** button and select **Create** to create antenna patterns. Various parameters of the antenna can be entered, as well as horizontal and vertical beamwidth values or ranges. For the selected horizontal and vertical values or ranges, the horizontal and vertical attenuations are set to 0, while all other attenuations are set to 1000. Based on the horizontal and vertical beamwidths, the horizontal and vertical antenna patterns are displayed.

Import/Export Antenna Files

Import Export Create

### Create a new antenna

Manufacturer:  
Cellular Expert

Model:  
Pyramid Horn 90deg

Frequency, MHz:  
900

Gain, dBi:  
10

Horizontal beamwidth:  
☒ Value, deg  
 90
 ☐ Range, deg  
 From 0 To 60

Vertical beamwidth:  
☐ Value, deg  
 15
☒ Range, deg  
 From 355 To 15

Horizontal Pattern

Horizontal Attenuation, dB

Vertical Pattern

Vertical Attenuation, dB

Input a new antenna's parameters

Create Antenna

**Manufacturer**

Antenna manufacturer (company or entity).

**Model**

Antenna identification or name.

**Frequency**

Antenna frequency value in MHz.

**Gain**

Antenna gain value in dBi.

**Horizontal beamwidth**

Antenna's horizontal beamwidth value or range in degrees. By default, the value is 60.

**Vertical beamwidth**

Antenna's vertical beamwidth value or range in degrees. By default, the value is 15.

**Create Antenna**

Creates the antenna pattern in the database.

## 7.11 Model Tuning



Click the **Model Tuning** button to open the **Model Tuning** dialogue.

Model Tuning is used to optimize prediction model parameters based on drive test points. These measurements must be placed on a custom feature class and bound to a cell by its cell ID.

The custom drive test feature layer (class) must have these fields:

- Field Strength (fs in the class table)
- Latitude
- Longitude
- Cell\_id (a field which binds the drive test to a cell network object)

|    | OBJECTID * | Shape * | fs  | latitude  | longitude | cell_id | workspace_id |
|----|------------|---------|-----|-----------|-----------|---------|--------------|
| 1  | 1          | Point   | -68 | 54.672215 | 25.244807 | 1       | 25           |
| 2  | 2          | Point   | -71 | 54.671777 | 25.243959 | 2       | 25           |
| 3  | 3          | Point   | -71 | 54.671679 | 25.243771 | 1       | 25           |
| 4  | 4          | Point   | -72 | 54.671629 | 25.243672 | 2       | 25           |
| 5  | 5          | Point   | -71 | 54.671608 | 25.243629 | 1       | 25           |
| 6  | 6          | Point   | -70 | 54.671599 | 25.243614 | 2       | 25           |
| 7  | 7          | Point   | -70 | 54.671592 | 25.243602 | 1       | 25           |
| 8  | 8          | Point   | -70 | 54.671585 | 25.243591 | 2       | 25           |
| 9  | 9          | Point   | -70 | 54.671578 | 25.243579 | 1       | 25           |
| 10 | 10         | Point   | -69 | 54.671556 | 25.243528 | 2       | 25           |
| 11 | 11         | Point   | -69 | 54.671475 | 25.243383 | 1       | 25           |
| 12 | 12         | Point   | -68 | 54.671377 | 25.243199 | 2       | 25           |

Upon the creation of the feature class, you should select the drive test layer. It is necessary to have cells on the map for the calculations to work. Their OBJECTID should correspond to the cell\_id present in the feature class's table.

Select a Drive Test Layer and the following properties should become visible:

**Model Tuning**

**Calculation Settings**

Drive Test Layer  
ModelTuning\_Grid1

Drive Test Fields:

Cell Identification:  
Cell\_ID

Field Strength:  
FS

Resolution, m  
10

**Prediction Model:** ITU-R P.452 (6GHz - 50GHz) - Default

- ITU-R P.452 (6GHz - 50GHz)
  - Default
- UniMacro (400MHz - 3GHz)
  - Default
- CEC ITU-R (100MHz - 6GHz)
  - Default
  - 1km radius
  - 2km radius
  - 3km radius
  - 5km radius
  - 7km radius

**Selected**  
5313 measurements

**Involved Cells**

☒ Toggle All

☒ Omni 2 (5313 measurements)

**Calculation Log**  
Getting Selection...  
Success.

Run

**Drive Test Layer**

The feature layer (class) that will be used to tune prediction model parameters.

**Cell Identification**

Cell name field name in drive test layer. The tool uses ObjectID value in Cells network objects to map drive test points.

**Field Strength**

Field Strength field name in drive test layer.

**Resolution**

Cell size for model tuning.

**Prediction Model**

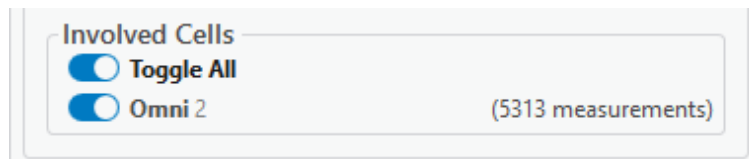
Prediction model name and type which will be used to tune its parameters.

**Selected**

The count of all measurement points.

**Involved Cells**

Information about the cell is given in this manner: cell name, cell ID, amount of measurements assigned to the cell. At the moment, measuring for a single cell is supported.

**Toggle All**

Include/Exclude all selected cells.

**Toggle Button By the Cell**

Include/exclude certain cells from the test calculations.

Select the Cells with which you want to do the drive test by clicking the checkboxes and pressing the **Run Calculation** button

Calculation results will appear in the dockpane, which contains the general, prediction model and clutter values of the calculations.



**Model Tuning**

Back Apply

**Totals**

**Reselect measurements by visibility**

**LOS**  
Reselect ☒  
Measurements 14

**OLOS**  
Reselect ☒  
Measurements 7

**NLOS**  
Reselect ☒  
Measurements 1

Apply selection

**Untuned statistics**

|                                    |       |
|------------------------------------|-------|
| Difference mean, dB:               | 10.27 |
| Difference standard deviation, dB: | 8.27  |
| Difference standard error          | 1.76  |

**Tuned statistics**

|                                    |      |
|------------------------------------|------|
| Difference mean, dB:               | 0.07 |
| Difference standard deviation, dB: | 4.57 |
| Difference standard error          | 0.97 |

> Statistics by visibility

> Statistics by clutter class

**Suggested prediction mode values**

|                                       |             |
|---------------------------------------|-------------|
| Offset coefficient, dB:               | Tuned 11.29 |
| Slope coefficient distance            | 20.23       |
| Slope coefficient distance obstructed | 40          |

> Suggested clutter class values

**Skipped measurements**

|                                   |   |
|-----------------------------------|---|
| Distance too high                 | 0 |
| Falls outside raster calculation  | 0 |
| Pixel standard deviation too high | 4 |

Measurement points can be reselected by visibility: LOS (line of sight), OLOS (obstructed line of sight), or NLOS (no line of sight). This allows for easy rerun of the tool with only relevant points to perform further tuning of the model.

### Untuned and Tuned statistics

The comparison between the untuned and tuned prediction models is presented by highlighting the differences in their mean values, standard deviations, and standard errors.

- **Difference Mean** during model tuning measures how the average prediction accuracy or performance improves after optimizing model parameters compared to the default settings. For example, tuning might reduce the average prediction error, leading to more accurate coverage or signal estimations.
- **Standard Deviation** reflects the variability in prediction performance. Comparing standard deviations between untuned and tuned models shows whether tuning makes predictions more consistent and reliable. For example, lower standard deviation after tuning indicates the model's predictions are less scattered and more stable.
- **Standard Error** during model tuning quantifies the precision of the mean prediction. A lower standard error in the tuned model suggests that the average prediction is more reliable and less influenced by sample variability, indicating improved model stability and confidence in its performance.

Statistics by visibility

LOS

Measurements:

49

Difference mean, dB:

7.5

Difference standard deviation, dB:

6.67

Tuned difference mean, dB:

-0.34

Tuned difference standard deviation, dB:

6.68

NLOS

OLOS

Statistics by clutter class

Transportation

Parks

Concrete building

Grassland

Medium dense forest

Suggested prediction mode values

Offset coefficient, dB:

29.07

Slope coefficient distance near

20.08

Slope coefficient distance far

21.86

A0

26.77

A1

62.01

A2

-5.49

A3

0.09

Suggested clutter class values

Statistics by visibility and by clutter class are also available. Untuned and tuned difference points are also now visualized on the map as part of the model tuning calculation result in separate layers.

### Suggested prediction model values

Model coefficients that are recommended to be changed.

- **Offset coefficient** – represents the offset in decibels added to the path loss grid. The default value is 32 dB.
- **Slope coefficient distance** – defines the slope based on the distance between the cell and the receiver location, with a default value of 20.
- **Slope coefficient distance obstructed** – represents the slope based on the obstructed distance between the cell and the receiver location. The default value is 40.

### Suggested clutter class values

Recommended clutter loss values

### Apply

Edit the tuned prediction model with suggested model and clutter class values.

### Back

Get back to the main Model Tuning calculation window.

## 8. Profile

### 8.1 Profile Tool

A Profile in wireless communication represents the geographical and environmental characteristics of the path between a transmitter and a receiver. It includes detailed information such as elevation data, terrain heights, and any obstacles (e.g., buildings, trees, or mountains) that might impact signal propagation.

All relevant calculations are already performed within the system. These include the power budget, which factors in the total power available for transmission and reception while accounting for system gains and losses, and the path loss, which measures the reduction in signal strength due to distance, frequency, and environmental obstructions. The profile also provides the calculated angles, including the elevation angle (vertical angle between the transmitter and receiver) and the azimuth angle (horizontal direction). Additionally, it determines whether a direct line-of-sight exists between the two points.

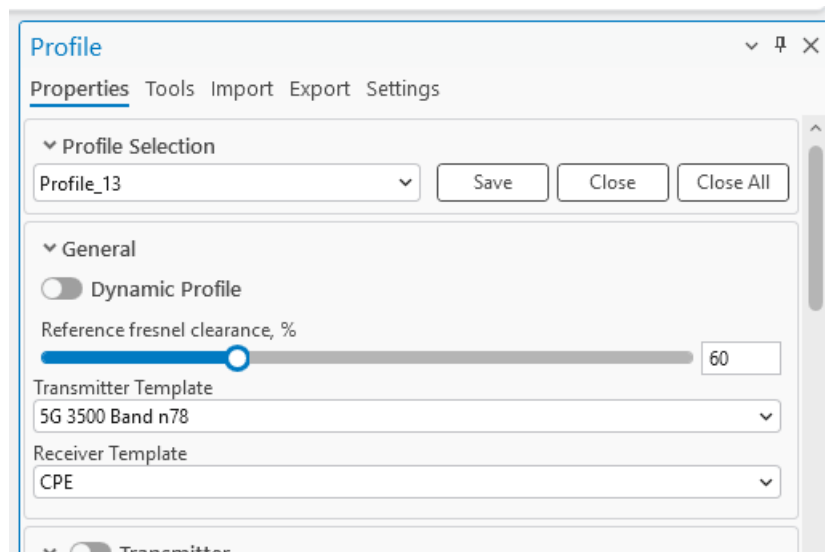
This comprehensive information is ready for use, enabling users to assess the feasibility and performance of a communication link for network planning, optimization, and troubleshooting.



Click the **Profile** button to open the **Profile** dialogue.

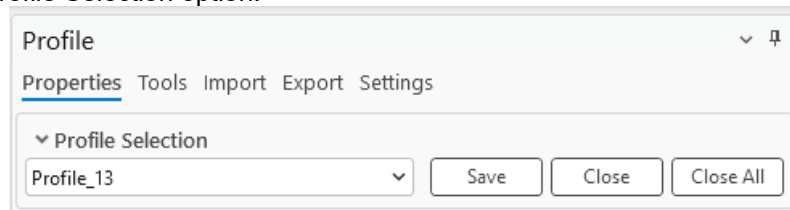
Profile tool enables you to determine the obstructions, elevation, and Fresnel zones between two points on a map.

#### 8.1.1 Properties



#### Profile: Profile Selection

Profile selection allows for quick switching between profiles while retaining the most recently defined parameters. This feature is especially useful for comparing and verifying calculations across different Tx and Rx configurations. When a new profile is created, it is automatically assigned a name and appears in the Profile Selection option.



Profile can be saved or removed from the list.

**Save**

The option saves the profile to Docs Manager tool with all defined parameters.

**Close**

Removes profile from the list in Profile Selection.

**Close All**

Removes all profiles from Profile Selection list.

**Profile: General****Fresnel Minimal Clearance, %**

Percentage by which the primary Fresnel zones will be scaled up or down thus creating a secondary Fresnel zone. The percentage must be in the range of 1 to 200. The value can be changed either by inputting it manually or by using the slider.

**Transmitter Template**

The template that is used for transmitter's default values.

**Receiver Template**

The template that is used for receiver's default values.

A screenshot of the 'Transmitter' configuration panel. At the top, there is a toggle switch labeled 'Transmitter' which is turned on. Below this, there are several input fields: 'Cell' with a dropdown menu showing 'None'; 'Latitude' with a text input field containing '53.48407311830833'; 'Longitude' with a text input field containing '9.92402763529321'; 'Height, m (min.: 1m)' with a text input field containing '23.00'. Below these fields are two toggle switches: 'Azimuth towards receiver' and 'Downtilt towards receiver', both of which are turned on. At the bottom, there is a text input field for 'El. Downtilt, deg.' containing the value '0'.

**Profile: Transmitter**

Toggling the switch to the left of **Transmitter** will enable the **Fixed Transmitter** functionality, which will modify only the receiver's positioning when drawing the profile on the map.

**Cell**

A cell from which the profile will be drawn. The parameters of the cell will be taken into calculation if the cell is snapped to by the profile tool.

**Latitude**

Decimal degrees Y type coordinate.

**Longitude**

Decimal degrees X type coordinate.

**Height, m**

Height above the ground in meters. The minimum value must be 1m.

**Azimuth towards receiver**

Enabled by default. When enabled, the transmitter's azimuth is towards the receiver. Disabling this option it would take azimuth value from the Cell object, and use it for FWA Power Budget calculations, it also allows the user to enter a custom azimuth value for the transmitter.

**Downtilt towards receiver**

Enabled by default. When enabled, the transmitter's tilt is towards the receiver. Disabling this option it would take tilt value from the Cell object, and use it for FWA Power Budget calculations, it allows the user to enter a custom tilt value for the transmitter.

**EI. Downtilt, deg.**

Electrical downtilt value for the transmitter, in degrees.

Antenna: 742215

View Antenna

| Object ID | Manufacturer   | Model               | Frequency, M | Gain, dBi |
|-----------|----------------|---------------------|--------------|-----------|
| 1         | COMMScope      | 10P-2L8M-B5-V2_Pc   | 1800         | 16.41     |
| 2         | WiFi           | Antenna 2.4GHz 360  | 900          | 8.1       |
| 3         | WiFi           | Antenna 5GHz 360    | 900          | 8.1       |
| 4         | COMMScope      | DB583-Y_00DT_090    | 903          | 5.11      |
| 5         | COMMScope      | HWX-6516DS1-VTM     | 2300         | 17.97     |
| 6         | COMMScope      | RRV4-65D-R6_Port 1  | 700          | 15.4      |
| 7         | COMMScope      | RRV4-65D-R6_Port 2  | 1800         | 15.8      |
| 8         | COMMScope      | RRV4-65D-R6_Port 3  | 2100         | 14.7      |
| 9         | COMMScope      | RRV4-65D-R6_Port 4  | 2620         | 16.1      |
| 10        | COMMScope      | SBNHH-1D65B_Port    | 704          | 14.41     |
| 11        | 3GPP TR 36 942 | Typical 33deg H and | 0            | 0         |
| 12        | 3GPP TR 36 942 | Typical 65deg H and | 0            | 0         |
| 13        | 3GPP TR 36 942 | Typical 90deg H and | 0            | 0         |
| 14        | 3GPP TR 36 942 | Typical 90deg H and | 0            | 0         |

Frequency, MHz

1800

Bandwidth, MHz

5

Power, dBm

43.00

Misc Loss, dBm

0.00

Tx mimo:

1

Rx mimo:

1

Subcarrier Spacing, kHz

15.00

The antenna for the transmitter can be selected from the table below the **El. Downtilt, deg** input. Its pattern can be viewed by clicking the **View Antenna** button on the right side. Frequency of the transmitter.

#### Bandwidth, MHz

Bandwidth of the transmitter.

#### Power, dBm

A power that the transmitter possesses.

#### Misc Loss, dB

Miscellaneous loss value in dB. Value is not required.

#### Tx mimo

Transmitter antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

#### Rx mimo

Receiver antenna count. Available values: 1, 2, 4, 8, 16, 32 and 64.

#### Subcarrier spacing

Value in kHz.

Profile

Properties

Tools

Import

Export

Settings

> General

> ☐ Transmitter

< Receiver

Latitude

54.9267011

Longitude

24.6388044

Height, m (min.: 1m)

1.50

☒ Azimuth towards transmitter

☒ Downtilt towards transmitter

< Antenna: 10P-2L8M-B5-V2\_Port 3 +45\_01DT\_1800

View Antenna

| Object ID | Manufactu      | Model           | Frequency | Gain, dBi |
|-----------|----------------|-----------------|-----------|-----------|
| 13        | 3GPP TR 36 942 | Typical 90deg H | 0         | 0         |
| 14        | 3GPP TR 36 942 | Typical 90deg H | 0         | 0         |
| 15        | 3GPP TR 36 942 | Typical Omni TR | 0         | 0         |
| 17        | COMMSCOPE      | 3X-V65S-C3-3XF  | 1695      | 12.51     |
| 18        | COMMSCOPE      | 3X-V65S-C3-3XF  | 1865      | 13.07     |
| 19        | COMMSCOPE      | 3X-V65S-C3-3XF  | 2050      | 13.53     |
| 20        | COMMSCOPE      | 3X-V65S-C3-3XF  | 2400      | 14.46     |
| 21        | COMMSCOPE      | 3X-V65S-C3-3XF  | 2690      | 14.24     |
| 22        | COMMSCOPE      | 4P-4L-B2        | 694       | 14.28     |
| 23        | COMMSCOPE      | 4P-4L-B2        | 849       | 15.31     |
| 24        | COMMSCOPE      | 4P-4L-B2        | 915       | 15.68     |
| 25        | COMMSCOPE      | 4P-4L-B2        | 960       | 15.72     |
| 26        | COMMSCOPE      | 8P-4L4M-A4      | 694       | 13.55     |
| 27        | COMMSCOPE      | 8P-4L4M-A4      | 768       | 14.27     |

Power, dBm

0.00

Misc. loss, dBm

0.00

> Prediction Model

Manual Profile

**Profile: Receiver**

**Latitude**  
Decimal degrees Y type coordinate.

**Longitude**  
Decimal degrees X type coordinate.

**Height, m**  
Height above the ground in meters. The minimum value must be 1m.

**Azimuth towards transmitter**  
Enabled by default. When enabled, the receiver’s azimuth is towards the transmitter. Disabling this option it would take azimuth value from the receiver (Cell) object, and use it for FWA Power Budget

calculations, it also allows the user to enter a custom azimuth value for the receiver.

#### **Downtilt towards transmitter**

Enabled by default. When enabled, the receiver's tilt is towards the transmitter. Disabling this option it would take tilt value from the receiver (Cell) object, and use it for FWA Power Budget calculations, it allows the user to enter a custom tilt value for the receiver.

#### **Power, dBm**

A power that the receiver possesses.

#### **Misc Loss, dB**

Miscellaneous loss value in dB. Value is not required.

#### **Profile: Prediction Model**

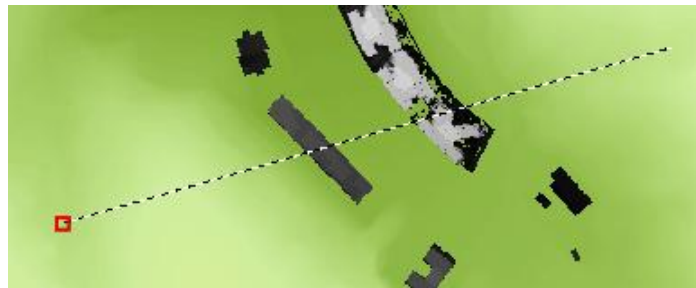
Double-click the prediction model from the list to select it for the profile.

### **8.1.2 Draw Profile**

When the Profile tool is selected and enabled, you will be able to select two points on the map in turn creating a Profile line.

The profile also lets you **snap to different Cellular Expert network objects**. If the object to which the user snaps is a cell, all parameters of that cell that are necessary to draw a profile will be read and included in the calculations. If the object is not a cell, then only the name and the object's coordinates will be taken.

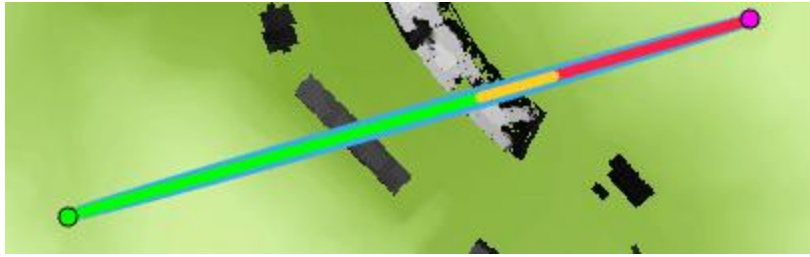
Most of these values will be displayed in the Profile pane shown above. If you change the values in the pane, these changes will be reflected in the Profile plot by redrawing it.



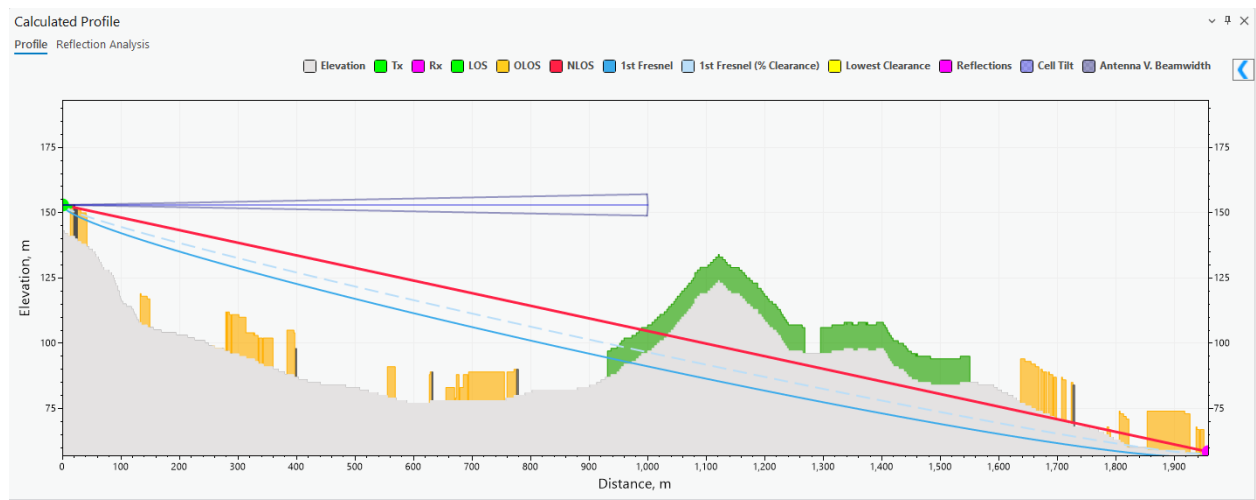
Upon the selection of a second point, these geometries will be created between the points:

- LOS (green) - the distance until the first obstruction in the profile's way
- OLOS (orange) – the distance until the first clutter LOS obstruction
- NLOS (red) – the obstructed path between the sender and receiver
- Lowest Clearance (yellow) – the point at which NLOS has the lowest clearance to the obstacle
- Rx (purple) – the receiver point
- Tx (green) – the transmitter point
- Fresnel (blue) – the Fresnel lines





The Profile plot illustrating these geometries, obstacles (buildings), and the Fresnel zone will appear in a dockpane below. You can inspect the values at particular points by moving the cursor around the plot. The cursor movement on the plot will be projected as a moving point on the map. If a cell is selected for the transmitter, the cell's tilt (the sum of mechanical and electrical tilts) and vertical beamwidth are displayed as additional symbols: a light blue line and a darker blue cone respectively. The length of the cell tilt and antenna vertical beamwidth symbols is the radius of the prediction model selected for the cell.



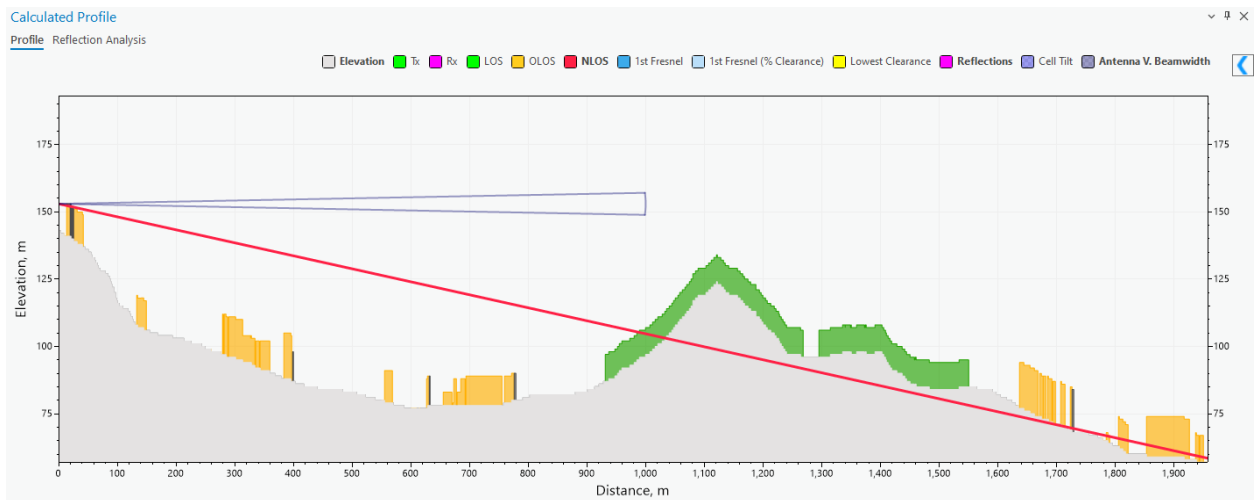
The  button allows you to see the Prediction Calculation results.

| General Calculations                |            |
|-------------------------------------|------------|
| Path type                           | LOS        |
| Visibility clearance                | 4.16 m     |
| Fresnel clearance                   | -1.01 m    |
| Clearance percentage                | 80.51 %    |
| Distance to NLOS                    | 1118.87 m  |
| Distance to OLOS                    | 1118.87 m  |
| Power Budget                        |            |
| Downlink FS                         | -63.12 dBm |
| Uplink FS                           | -89.7 dBm  |
| FWA downlink RSL                    | -4.61 dBm  |
| FWA uplink RSL                      | -48.4 dBm  |
| Path Loss                           |            |
| Total path loss                     | 98.08 dB   |
| Basic loss                          | 98.08 dB   |
| Multipath focusing loss             | 0 dB       |
| Terrain / building diffraction loss | 0 dB       |
| Enclosed receiver attenuation       | 0 dB       |
| Enclosed receiver diffraction loss  | 0 dB       |
| Angles                              |            |
| Transmitter azimuth                 | 34 °       |
| Transmitter tilt                    | 1.01 °     |
| TX-RX azimuth                       | 34 °       |
| TX-RX tilt                          | 1.01 °     |
| Receiver azimuth                    | 214 °      |
| Receiver tilt                       | -1.01 °    |
| RX-TX azimuth                       | 214 °      |
| RX-TX tilt                          | -1.01 °    |

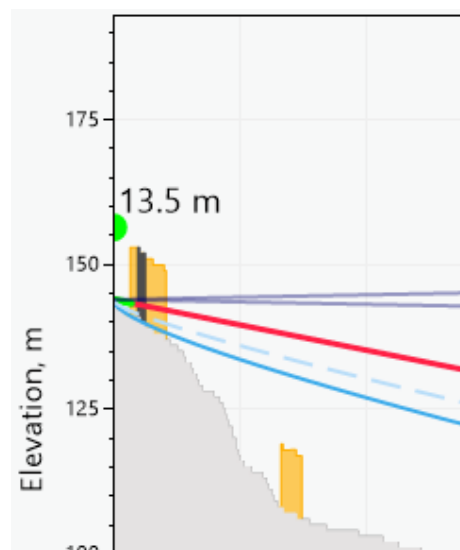
You can change the colors of the profile by clicking the colored squares near the names of the parameters.

☐ Elevation 
 ☒ Tx 
 ☒ Rx 
 ☒ LOS 
 ☒ OLOS 
 ☒ NLOS 
 ☒ 1st Fresnel 
 ☒ 1st Fresnel (% Clearance) 
 ☒ Lowest Clearance 
 ☒ Reflections 
 ☒ Cell Tilt 
 ☒ Antenna V. Beamwidth

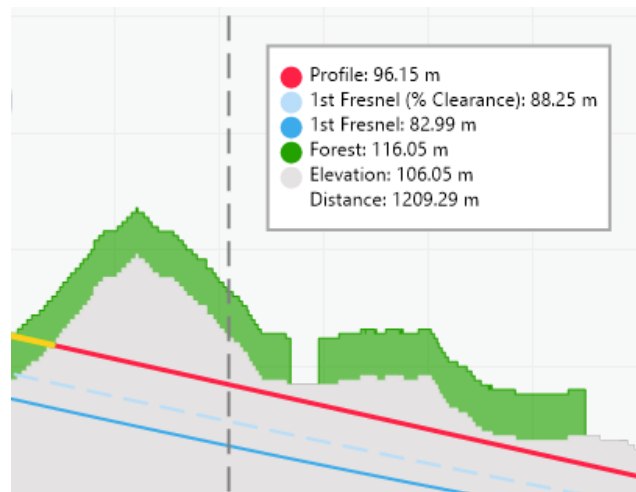
The colors will be updated automatically. You can toggle the visibility of each separate parameter of the profile by clicking on the name of the element. Enabled elements are indicated by **bold** text, and disabled elements are indicated by regular text.



You can change the height of the transmitter/receiver points by dragging their ends on the plot.

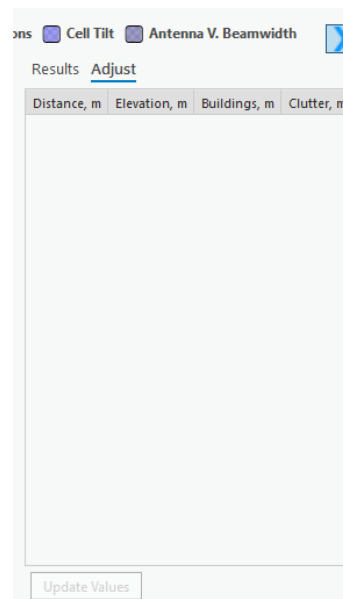


Hovering the cursor over the plot displays a tooltip with the meter values for profile, building, clutter, elevation, distance, etc., as well as their representations in colors.

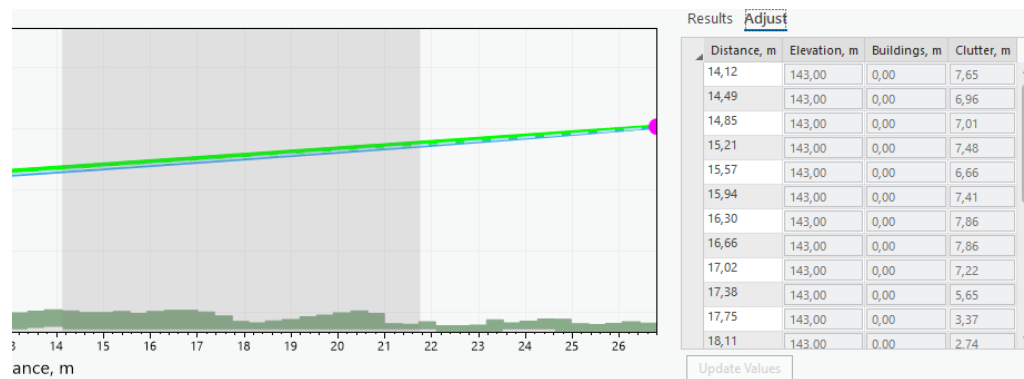


#### 8.1.2.1 Adjust Data

Adjust data is found on the Profile Plot dockpane near the Results. The tool lets you change the elevation, building, and clutter data of the area visible in the profile plot.



When the **Adjust** tab is opened, select a desirable range for the data adjustment on the plot.

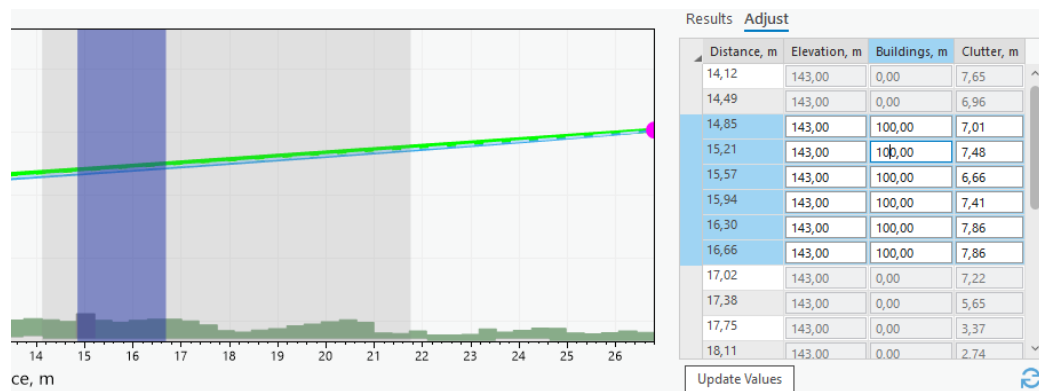


You can make slight changes to the range of the selection area by hovering over the area's edges and dragging them. To adjust the values, click on one of the text boxes and insert the value.

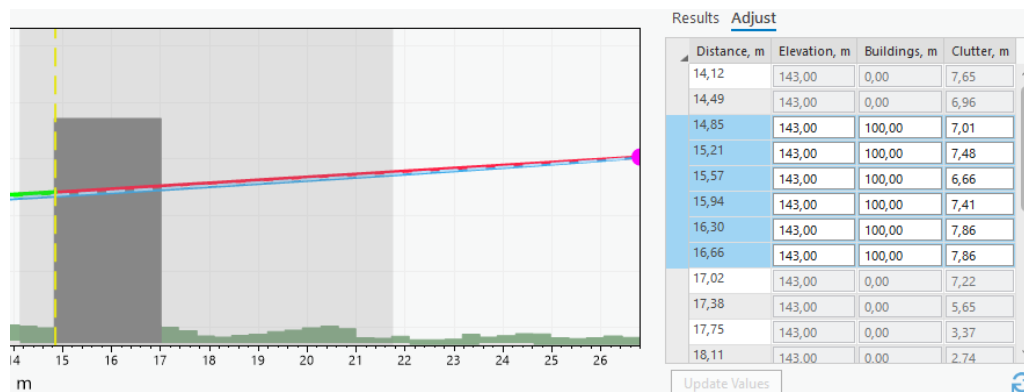
Results Adjust

| Distance, m | Elevation, m | Buildings, m | Clutter, m |
|-------------|--------------|--------------|------------|
| 14,12       | 143,00       | 0,00         | 7,65       |
| 14,49       | 143,00       | 0,00         | 6,96       |
| 14,85       | 143,00       | 10,00        | 7,01       |
| 15,21       | 143,00       | 0,00         | 7,48       |
| 15,57       | 143,00       | 0,00         | 6,66       |

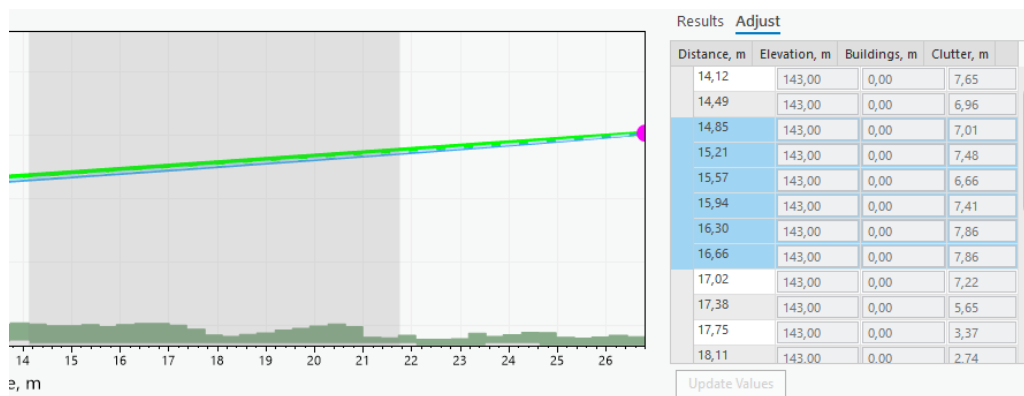
To change multiple values simultaneously, drag across the adjustment table and select multiple rows. Changing the value of a single text box will also change all the other chosen rows' values in that text box. The selected area will be highlighted on the profile plot.



To update the values, either select an unselected row or press the  button.



To reset the adjusted values to defaults, click the **Refresh button** .



## Manual Profile

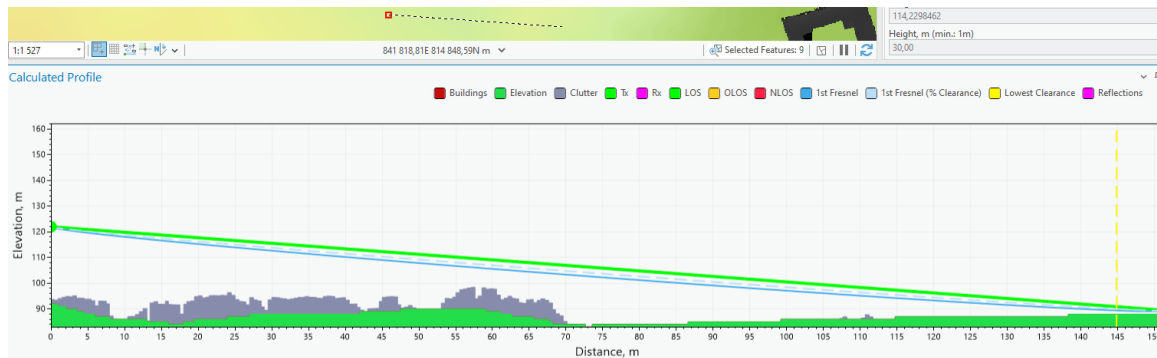
If you want to insert specific coordinates and draw a profile that way, you can insert these values in the Profile pane and click the **Manual Profile** button.

## Dynamic Profile

The  **Dynamic Profile** button toggles the Dynamic Profile.

**Dynamic Profile** lets you see Profile calculations and the plot be drawn in real-time as the cursor moves. Whenever the cursor stops momentarily – the profile gets drawn. The transmitter point needs to be entered for the dynamic profile to be activated. The previous transmitter point will be chosen if the point is not currently entered.

If a second point is selected while the profile is being drawn, the dynamic profile will be disabled, and the LOS lines will appear on the map.



### 8.1.3 Tools

Reflections are considered in profile and RF calculations to assess how radio waves bounce off surfaces, affecting signal path and strength.

To enable reflections select either Single or Multipath Reflection. Be aware that if reflections are not visible you may need to adjust the height of the receiver/transmitter accordingly.

It is recommended to disable the Step Plot of the profile before using Reflections (see *Settings*).

▼ Reflections

☒ Use Single Reflection

☐ Use Multipath Reflections

☐ Use Divergence Factor

Step Size, m:

Polarizations

Horizontal

☒ Use Clutter Classes

| Clutter      | Conductivity, S/m | Permittivity |
|--------------|-------------------|--------------|
| Buildings    | 0.001             | 5            |
| Forest       | 0.004             | 13           |
| Dense forest | 0.004             | 13           |
| Urban        | 0.001             | 5            |
| Dense urban  | 0.001             | 5            |
| Bare ground  | 0.002             | 10           |
| Crops        | 0.03              | 20           |
| Water        | 1                 | 80           |
| Road         | 0.002             | 10           |

☒ Select Reflection Range

Reflection Incident Area, m

From:  To:

Select a selection area on the Profile plot

## Use Single Reflection

Enable a reflection that will reflect straight from the transmitter to the receiver point with the smallest angle.

**Use Multipath Reflections**

Enable all reflections that happen along the profile line.

**Use Divergence Factor**

Quantifies the extent to which a reflected signal spreads out, affecting its strength and coverage.

**Step Size**

The length period by which the reflection will be calculated. The step size is determined by taking the average cell size of the elevation raster.

**Polarizations**

The polarization of reflections (vertical or horizontal).

**Use Clutter Classes**

Enables the use of default clutter classes for the reflection calculations (see *Clutter Classes*). If disabled, custom values may be used for conductivity and permittivity.

**Select Reflection Range**

Lets you set a range in which the reflection calculations should happen. This will affect both Single and Multipath reflections and help highlight specific important areas along the profile as well as speed up the calculation process.

The range will be enabled as soon as a profile is drawn.

You can select this range on the plot, by holding down the left mouse button and dragging across the screen.



Reflection results will appear in the Profile Results table.



|                                     |               |
|-------------------------------------|---------------|
| <b>Results</b>                      | <b>Adjust</b> |
| FWA uplink RSL                      | -33.25 dBm ^  |
| <b>Path Loss</b>                    |               |
| Total path loss                     | 82.93 dB      |
| Basic loss                          | 82.93 dB      |
| Multipath focusing loss             | 0 dB          |
| Terrain / building diffraction loss | 0 dB          |
| Enclosed receiver attenuation       | 0 dB          |
| Enclosed receiver diffraction loss  | 0 dB          |
| <b>Angles</b>                       |               |
| Transmitter azimuth                 | 49 °          |
| Transmitter tilt                    | 5.06 °        |
| TX-RX azimuth                       | 49 °          |
| TX-RX tilt                          | 5.06 °        |
| Receiver azimuth                    | 229 °         |
| Receiver tilt                       | -5.06 °       |
| RX-TX azimuth                       | 229 °         |
| RX-TX tilt                          | -5.06 °       |
| <b>Reflections</b>                  |               |
| Clutter Class                       | buildings     |
| Conductivity                        | 0.001 S/m     |
| Permittivity                        | 5             |
| Received Signal                     | 14.22 dB      |
| Reflection Distance                 | 151.3 m       |
| Delay                               | 6.63 ns       |
| Grazing Angle                       | 180.66 mrad   |
| Divergence Factor                   | 1             |
| Terrain Roughness                   | 1.46 m        |
| Inclination                         | 88.5 mrad v   |

#### 8.1.3.1 Reflection Analysis

The **Reflection Analysis** tool for Profile is designed to help analyze and visualize signal reflections based on the changes in various profile parameters like frequency, transmitter height, receiver height, and K-factor. Reflections must be enabled to perform analysis, and the Single Reflection option is automatically enabled when the tool is selected.

▼ Reflection Analysis

Dependency on

☒ Frequency

☐ Receiver height

☐ Transmitter height

☐ K-Factor

View

☒ K-Factor

☐ ArcTan(K)

☐ Radius

Transmitter height, m

Receiver height, m

Height

30

1.5

Multiplier

Radius, km

K-Factor

1

6400

From

To

Frequency, GHz

0.9

1

| ID  | From frequency, GHz | To frequency, GHz | Step, GHz | Tr. height, m | Rec. height, m |
|-----|---------------------|-------------------|-----------|---------------|----------------|
| 124 | 0.9                 | 1                 | 0.001     | 30            | 1.5            |

<

>

Add

Remove

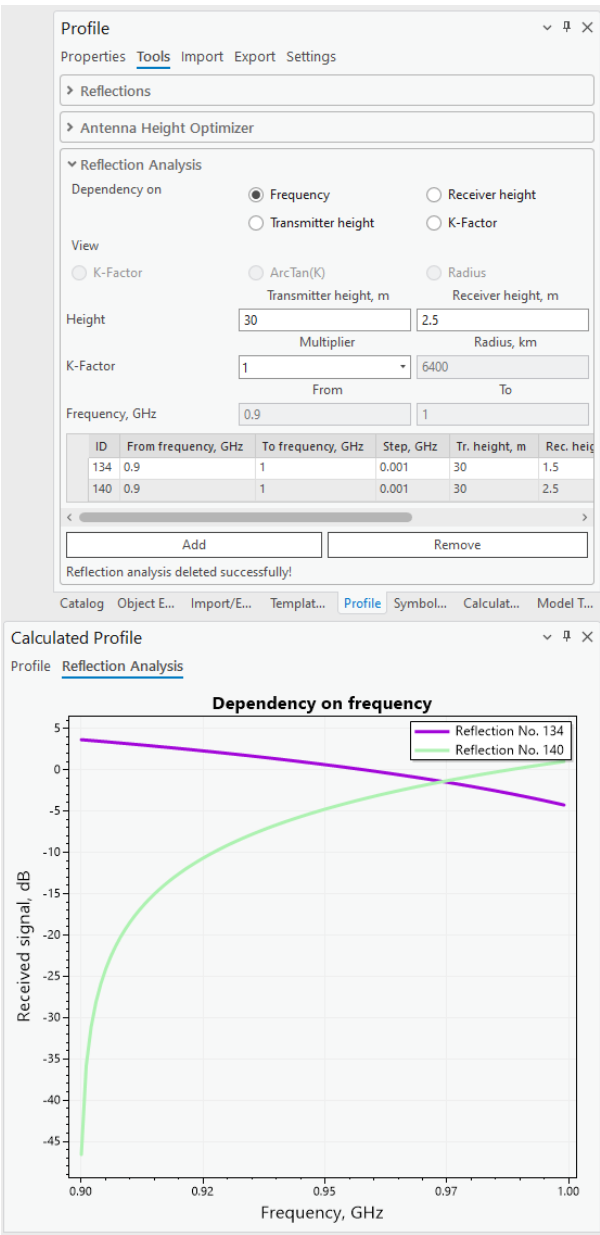
New reflection analysis created successfully!

**Dependency on Frequency**  
Calculate reflection analysis based on frequency range (from GHz to GHz)

**Dependency on Receiver height**  
Calculate reflection analysis based on receiver height range (from m to m)

**Dependency on Transmitter height**  
Calculate reflection analysis based on transmitter height range (from m to m)

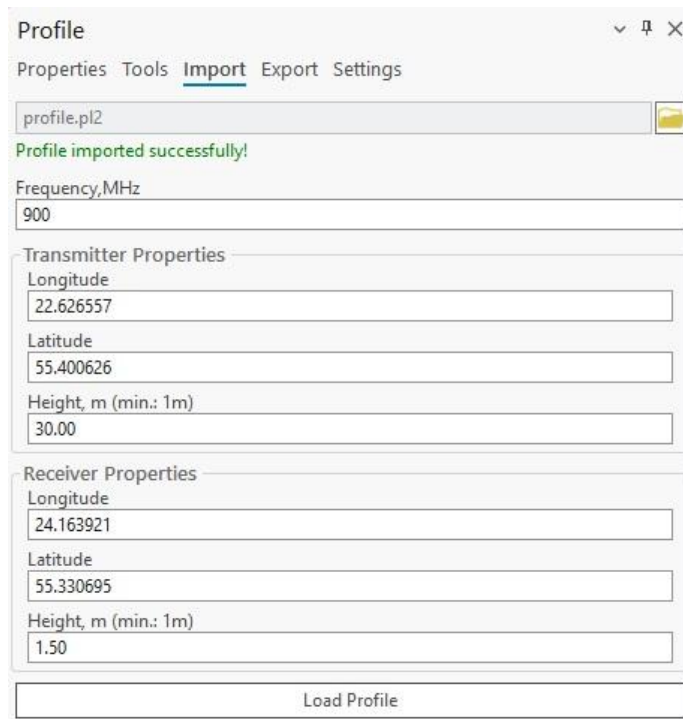
**Dependency on K-Factor**  
Calculate reflection analysis based on K-factor range (from radius, km to radius, km)



The reflection analysis results for each type of dependency are displayed in the **Reflection Analysis** tab of the **Calculated Profile** window.

#### 8.1.4 Import

Import a profile by selecting a profile file in the Import section. Supported formats: *.pl2 (path loss file)*. Once imported successfully, the profile data may then be customized.



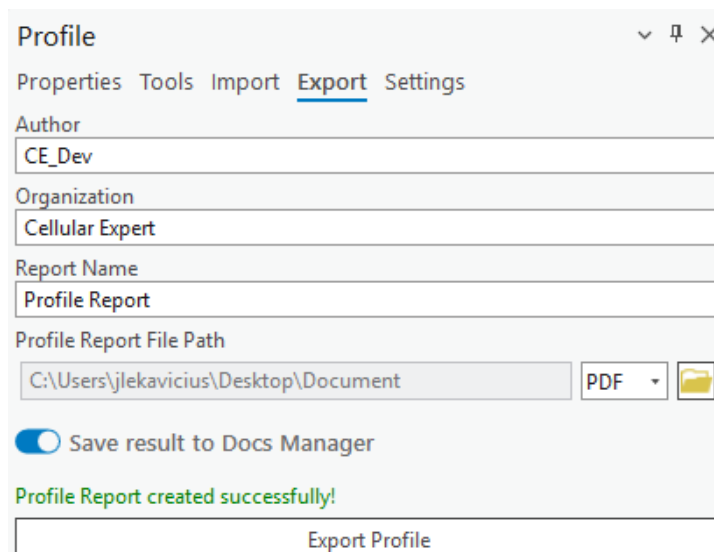
The screenshot shows the 'Profile' window with the 'Import' tab selected. The window has a title bar with a dropdown arrow, a pin icon, and a close button. Below the title bar are tabs for 'Properties', 'Tools', 'Import', 'Export', and 'Settings'. The 'Import' tab contains a text field with 'profile.pl2' and a folder icon button. Below this is a green status message: 'Profile imported successfully!'. There are two main sections: 'Transmitter Properties' and 'Receiver Properties'. The 'Transmitter Properties' section includes fields for 'Frequency, MHz' (900), 'Longitude' (22.626557), 'Latitude' (55.400626), and 'Height, m (min.: 1m)' (30.00). The 'Receiver Properties' section includes fields for 'Longitude' (24.163921), 'Latitude' (55.330695), and 'Height, m (min.: 1m)' (1.50). At the bottom is a 'Load Profile' button.

### Load Profile

Creates the profile with the provided data.

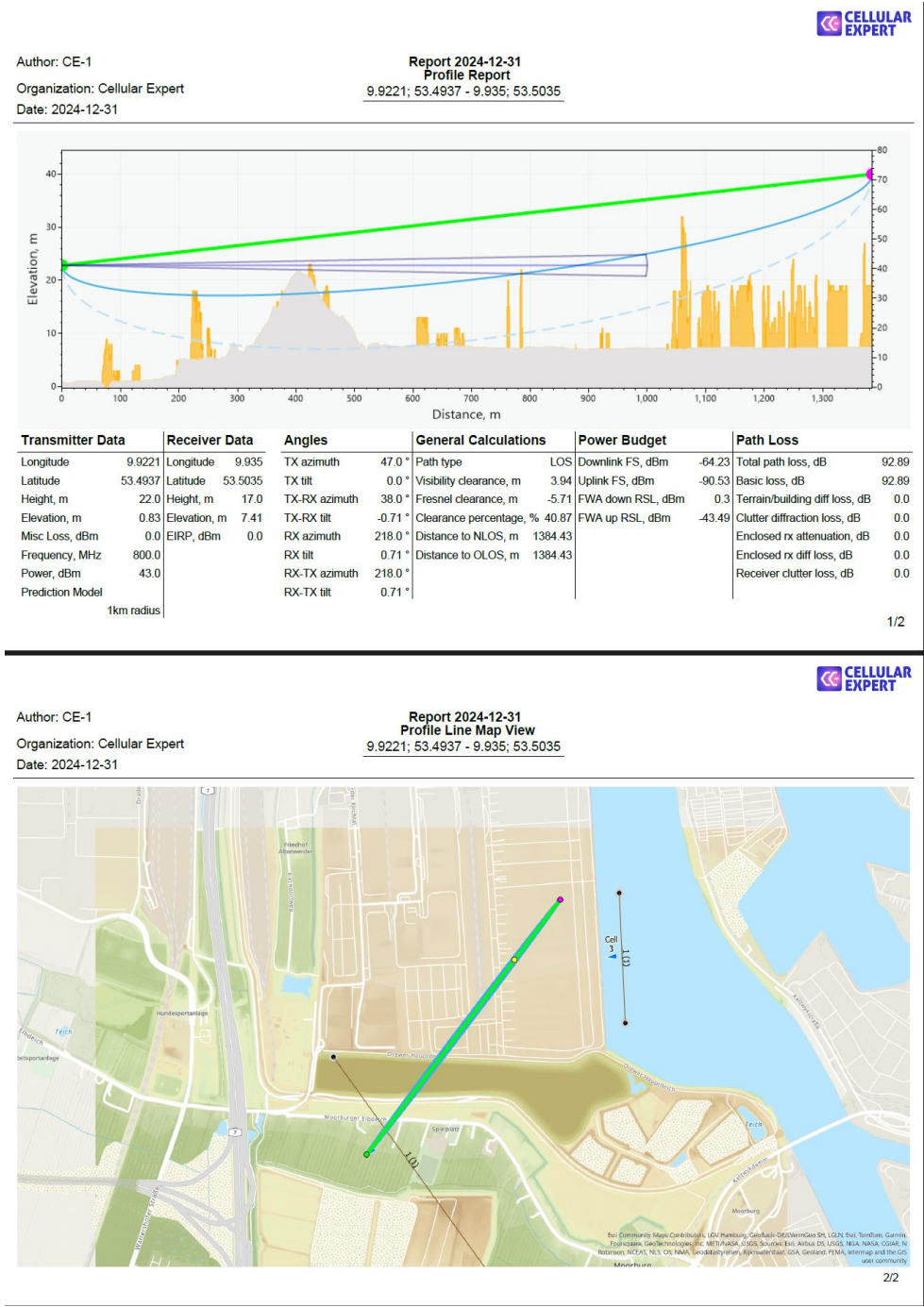
#### 8.1.5 Export (Profile Report)

The input data and calculation results can be automatically transferred into a **Profile Report**. This report will show transmitter/receiver input data, calculation results as well as the Profile plot and map view in which the profile was drawn. The report can be exported in PDF and PL2 formats.



The screenshot shows the 'Profile' window with the 'Export' tab selected. The window has a title bar with a dropdown arrow, a pin icon, and a close button. Below the title bar are tabs for 'Properties', 'Tools', 'Import', 'Export', and 'Settings'. The 'Export' tab contains fields for 'Author' (CE\_Dev), 'Organization' (Cellular Expert), 'Report Name' (Profile Report), and 'Profile Report File Path' (C:\Users\jlekavicius\Desktop\Document). There is a 'PDF' dropdown menu and a folder icon button. Below these is a toggle switch for 'Save result to Docs Manager' which is turned on. A green status message reads: 'Profile Report created successfully!'. At the bottom is an 'Export Profile' button.

The resulting Profile report will look similar to this example:



CELLULAR  
EXPERT

Author: CE-1

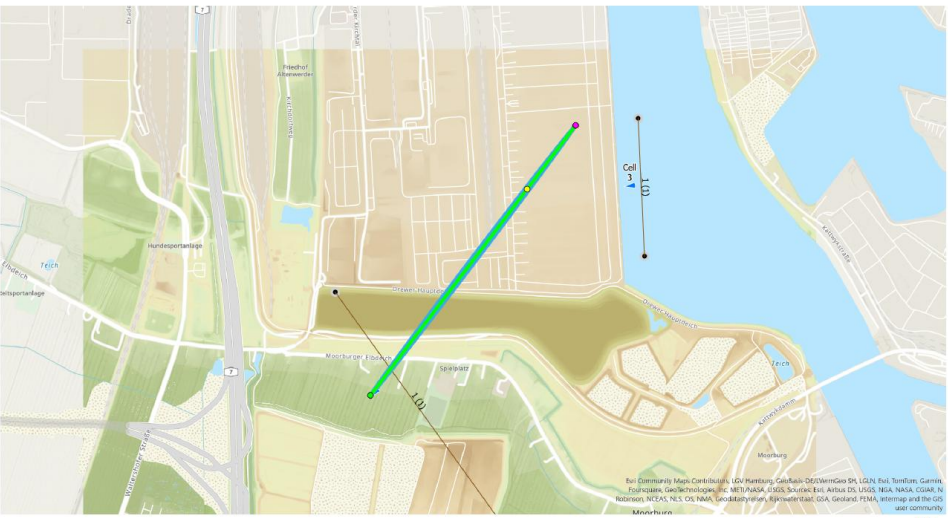
Organization: Cellular Expert

Date: 2024-12-31

Report 2024-12-31

Profile Line Map View

9.9221; 53.4937 - 9.935; 53.5035

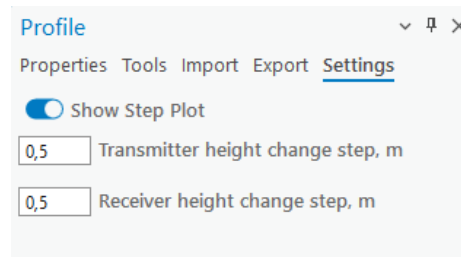


The map shows a green line representing the signal path between two points. The path starts at a point labeled 'Cell 3' and ends at a point labeled 'Cell 1'. The map includes a scale bar indicating 1.00 km. The terrain is shown in shades of green and brown, with a blue area representing water. The map is labeled with various street names and landmarks.

2/2

### 8.1.6 Settings

Currently, you can configure the profile's visual properties and controls in profile settings. The settings can be changed once a profile is drawn.



#### Show Step Plot

Display data points as a series of steps rather than a smooth line or individual points.

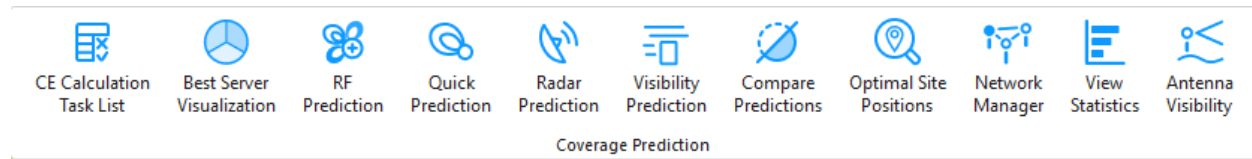
#### Transmitter height change step

The step (in meters) for changing the transmitter height when dragging it in the Profile graph. The default value is 0.5 meters.

#### Receiver height change step

The step (in meters) for changing the receiver height when dragging it in the Profile graph. The default value is 0.5 meters.

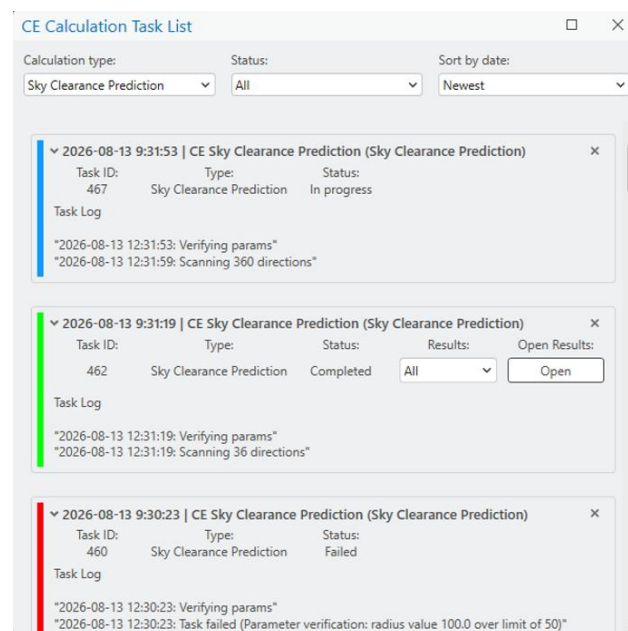
## 9. Coverage Prediction



### 9.1 CE Calculation Task List

All results of the prediction calculations can be found in the **CE Calculation Task List** tab. This includes failed and successful calculations.

Click on the  button to open the **CE Calculation Task List** dialogue.



The task list refreshes automatically once calculation tasks are run. The task status is indicated by three main colors:  blue (in progress),  green (completed), and  red (failed). Calculation tasks can be deleted from the task list by clicking **x** on the right side of the task. To open a result raster, select it from the results dropdown and click **Open Results**.

Filtering by calculation spans these types: Antenna Visibility Prediction, EMF Calculation, Link Prediction, Model Tuning, Optimal Site Positions Calculation, RF Prediction, Radar Prediction, Siren Sound Prediction, Visibility Prediction, and Sky Clearance Prediction.

Filtering by status spans these types: All, in progress, completed, failed.

The tasks can also be sorted by date, either newest or oldest first.

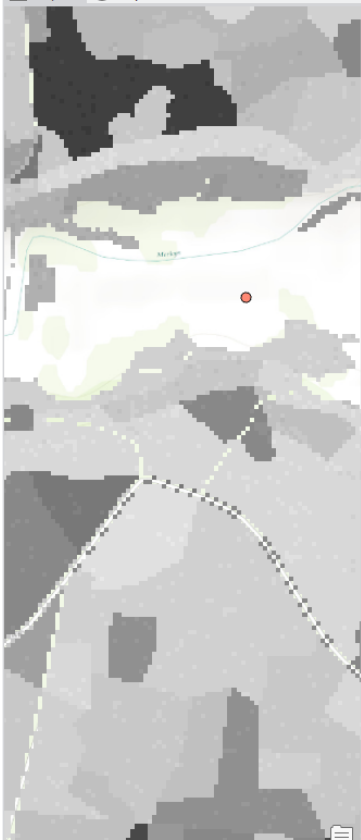
9.2 Quick SAT Link Budget

Click the  button to open the **Quick SAT Link Budget** dialogue.

CE Quick SAT Link Budget tool performs downlink link budget calculations for satellite-to-terminal paths. The calculation includes free-space path loss, atmospheric absorption according to ITU-R P.676, rain attenuation according to ITU-R P.618, satellite EIRP and ground terminal G/T, carrier-to-noise ratio (C/N), and link margin. Results are presented in a dedicated satellite link budget results panel that shows all intermediate values of the budget.

Map

Map\_3D



Quick SAT Link Budget

Calculation settings

Calculation name

Quick SAT Link Budget

Latitude

54.14448

Longitude

24.2221

Ground altitude, m

75

Satellite

Template

Eutelsat 9B

Longitude

9

EIRP, dBW

53

Band

Ku

Ground Terminal

> Antenna: UHX8-127

View Antenna

Terminal G/T, dB/K

20

Link

Carrier frequency, MHz

30000

Polarization

Vertical

Target availability, %

99.5

Receiver bandwidth, MHz

30

Required C/N, dB

7

Calculation Log

Quick SAT Link Budget Results

Geometry

Azimuth, deg:

198.57

Elevation, deg:

26.68

Slant range, km:

38906

Clear-Sky Budget

Satellite EIRP, dBW:

53.0

Free-space loss, dB:

213.8

Gaseous attenuation, dB:

0.78

Ground G/T, dB/K:

20.0

Clear-sky C/N<sub>0</sub>, dBHz:

87.0

Atmospheric (P.618)

Rain attenuation, dB:

1.68

Cloud attenuation, dB:

1.72

Scintillation, dB:

0.49

Total atmospheric, dB:

4.24

Worst-month equiv, %:

98.44

Result

Available C/N, dB:

8.79

Required C/N, dB:

7.00

Margin, dB:

1.79

Status:

LINK CLOSES

9.3 Sky Clearance

Click the  button to open the **Sky Clearance** dialogue.



The Sky Clearance tool determines whether a ground station at a given location can see the required sky, accounting for terrain, buildings, and vegetation. An analysis point is set by clicking the map or by selecting an existing ground station, and results are produced as rasters over a configurable radius around it. Obstruction is derived from elevation and clutter data, and the search distance for blocking terrain is automatically determined by the surrounding relief and the satellite elevation angle.

When the Sky Clearance button is pressed, a map tool will be activated which will let you select a point on the map. Upon selecting this point and pressing the **Run** button, a visual representation of the calculation will be rendered on the map and the relevant data presented in the **CE Calculation Task List**.

**Sky Clearance**

**Calculation settings**

Calculation name  
CE Sky Clearance Prediction

Resolution, m  
10

Radius, km  
1

Check step, deg ⓘ  
10

Ground station template  
Starlink Standard

**Satellite type**  
GNSS

All directions around each point. Result: the angle the antenna must clear to see sky all round.

**Ground Station Settings**

Selected Ground Station

X  
0

Y  
0

Z  
0

Longitude  
0

Latitude  
0

Height above ground, m  
3

Terminal G/T, dB/K  
11.4

Polarization  
Circular

Minimum elevation, deg  
25

Calculation Log

Run

## **Sky Clearance Parameters**

### **Calculation Name**

Name of the calculation that will be displayed in the CE Calculation Task List

### **Resolution**

Resolution for raster calculations. The output rasters will be produced with the indicated cell size.

### **Check step**

Out of 360 degrees, the step used (e.g. every 10 degrees) for obstacle analysis. Lower step results in more accurate calculations but longer calculation time. Value in degrees.

**Ground station template**

Ground Station Template will be used in the prediction calculations.

**Satellite type**

Possible options: GNSS, GEO, LEO. Different type selection results in different calculation types

**Antenna**

Antenna that will be used in the prediction calculations.

**Selected Ground Station (Optional)**

A ground station from which the calculation will be done.

**X**

Coordinate in the projected coordinate system.

**Y**

Coordinate in the projected coordinate system.

**Z**

Coordinate in the projected coordinate system.

**Latitude**

Decimal degrees Y type coordinate in the WGS 1984 geographical coordinate system.

**Longitude**

Decimal degrees X type coordinate in the WGS 1984 geographical coordinate system.

**Height above ground**

Ground station height above the ground in meters.

**Antenna**

Define antenna patterns for the Ground Station object.

**Terminal G/T**

Ratio of antenna gain to system noise temperature. Determines the terminal's receive sensitivity. Value in dB/K.

**Minimum elevation**

Lowest elevation angle at which the terminal can use a satellite. Value in degrees.

**Polarization**

Signal polarization, options: Horizontal, Vertical, or Circular.

**Band**

Frequency band of the terminal, such as Ka, Ku, or C.

**EIRP**

Effective isotropic radiated power of the terminal. Value in dBW.

**Frequency**

Transmit (uplink) frequency. Value in MHz.

**Bandwidth**

Channel bandwidth Value in MHz.

### Required C/N

Minimum carrier-to-noise ratio, needed for the link to close. Value in dB.

### Constellation altitude (LEO)

Orbital height of the LEO constellation above the Earth's surface. Used to determine the slant range to the satellites. Typical values are around 550 km for Starlink and 1200 km for OneWeb. Value in km.

### Target availability

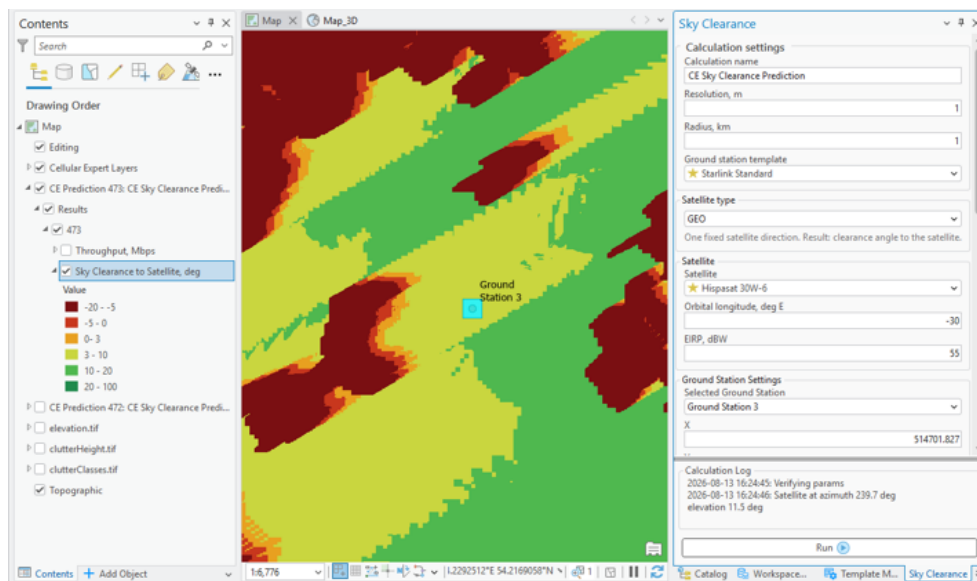
Percentage of an average year for which the link must remain available. Determines how much rain and atmospheric fading the budget allows for: higher availability requires a larger margin and produces a more conservative result.

### Orbital longitude (GEO)

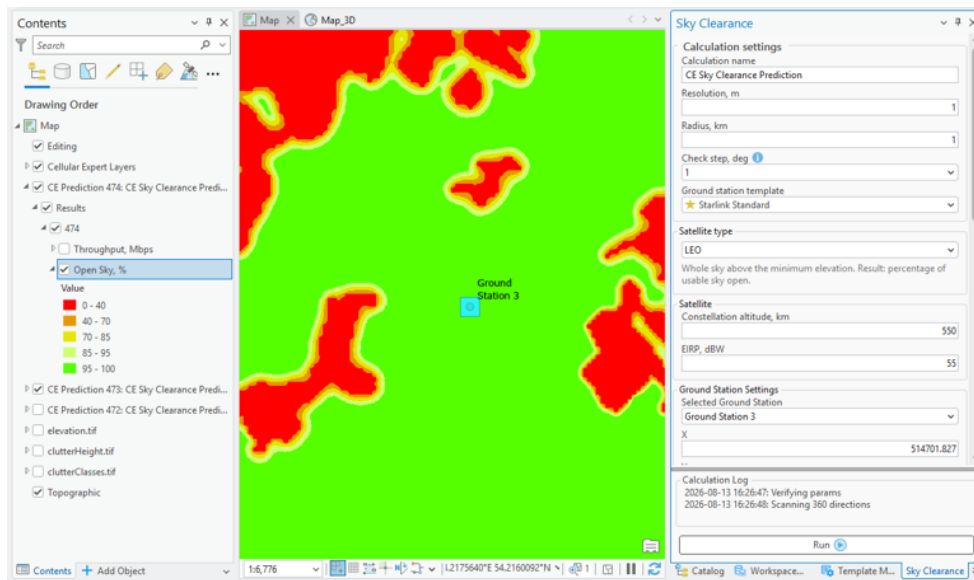
Longitude of the geostationary satellite's orbital slot. Together with the ground station location it determines the elevation and azimuth look angles to the satellite. Negative values indicate positions west of the Greenwich meridian. Value in degrees east.

Results depending on selected mode:

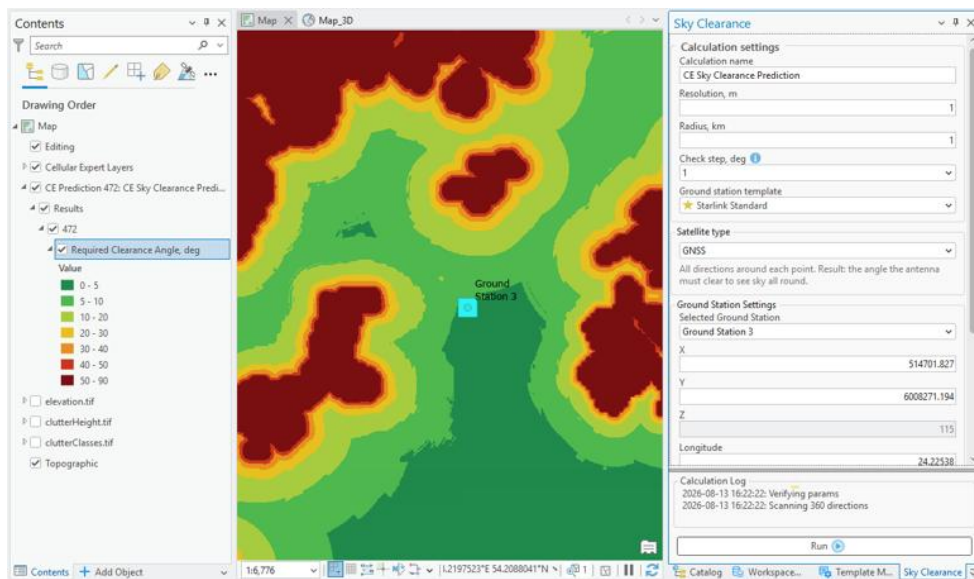
- **GEO:** a geostationary satellite occupies a single fixed direction in the sky. The tool reports the clearance angle to that satellite at every point: positive where the dish sees it, negative where terrain or clutter blocks the line of sight.



- **LEO:** constellations such as Starlink and OneWeb use the whole sky above a minimum elevation angle. The tool reports the percentage of that usable sky which is unobstructed.

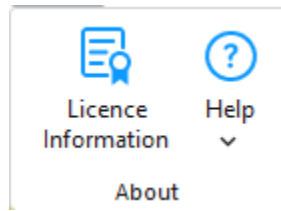


- **GNSS:** for satellite navigation, the tool sweeps every direction around each point and reports the required clearance angle: the elevation above which the sky is open in all directions.



Indicative throughput is also produced for the GEO and LEO modes, combining the link budget with the local obstruction. All rasters are added to the map with Cellular Expert result symbology.

## 10. About



### 10.1 License information



Click the  button to open the License Information dialogue.

License information is a useful resource to see your current version of Cellular Expert for ArcGIS Pro, user key, currently active licenses, and their expiration dates. The license information window is also used when a user is enabling the *CE for ArcGIS Pro* extension on their computer. For more information about license activation see *Activation*.

A screenshot of the 'Cellular Expert License' window. The title bar says 'Cellular Expert License' with standard window controls. Below the title bar, it says 'Cellular Expert 5.0 for ArcGIS Pro'. The window contains several sections: 'Your User Key:' followed by a text input field; 'Your Cellular Expert RCP License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; 'Your Cellular Expert RLP License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; 'Your Cellular Expert EMF License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; 'Your Cellular Expert Sound License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; 'Your Cellular Expert Indoor License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; 'Your Cellular Expert SAT License' with 'Status: VALID' and 'Expiration Date: 2027-07-28 0:00:00'; and 'Your License Activation Key' followed by a large text input field. At the bottom, there is an 'Activate License' button.

## 10.2 Help

### 10.2.1 Documentation

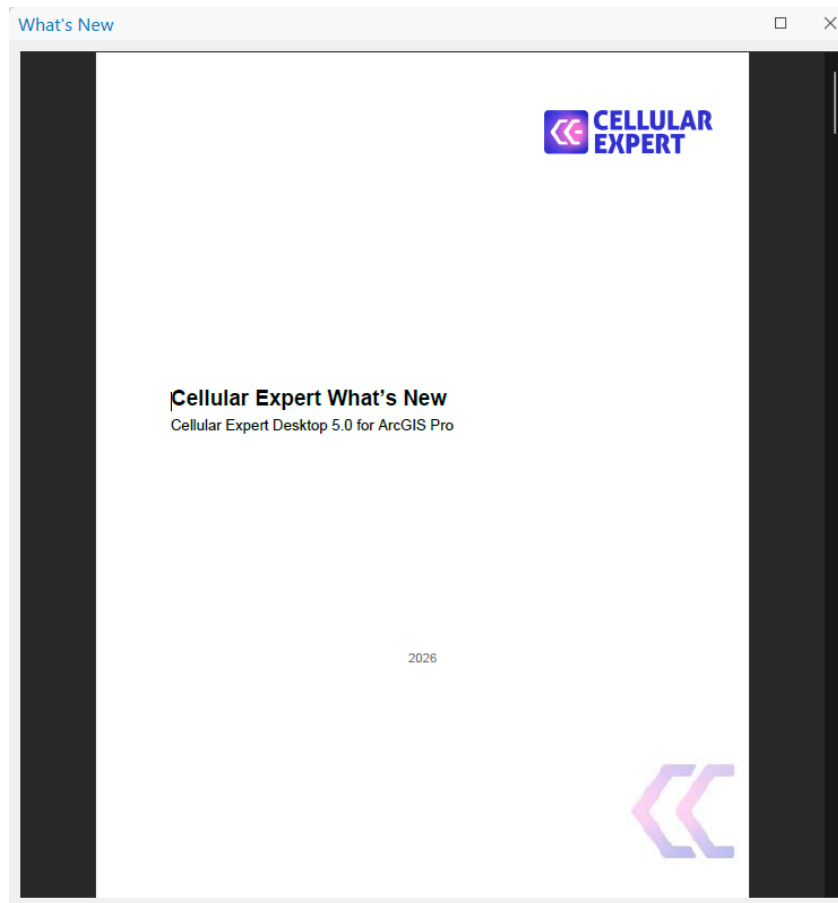
Click the  **Help** button in the **Help** dropdown list to open the **Documentation** of *CE for ArcGIS Pro*.

Here you will find extensive documentation of the add-on. Also, this should be the first place you check when trying to figure out a problem.



### 10.2.2 What's New

Click the  **What's New** button in the **Help** dropdown list to open the **What's New** document. This document is updated for each new release of Cellular Expert for ArcGIS Pro, and here you will find the changelog for the current installed version. This document serves as the introduction of added new features, enhancements, bug fixes, and other changes.



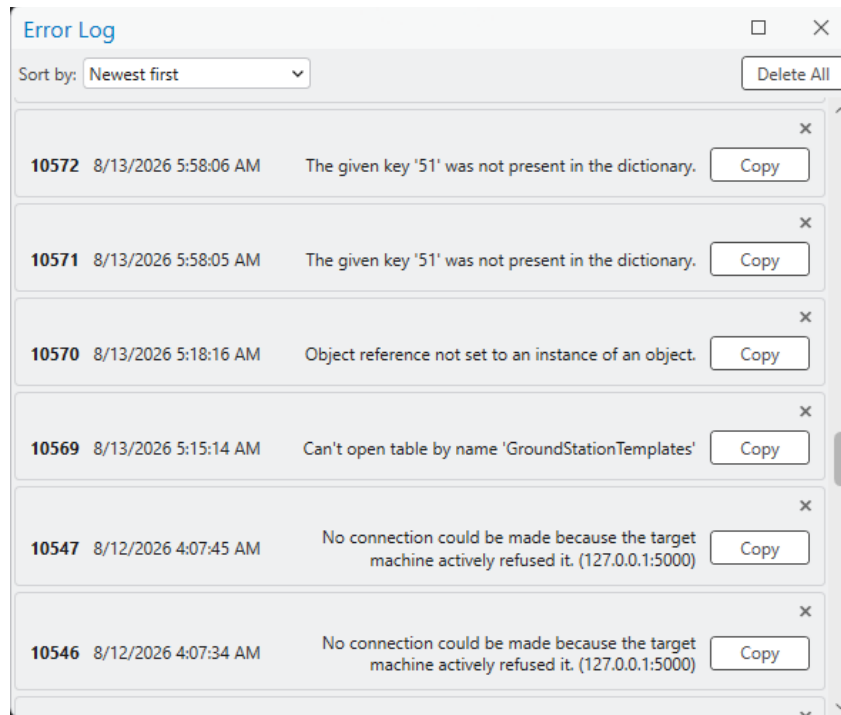
### 10.2.3 Technical Support

Click the  **Technical Support** button in the **Help** dropdown list to open the **Technical Support** page in your default Internet browser. In the Cellular Expert Support page, you can browse the Knowledge base for instructions and solutions related to Cellular Expert for ArcGIS Pro, such as how to prepare geodata for a project. Additionally, you can log in or sign up to submit a new support ticket.

### 10.2.4 Error log

Click the  **Error Log** button in the **Help** dropdown list to open the **Error Log** dialogue.

Here you will find information about errors that have occurred during certain processes of the *CE for ArcGIS Pro* extension's lifetime. These logs are crucial to improving the overall quality of the user experience. When an error happens and you decide to contact a Cellular Expert, you will be asked to send these logs so that the problems you encounter can be patched as soon as possible. The error logs can be sorted by either newest first (default option) or lowest first.

**Copy**

Copy the necessary error information into your clipboard and send it to Cellular Expert. A notification appears on the screen indicating that the information was copied to the clipboard successfully.

**Delete All**

If the error log ever gets too crowded, you can delete all errors from it.



## 11. Technical Support

For any issues, e-mail support ([support@cellular-expert.com](mailto:support@cellular-expert.com)) and we will register your ticket.

When you contact us, please describe your issue precisely and completely. Please mention which steps you already tried, when the problem appears, etc. Attach a screenshot of the Cellular Expert log files.

**Note:** all provided information remains confidential and is used only for solving your issue. Other users cannot see your ticket and it is not published in any other way.

Thank you for using our software.  
Cellular Expert Team

[www.cellular-expert.com](http://www.cellular-expert.com)